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650 Route des Lucioles
F-06921 Sophia Antipolis Cedex - FRANCE

Tel.: +33 4 92 94 42 00 Fax: +33 4 93 65 47 16

Siret N° 348 623 562 00017 - APE 7112B
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Foreword

This Technical Report has been produced by the 3rd Generation Partnership Project (3GPP).

The present document is part of a multi-part TR as described below:

- Part 1: LI for IMS based services. See TR 33.929-1 [10].
- Part 2: LI for IMS based STIR/SHAKEN. See TR 33.929-2 [11].
- Part 3: LI for messaging services. See TR 33.929-3 [12].
- Part 4: LI for data in 5G core. See TR 33.929-4 [13].
- Part 5: LI for ID association caching. See TR 33.929-5 [14].
- Part 6: LI for IMS based RCS. See TR 33.929-6 [15].
- Part 7: LI location acquisition capabilities. See TR 33.929-7 [16].
- Part 8: LI for MMS (present document).
- Part 9: LI for PTC. See TR 33.929-9 [17].

In Part 1, the illustrations of LI for IMS-based services focus on the LI aspects of IMS sessions which include the architecture topologies and the call flows covering basic sessions, redirected sessions, target non-local ID, conferencing, roaming (local break-out and home-routed roaming).

In Part 2, the illustrations of LI for IMS based STIR/SHAKEN focus on various STIR/SHAKEN related LI reporting scenarios.

In Part 3, the illustrations of LI for messaging services focus on SMS over NAS and SMS over IP.

In Part 4, the illustrations of LI for data in 5G core focus on the LI aspects of data interception focussing on the PDU session related events including the handover scenarios.

In Part 5, the illustrations of LI for ID association caching focus on the capabilities used to provide the temporary identity to permanent identity (and vice-versa) associations known to the CSP network. The illustrations include the architectural concepts and few examples of use-cases encountering various timing scenarios.

In Part 6, the illustrations of LI for IMS based RCS focus on the architecture topologies and the call flows when the RCS service is offered by the CSP or by a Third Party Provider.

In Part 7, the illustrations of LI location acquisition capabilities focus on the conceptual overview and the flow diagrams for location reporting, LALS and location acquisition.

In Part 8 (present document), the illustrations of LI for MMS focus on the conceptual overview and the flow diagrams for MMS.

In Part 9, the illustrations of LI for PTC focus on the conceptual overview and the flow diagrams for MCPTT and PoC, the two commonly referred to as Push to Talk over Cellular (PTC).

Introduction

Unlike the previous generation of LI technical specifications, the latest LI technical specifications (TS 33.126 [2], TS 33.127 [3], TS 33.128 [4]) contain only the normative part of the requirements. However, the implementers may need additional information such as architectural diagrams, conceptual scenarios, flow-diagrams and additional background information to better understand the requirements in those specifications. The present document collects the relevant informative annexes from the previous generation of LI technical specifications and then expands them to include the similar level of details as for latest technologies such as 5G.

1 Scope

The present document provides architectural diagrams, conceptual scenarios, flow-diagrams, examples, and other background information which can be useful to implement the LI functions defined in TS 33.126 [2], TS 33.127 [3] and TS 33.128 [4].

The present document covers the LI for MMS that include the illustrations covering the conceptual overview and the flow diagrams for MMS.

2 References

The following documents contain provisions which, through reference in this text, constitute provisions of the present document.

- References are either specific (identified by date of publication, edition number, version number, etc.) or non-specific.
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[1] 3GPP TR 21.905: "Vocabulary for 3GPP Specifications".

[2] 3GPP TS 33.126: "Lawful Interception requirements".

[3] 3GPP TS 33.127: "Lawful Interception (LI) architecture and functions".

[4] 3GPP TS 33.128: "Protocol and procedures for Lawful Interception (LI)".

[5] 3GPP TS 24.147: "Conferencing using the IP Multimedia (IM) Core Network (CN) subsystem; Stage 3".

[6] 3GPP TS 23.501: "System architecture for the 5G System".

[7] 3GPP TS 24.174: "Support of multi-device and multi-identity in the IP Multimedia Subsystem (IMS); Stage 3".

[8] 3GPP TS 23.228: "IP Multimedia Subsystem (IMS), Stage 2".

[9] 3GPP TR 33.928: "ADMF logic for provisioning Lawful Interception (LI)".

[10] 3GPP TR 33.929-1: "Lawful Interception (LI) implementation guidance; LI for IMS-based services".

[11] 3GPP TR 33.929-2: "Lawful Interception (LI) implementation guidance; LI for IMS based STIR/SHAKEN".

[12] 3GPP TR 33.929-3: "Lawful Interception (LI) implementation guidance; LI for messaging services".

[13] 3GPP TR 33.929-4: "Lawful Interception (LI) implementation guidance; LI for data in 5G core".

[14] 3GPP TR 33.929-5: "Lawful Interception (LI) implementation guidance; LI for ID association caching".

[15] 3GPP TR 33.929-6: "Lawful Interception (LI) implementation guidance; LI for IMS based RCS".

[16] 3GPP TR 33.929-7: "Lawful Interception (LI) implementation guidance; LI location acquisition capabilities".

[17] 3GPP TR 33.929-9: "Lawful Interception (LI) implementation guidance; LI for PTC".

[18] 3GPP TS 23.140: "Multimedia Messaging Protocol. Functional Description. Stage 2".

[19] OMA-TS-MMS_ENC-V1_3-20110913-A: "Multimedia Messaging Service Encapsulation Protocol".

3 Definitions of terms, symbols and abbreviations

3.1 Terms

For the purposes of the present document, the terms given in 3GPP TR 21.905 [1] and the following apply. A term defined in the present document takes precedence over the definition of the same term, if any, in 3GPP TR 21.905 [1].

3.2 Symbols

For the purposes of the present document, the following symbols apply:

None.

3.3 Abbreviations

For the purposes of the present document, the abbreviations given in 3GPP TR 21.905 [1] and the following apply. An abbreviation defined in the present document takes precedence over the definition of the same abbreviation, if any, in 3GPP TR 21.905 [1].

3GPP	3rd Generation Partnership Program
ADMF	Administration Function
AS	Application Server
CC	Content of Communication
CSP	Communication Service Provider
GPSI	Generic Public Subscription Identifier
ID	Identity
LEA	Law Enforcement Agency
LEMF	Law Enforcement Monitoring Facility
LI	Lawful Interception
LICF	Lawful Interception Control Function
LI_HI1	LI_Handover Interface 1
LI_HI2	LI_Handover Interface 2
LI_HI3	LI_Handover Interface 3
LI_T1	Lawful Interception Triggering Interface 1
LI_T3	Lawful Interception Triggering Interface 3
LI_X1	Lawful Interception Internal Interface 1
LI_X2	Lawful Interception Internal Interface 2
LI_X3	Lawful Interception Internal Interface 3
LIPF	LI Provisioning Function
LTE	Long Term Evolution
MCPTT	Mission Critical Push To Talk
MDF	Mediation and Delivery Function
MDF2	Mediation and Delivery Function 2
MDF3	Mediation and Delivery Function 3
MMBox	Multimedia Mailbox
MMS	Multimedia Message Service
MMS VAS	MMS Value Added Service
MSCF	Messaging Service Control Function
PLMN	Public Land Mobile Network
PoC	Push to talk over Cellular
POI	Point Of Interception

PTC	Push to Talk over Cellular
RCS	Rich Communication Suite
SHAKEN	Signature-based Handling of Asserted information using toKENs
SUPI	Subscriber Permanent Identifier
UDM	Unified Data Management
VPLMN	Visited PLMN
xCC	LI_X3 Communications Content
xIRI	LI_X2 Intercept Related Information

4 Illustration of LI for MMS

4.1 General

Multimedia Messaging Service (MMS) is a session-based messaging service handled by the Network Functions (NFs) referred to as MMS Relay and MMS Server together referred to as MMS Relay/Server in the present document.

An MMS client can interact with variety of other clients via the Multimedia (MM) messages. The details of MMS architecture are described in TS 23.140 [18].

In the subsequent clauses, the application within MMS client's device is referred to as:

- MMS User Agent when the MM message originated from, or received at with, a phone number (e.g. MSISDN) as the identity.
- Email Client when the MM message originated from, or received at with, an Email Address as the identity.

NOTE: The email message may be treated as an MM message in the illustrations of the present document.

The LI aspects of MMS are defined in TS 33.127 [3], TS 33.128 [4] and TR 33.928 [9]. The term MMS LI used in the present document refers to the LI functions provided by the MMS Relay/Server. The OMA specification [19] and TS 33.128 [4] refer to the MMS Relay/Server as MMS Proxy-Relay.

An overview of the MMS LI architecture is depicted in figure 4.1-1 below.

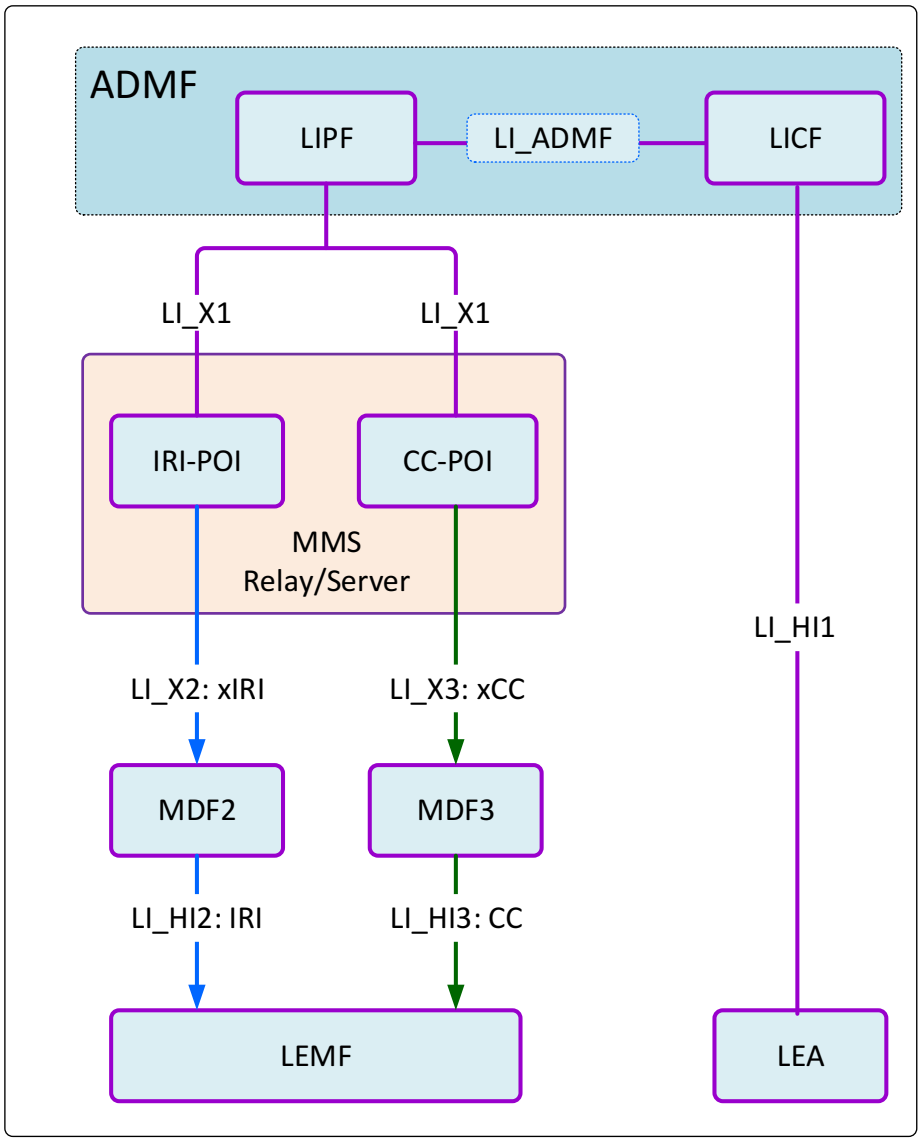


Figure 4.1-1: An overview of MMS LI architecture

With a warrant to intercept the MMS related communications of a target, the LEA submit an intercept request to the CSP (i.e. to the ADMF) over the LI_HI1 interface. The LIPF within the ADMF provisions the target identities to IRI-POI and the CC-POI present in the MMS Relay/Server. The target identity can be local to the MMS Relay/Server or can be a non-local ID.

The IRI-POI generates the xIRIs from the MM1/MM4 messages (see clause 4.2.3) and the send the same over the LI_X2 interface to the MDF2. The MDF2 forwards the same to the LEMF over the LI_HI2 interface as IRI messages.

The CC-POI generates the xCC from the MM1/MM4 messages that contain the media information and the send the same over the LI_X3 interface to the MDF3. The MDF3 forwards the same to the LEMF over the LI_HI3 interface as IRI messages.

The illustrations given in the present document focus on scope of target identifiers (see clause 4.3) and scope of MMS messages (see clause 4.4) for generating the xIRIs. No further illustration of xCC generation is presented.

4.2 MMS background with LI scope

4.2.1 Background

4.2.1.1 MMS functional elements and the interfaces

Figure 4.2.1.1-1 below gives an overview of the MMS concepts and the interfaces based on the architecture defined in TS 23.140 [18]. To illustrate the concept, two MMS User Agents (A and B) are shown.

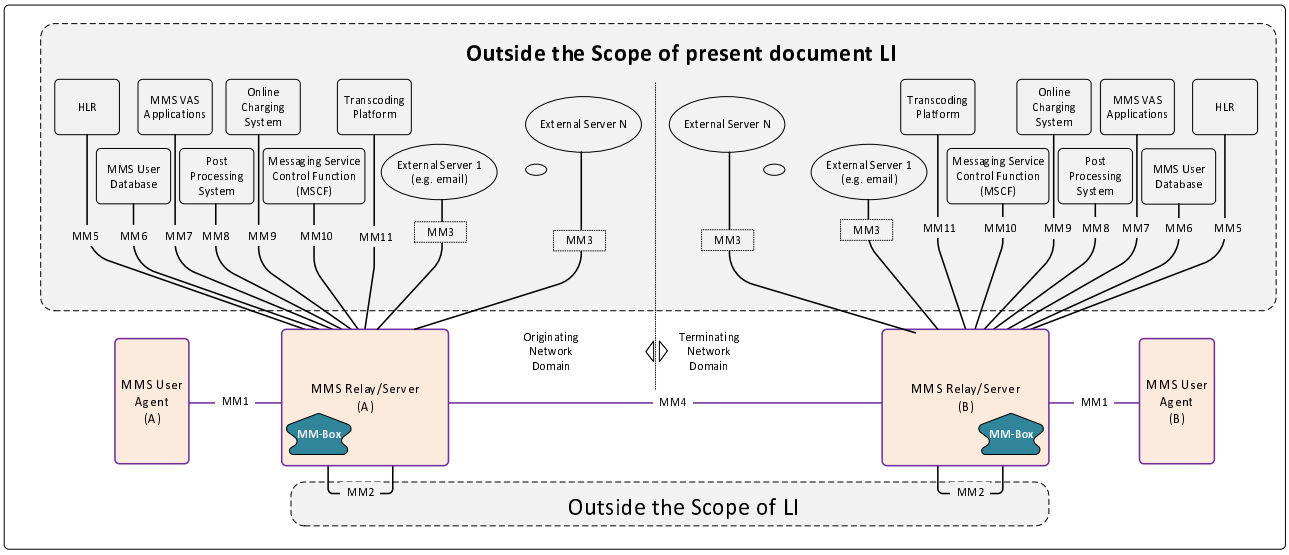


Figure 4.2.1.1-1: MMS concepts per TS 23.140

NOTE: MMS User Agent (A) and MMS User Agent (B) may also be served by the same MMS Relay/Server.

The MM messages to or from an MMS User Agent flow through the MMS Relay Server.

To send an MM message, the originating MMS User Agent submits the MM message to the MMS Relay/Server over the MM1 interface. The MM message may be stored at the MMBox.

The terminating MMS Relay/Server would store the MM message and notify the terminating MMS User Agent with the link for the stored MM message over MM1 interface. The terminating MMS User Agent would then (can also be automatic) retrieve the stored MM message. The MM message may also be stored/retrieved using an MMBox.

If two different MMS Relay/Servers serve the two involved MMS User Agents, then the MM messages are transferred between the two MMS Relay/Servers over the MM4 interface.

Once the message is retrieved (at the terminating end), a delivery report may be sent from the terminating MMS Relay/Server to the originating MMS User Agent via originating MMS Relay/Server.

A read report may be sent from the terminating MMS User Agent to the originating MMS User Agent via MMS Relay/Servers.

Alternatively, the terminating MMS User Agent may forward or delete the MM message stored at the MMS Relay/Server without retrieving. Also, the terminating MMS Relay/Server may cancel the retrieved MM message at the terminating MMS User Agent.

The MM message can also be stored, uploaded, viewed, retrieved through the MMBox.

Other functional entities and the interfaces may come into play when an MM message cannot be transferred over the MM4 interface. For example, the MM message may also be initiated by an MMS VAS application which would be handled in the same except that MMS application would submit the message via MM7 interface. Likewise, if the MM message happens to be an email message, then the originating MMS Relay/Server would forward that email message via the External Server (i.e. email server) over the MM3 interface. Likewise, a terminating MMS Relay/Server may

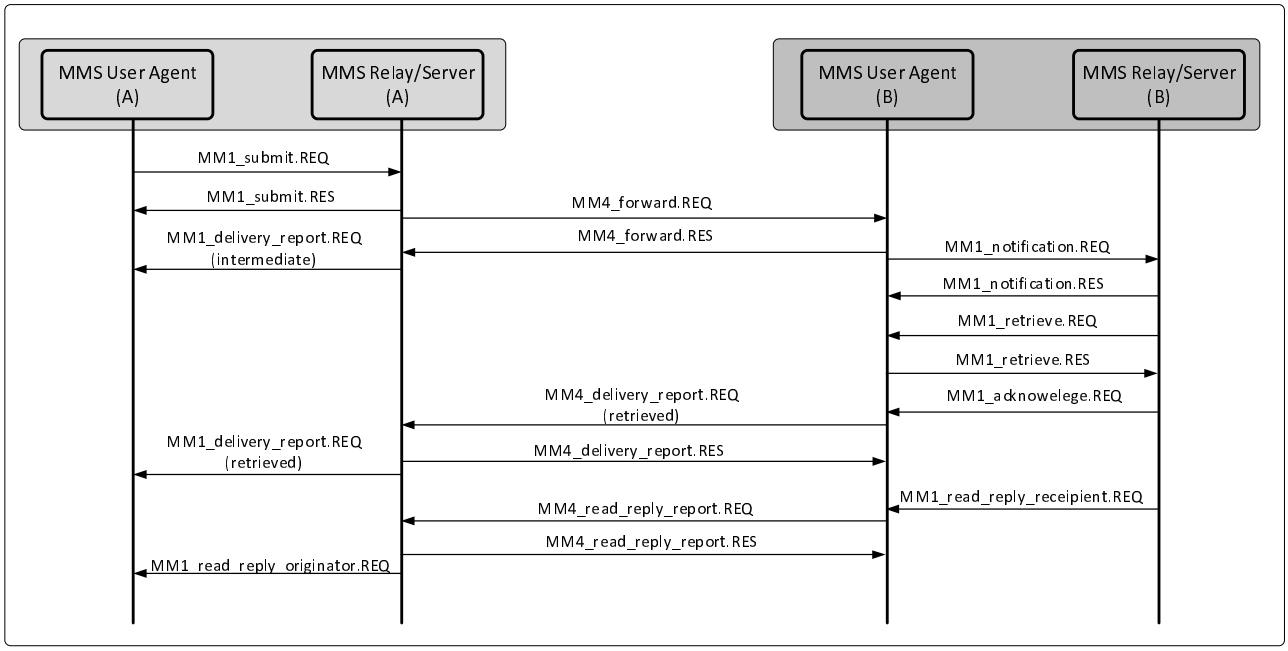
receive an email message destined to an MMS User Agent via the External Server (i.e. Email Server) over the MM3 interface.

For LI, however, the focus is on the message exchanges between the MMS Relay/Server and the MMS User Agent over the MM1 interface and the message exchanges between the two MMS Relay/Servers over the MM4 interface. Therefore, all interfaces and functional entities shown in figure 4.2.1.1-1 except MM1, MM4, MMS User Agent, MMS Relay/Server are outside the scope of present document even though the MM3 has an indirect influence and hence, referenced along with the External Server (Email).

4.2.1.2 Flow diagrams

4.2.1.2.1 MMS User Agent A to MMS User Agent B

An abstract level flow diagram depicting the flow of an MM message between the two MMS User Agents served by two different MMS Relay/Servers is shown in figure 4.2.1.2.1-1 (based of TS 23.140 [18]).



4.2.1.2.1-1: MMS flow (A-to-B) at an abstract level (TS 23.140 [18] based)

Brief description of the flow

The MMS User Agent (A) submits the MM message over the MM1 interface using the MM1_submit.REQ message. The originating MMS Relay/Server (A) forwards the MM message to the terminating MMS Relay/Server (B) over the MM4 interface using the MM4_forward.REQ message. An intermediate delivery report is sent to the MMS User Agent (A) using the MM1_delivery_report (intermediate) message.

The MMS Relay/Server (B) notifies the MMS User Agent (B) using the MM1_notification.REQ message. The MMS User Agent (B) retrieves the MM message using the MM1_retrieve.REQ message.

A delivery report is sent by the MMS Relay/Server (B) when it receives a confirmation of the retrieval (MM1_acknowledge.REQ) using the MM4_delivery_report.REQ (retrieved) message. The MMS Relay/Server (A) sends the MM1_delivery_report.REQ (retrieved) message to the MMS User Agent (A).

The MMS User Agent (B) sends a read-reply report using the MM1_read_reply_receipient.REQ to the MMS Relay/Server (B). The MMS Relay/Server (B) forwards the same to the MMS Relay/Server (A) using the MM4_read_reply_report.REQ message. The MMS Relay/Server (A) forwards the read-reply report to the MMS User Agent (A) using the MM1_read_reply_originator.REQ message.

Not elaborated or not shown

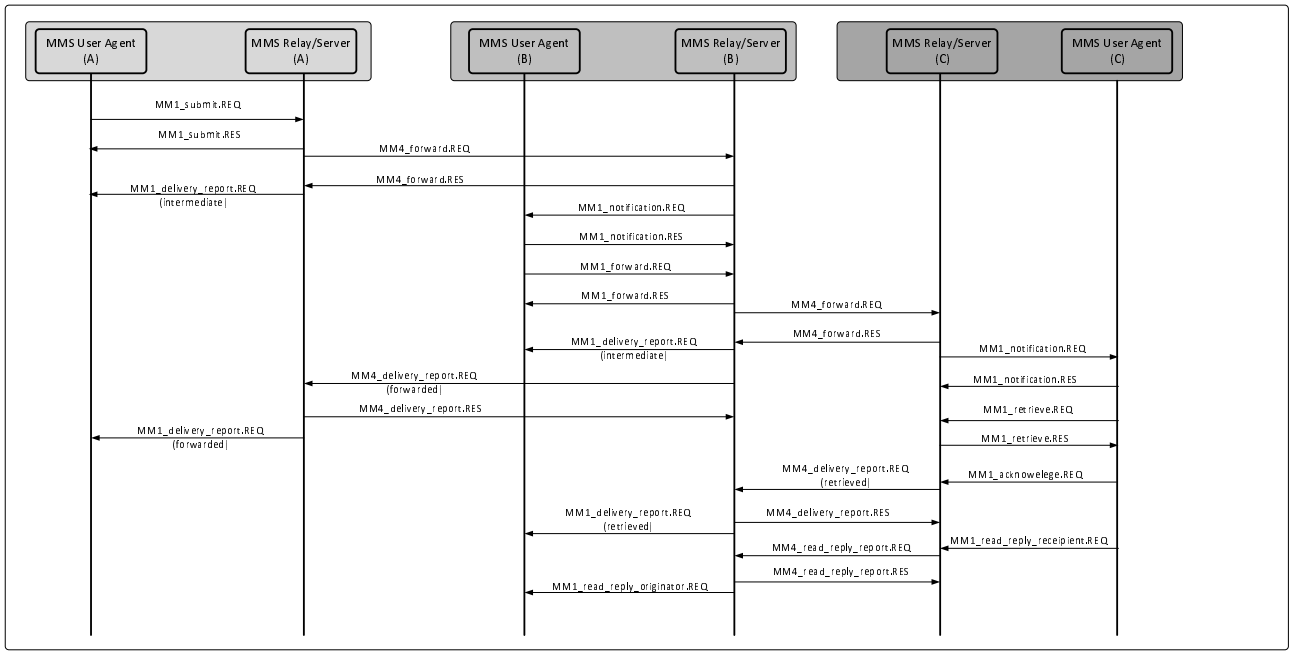
Sending of delivery reports are conditional, in the sense, sent only when requested so by the MMS User Agent (A) and permitted (except the intermediate) by the MMS User Agent (B). Sending of the read-reply report is also conditional in the sense, sent by the MMS User Agent (B), if requested by the MMS User Agent (A).

The retrieval of the MM message at the terminating end can also be automatic, in the sense, without any user action. When it is not automatic, other alternate steps mentioned in clause 4.2.1.1 are possible (e.g. forward).

The MMBox, at either end of the flow (i.e. within the MMS Relay/Servers) may be used. The MMBox related steps are not shown in figure 4.2.1.2.1-1.

4.2.1.2.2 MMS User Agent A to MMS User Agent B forwarded to MMS User Agent C

An abstract level flow diagram depicting the flow of an MM message between the three MMS User Agents served by three different MMS Relay/Servers is shown in figure 4.2.1.2.2-1.



4.2.1.2.2-1: MMS flow (A-to-B-to-C) at an abstract level (TS 23.140 [18] based)

Brief description of the flow

The MMS User Agent (A) submits the MM message over the MM1 interface using the MM1_submit.REQ message. The originating MMS Relay/Server (A) forwards the MM message to the MMS Relay/Server (B) over the MM4 interface using the MM4_forward.REQ message. An intermediate delivery report is sent to the MMS User Agent (A) using the MM1_delivery_report (intermediate) message.

The MMS Relay/Server (B) notifies the MMS User Agent (B) using the MM1_notification.REQ message. The MMS User Agent (B) initiates the forwarding to MMS User Agent (C) using the MM1_forward.REQ message. The MMS Relay/Server (B) forwards the MM message to the MMS Relay/Server (C) over the MM4 interface using the MM4_forward.REQ message. An intermediate delivery report is sent to the MMS User Agent (B) using the MM1_delivery_report (intermediate) message. A forwarded delivery report is sent toward the MMS User Agent (A) by the MMS Relay/Server (B) using the MM4_delivery_report.REQ (forwarded). The MMS Relay Server (A) sends the MM1_delivery_report.REQ (forwarded) to the MMS User Agent (A).

The MMS Relay/Server (C) notifies the MMS User Agent (C) using the MM1_notification.REQ message. The MMS User Agent (C) retrieves the MM message using the MM1_retrieve.REQ message.

A delivery report is sent by the MMS Relay/Server (C) when it receives a confirmation of the retrieval (MM1_acknowledge.REQ) using the MM4_delivery_report.REQ (retrieved) message. The MMS Relay/Server (B) sends the MM1_delivery_report.REQ (retrieved) message to the MMS User Agent (B).

The MMS User Agent (C) sends a read-reply report using the MM1_read_reply_receipient.REQ to the MMS Relay/Server (C). The MMS Relay/Server (C) forwards the same to the MMS Relay/Server (B) using the

MM4_read_reply_report.REQ message. The MMS Relay/Server (B) forwards the read-reply report to the MMS User Agent (B) using the MM1_read_reply_originator.REQ message.

Not elaborated or not shown

Sending of delivery reports are conditional, in the sense, for A-to-B leg, sent only when requested so by the MMS User Agent (A) permitted (except the intermediate) by the MMS User Agent (B). Likewise, for B-to-C leg, sent only when requested so by the MMS User Agent (B) permitted (except the intermediate) by the MMS User Agent (C). Sending of the read-reply report (for B-to-C leg) is also conditional in the sense, sent by the MMS User Agent (C), if requested by the MMS User Agent (B).

The retrieval of the MM message at the terminating end (MMS User Agent (C) can also be automatic, in the sense, without any user action. When it is not automatic, other alternate steps mentioned in clause 4.2.1.1 are possible (e.g. further forward).

The MMBox, at all three ends of the flow (i.e. within the MMS Relay/Server), may be used. The MMBox related steps are not shown in figure 4.2.1.2.2-1.

4.2.2 LI scope

For LI, the interfaces MM1, MM4 and the functional elements MMS Relay/Server (with the name MMS Relay-Proxy) and MMS User Agent are referenced in TS 33.128 [4]. All other interfaces and the functional elements depicted in figure 4.2.1.1-1 are out of scope even though the MM messages to and from the MMS User Agents may traverse through other interfaces to other functional entities (e.g. MM3 interface to/from External Email Server, MM7 interface from VAS applications).

Within the CSP's MMS domain, the MMS Relay/Server provides the IRI-POI and CC-POI functions. The IRI-POI and CC-POIs are provisioned with the target identifiers and deliver the xIRIs and the xCC to the MDF2 and MDF3 respectively. The MDF2 delivers the IRI messages to the LEMF over the LI_HI2 interface and MDF3 delivers the CC over the LI_HI3 interface to the LEMF.

The following aspects are applicable to MMS LI, partly, as specified in TS 33.127 [4] and TS 33.128 [4]:

- The target identities can be an E.164 number, an Email Address, GPSI (MSISDN, NAI), IMSI and SUPI (IMSI, NAI). See NOTE 1.
- The Email Address can only be used as a target non-local ID.
- The MM message originated from an MMS User Agent or destined to an MMS User Agent are routed through the MMS Relay/Server.
- The MM message originated from an Email Client or destined to an Email Client may not go through the MMS Relay/Server.
- An incoming MMS session can be forwarded by the receiving MMS User Agent, or by the receiving Email Client, or can be redirected by the receiving MMS Relay/Server that serves the MMS User Agent.
- MM messages can be stored at, uploaded to, retrieved from, or deleted from the MMBox for the MMS User Agent.

The originating, redirecting, or redirected-to MMS User Agent can be a target or target non-local ID in the immediate prior or next network. When MMS User Agent is roaming, the MMS LI is applicable only to the HPLMN.

NOTE 1: As per TS 23.140 [18], only a phone number and an Email Address are the identities used in the MM messages. In the list, MSISDN, GPSI (MSISDN) and E.164 numbers represent the phone number.

The scope of LI based on the nature of the target is illustrated in figure 4.2.2-1 below.

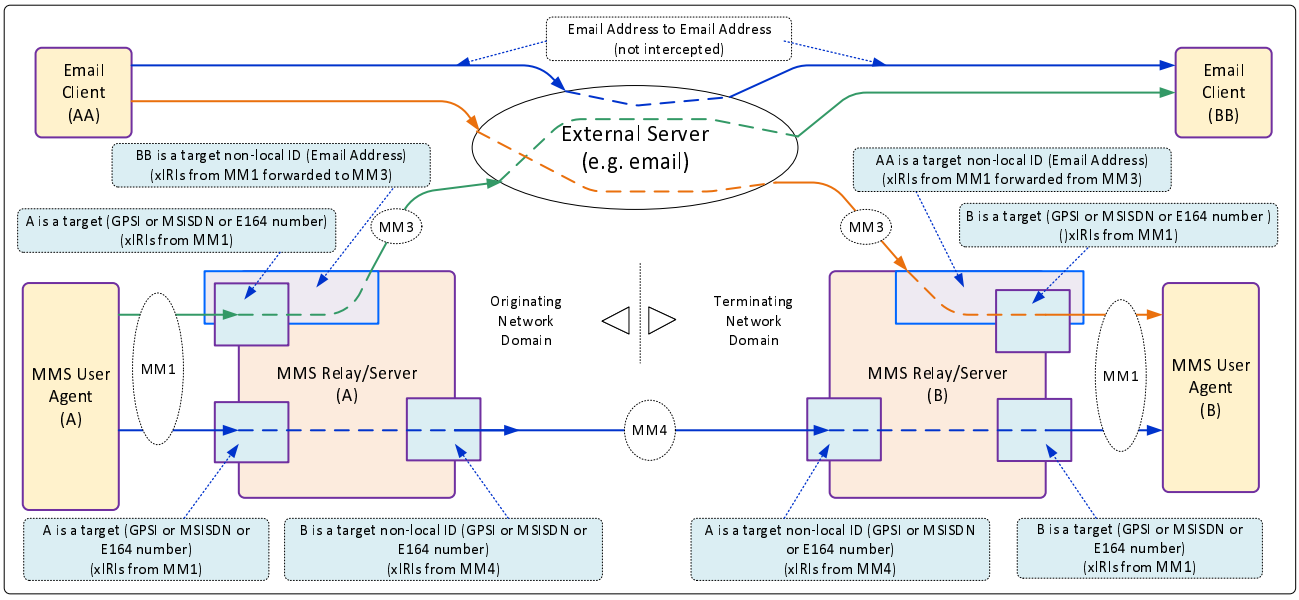


Figure 4.2.2-1: Scope of LI within the MMS Relay/Server

NOTE 2: The Email Server shown in figure 4.2.2-1 can be different for Email Client AA and Email Client BB.

NOTE 3: MMS User Agent (A) and MMS User Agent (B) may be served by the same MMS Relay/Server.

When the target is identified with a phone number (GPSI (MSISDN), MSISDN or E.164 number) and is local to the MMS Relay/Server, the LI functions use the messages from the MM1 interface for generating the xIRIs.

When the target is identified with a phone number (GPSI (MSISDN), MSISDN or E.164number) and is non-local to MMS Relay/Server, the LI functions use the messages from the MM4 interface for generating the xIRIs.

When the target non-local ID is identified with an Email Address, the LI functions use the messages from MM1 interface transferred to or from the MM3 interface.

The Email Client interacts with the Email Server (shown as External Server in figure 4.2.2-1) to send or receive an email message. If an MM message does not pass through the MMS Relay/Server, then such an MM message cannot be intercepted. For example, as shown in figure 4.2.2-1, an email message transferred between the two Email Clients does not hit the MMS Relay/Server at either end of the flow, and therefore, such email messages are not intercepted.

When either end of an email message is an MMS User Agent, then that MM message does pass through the MMS Relay/Server and hence, can be intercepted.

TS 33.128 refers to the OMA specification [19] for the MM1 message details and refer to TS 23.140 [18] for the MM4 message details. The MM1 messages may also be uploaded to, stored at, or retrieved from the MMBox (not shown in figure 4.2.2-1).

4.3 Scope of target identities

4.3.1 General

As described in clause 4.2 and as illustrated in figure 4.2-2, the scope of MMS LI depends on the source and destination identity of the MM message.

For the purpose of illustrations given in the present document, a phone number is referenced either as a "phone number" or as an "MSISDN" that may be any of the following:

- MSISDN.
- GPSI (MSISDN).
- E.164 number.

Furthermore, in the illustrations, two forms of email addresses are referenced:

- Generic Email Address.
- MSISDN based Email Address.

The generic Email Address is used by the Email Client. The MSISDN based Email Address is a specific case where the MSISDN (i.e. any of E.164 number, GPSI (MSISDN), or MSISDN) of the remote MMS User Agent is in the user-id part of that Email Address. The MSISDN based Email Address is used to address the remote MMS User Agent in an email format and allows an Email Client to send an MM message to an MMS User Agent and vice-versa. The domain name of the MSISDN based Email Address may belong to the CSP and may be helpful in finding the MMS Relay/Server that serves the MMS User Agent.

An IMSI or SUPI (IMSI) can be the targets for MMS LI in the following circumstances:

- During the initial authorization, the MMS/Relay Server is able to associate the target identities IMSI or SUPI (IMSI) to an MSISDN. In this case, all MM messages using that MSISDN are intercepted.

- When the MMS Relay/Server as a part of discovering the destination MMS/Relay Server gets the IMSI or SUPI (IMSI) of the terminating MMS User Agent. In that case, the MM messages destined to the associated MSISDN are intercepted. In this case, such an IMSI or GPSI (IMSI) are treated as target non-local ID.

For the purpose of illustrations, the NAI (SUPI or GPSI) are treated as a generic Email Address. The present document limits the discussions of NAI (SUPI or GPSI), IMSI, or SUPI (IMSI) based target identities to this clause.

The subsequent clauses focus only on phone number (E.164 number, MSISDN, GPSI (MSISDN)) and the Email Address (generic or MSISDN based) based target identities.

In the illustrations, the indicated addresses are at the SMTP layer for the messages transferred over the MM3 interface (between MMS Relay/Server and the External Server) and are at the application layer for the messages transferred over the MM1 (between MMS User Agent and MMS Relay/Server) and the MM4 (between the two MMS Relay/Servers) interfaces.

4.3.2 Topology views - without redirection

4.3.2.1 General

In the illustrations shown in the following sub-clauses, the MMS User Agent (A) or Email Client (AA) sends an MM message to MMS User Agent (B) or to an Email Client (BB). The two MMS User Agents may be served by different MMS Relay Servers or by the same MMS Relay Server. Likewise, the two Email Clients may have the same or different Email domains.

4.3.2.2 Multiple MMS Relay/Servers

4.3.2.2.1 Overview

Figure 4.3.2.2.1-1 illustrates the scenarios of different identities that are applicable for MMS LI when there is no redirection and two MMS Relay/Servers are involved in the MMS flow.

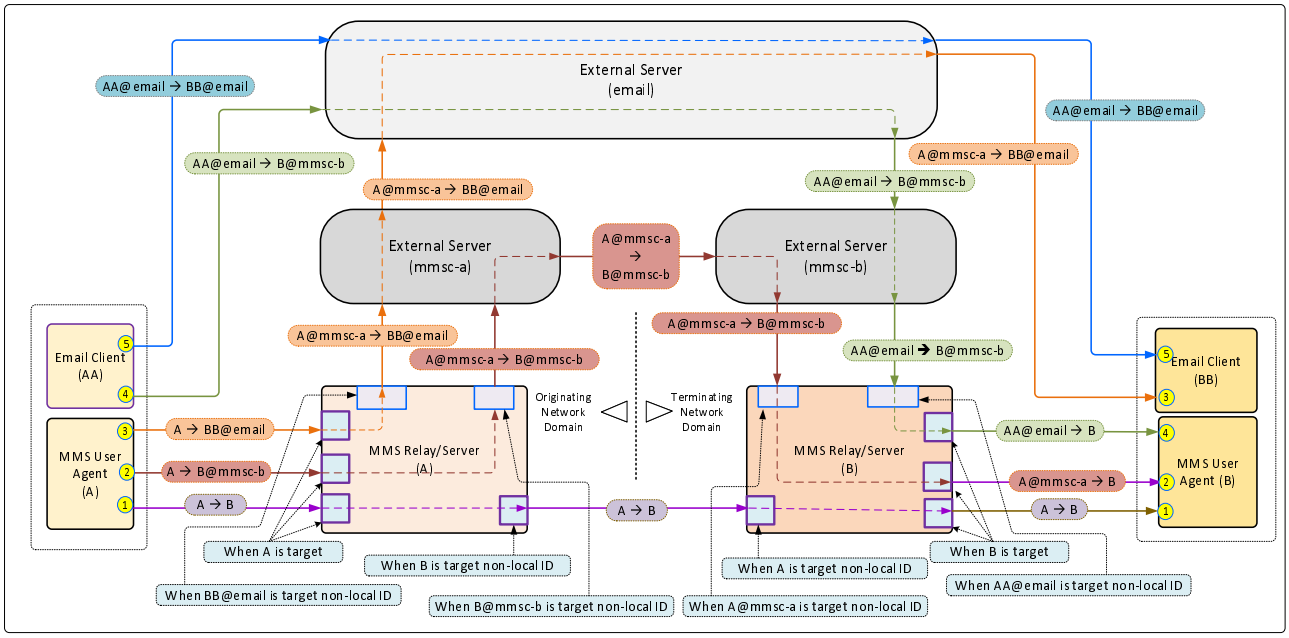


Figure 4.3.2.2.1-1: Scope of target identities without redirection (multiple Relay/Servers)

As shown in figure 4.3.2.2.1-1, MM messages originated from an Email Client (i.e. generic Email Address) do not hit the MMS Relay/Server at the originating end. Likewise, MM messages destined to an Email Client (i.e. generic Email Address) do not hit the MMS Relay Server at the terminating end. Therefore, the MM messages from a generic Email Address to a generic Email Address are outside the scope of MMS LI.

However, MM messages do hit the MMS Relay Server when at least one end of the MM message is a phone number allowing the Email Address to be the identity of the remote user. Because of this, an Email Address can only be a target non-local ID.

The phone number may be in an email format within the MM message (i.e. MSISDN based Email Address). In this case, the local MMS Relay/Server (originating end or terminating end) adds or removes the domain name. The email domain name could identify the MMS Relay/Server.

Figure 4.3.2.2.1-1 shows five scenarios with permutations of different identities used at either end of the MM message flow. These five scenarios are further illustrated in the clause 4.3.2.2.2.

4.3.2.2.2 Scenario descriptions

4.3.2.2.2.1 General

Figure 4.3.2.2.1-1 illustrates the following five scenarios:

- MM message originated from a phone number, destined to a phone number, terminated at a phone number. This is referenced as MSISDN to MSISDN – case 1.
- MM message originated from a phone number, destined to an MSISDN based Email Address, terminated at a phone number. This is referenced as MSISDN to MSISDN – case 2.
- MM message originated from a phone number, destined to a generic Email Address, terminated at an Email Client (generic Email Address). This is referenced as MSISDN to generic Email Address.
- MM message originated from at an Email Client (generic Email Address), destined to an MSISDN based Email Address, terminated at a phone number. This is referenced as generic Email Address to MSISDN.
- MM message originated from at an Email Client (generic Email Address), destined to a generic Email Address, terminated at an Email Client (generic Email Address). This is referenced as generic Email Address to generic Email Address.

In MSISDN to MSISDN (case 2), the termination end of an MSISDN based Email Address is a phone number (MSISDN, GPSI (MSISDN), E.164 number). From the originating end perspective, the destination is an MSISDN

based Email Address. From the terminating end's perspective, the originating point is an MSISDN based Email Address.

4.3.2.2.2.2 Scenario 1: MSISDN to MSISDN - case 1

MMS User Agent (A) uses the own phone number (MSISDN, GPSI (MSISDN), E.164 number) and addresses the destination (MMS User Agent (B)) also with a phone number (MSISDN, GPSI (MSISDN), E.164 number).

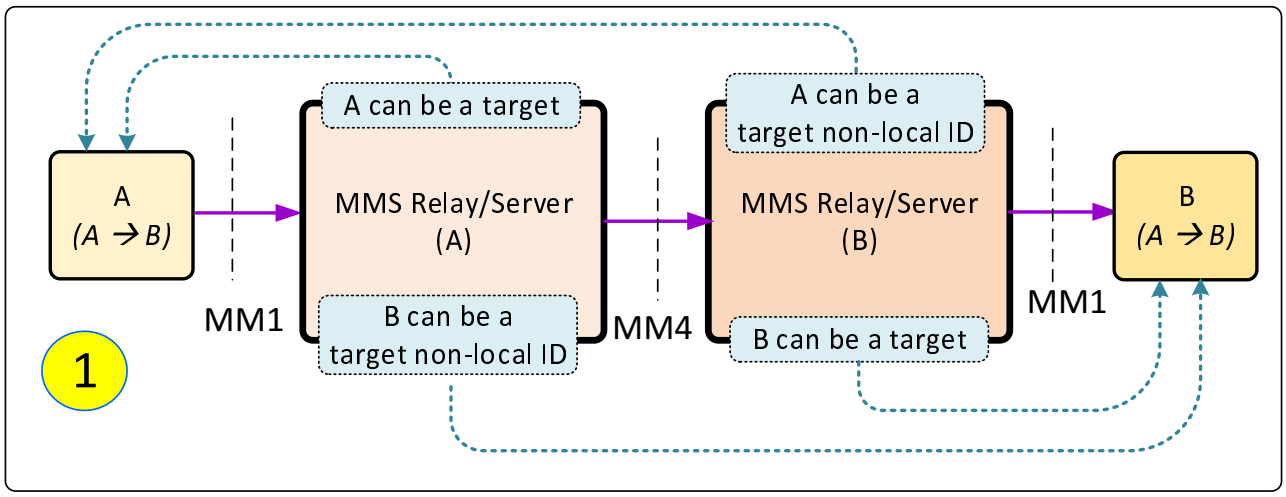


Figure 4.3.2.2.2.2-1: MM message from MSISDN (A) to MSISDN (B) – case 1

In this scenario, phone number A can be a target in the MMS Relay/Server (A) and can be a target non-local ID in the MMS Relay/Server (B). Likewise, phone number B can be a target in the MMS Relay/Server (B) and can be a target non-local ID in the MMS Relay/Server (A).

4.3.2.2.2.3 Scenario 2: MSISDN to MSISDN – case 2

MMS User Agent (A) uses the own phone number (MSISDN, GPSI (MSISDN), E.164 number) and addresses the destination (MMS User Agent (B)) with an MSISDN based Email Address (shown as B@mmsc-b).

The MMS Relay/Server (A) adds the domain name (shown as @mmsc-a) to the A's identity (shown as A@mmsc-a) before forwarding the MM message to the External Server serving the domain mmsc-a.

The MMS Relay/Server (B) would remove the domain name (shown as @mmsc-b) from B@mmsc0b, at the terminating end from B@mmsc-b.

The phone number B is seen as an MSISDN based Email Address in the MMS Relay/Server (A) and the phone number A is seen as an MSISDN based Email Address in the MMS Relay Server (B).

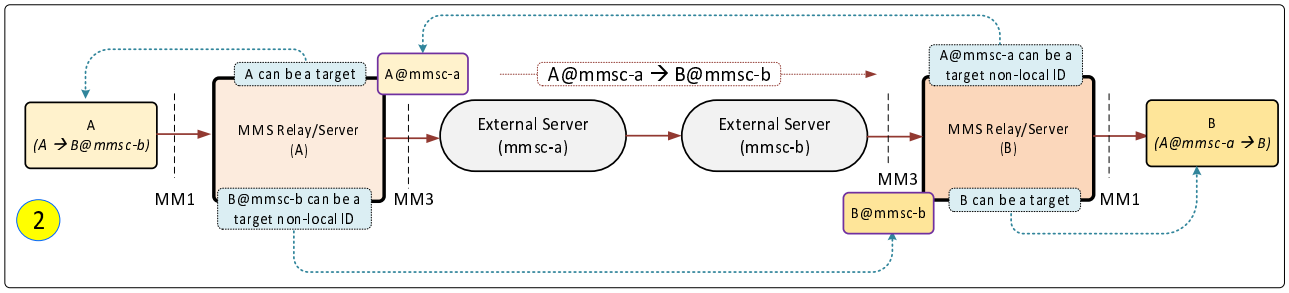


Figure 4.3.2.2.2.3-1: MM message from MSISDN (A) to MSISDN (B) – case 2

In this scenario, the phone number A can be a target in the MMS Relay/Server (A) and A@mmsc-a can be a target non-local ID in the MMS Relay/Server (B). Likewise, phone number B can be a target in the MMS Relay/Server (B) and B@mmsc-b can be a target non-local ID in the MMS Relay/Server (A).

4.3.2.2.2.4 Scenario 3: MSISDN to generic Email Address

MMS User Agent (A) using the own phone number (MSISDN, GPSI (MSISDN), E.164 number) sends an MM message to a generic Email Address (shown as BB@email). This MM message goes from the external server (email) to the Email Client BB.

The MMS Relay/Server (A) adds the domain name (shown as @mmsc-a) to the A's identity (shown as A@mmsc-a) before forwarding the MM message to the External Server serving the domain mmsc-a.

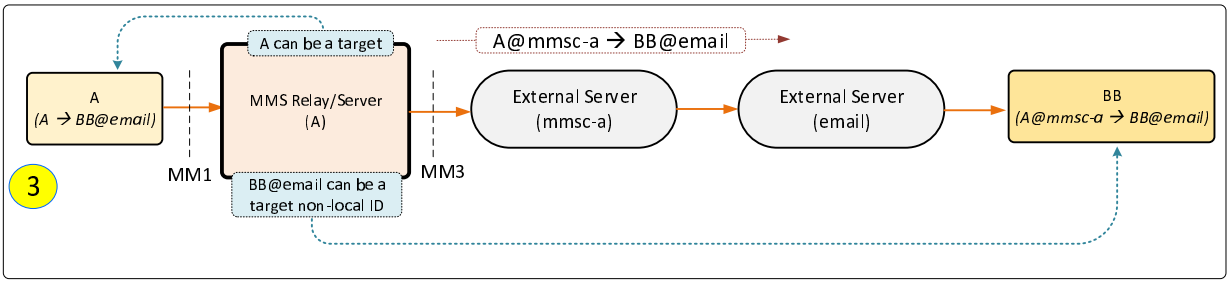


Figure 4.3.2.2.2.4-1: MM message from MSISDN (A) to generic Email Address (BB)

In this scenario, the phone number A can be a target in the MMS Relay/Server (A). The Email Address BB@email can be a target non-local ID in the MMS Relay/Server (A).

4.3.2.2.2.5 Scenario 4: Generic Email Address to MSISDN

Email Client (shown as AA) sends an MM message to a MSISDN based Email Address (shown as B@mmsc-b).

The MMS Relay/Server (B) would remove the domain name (shown as @mmsc-b) at the terminating end.

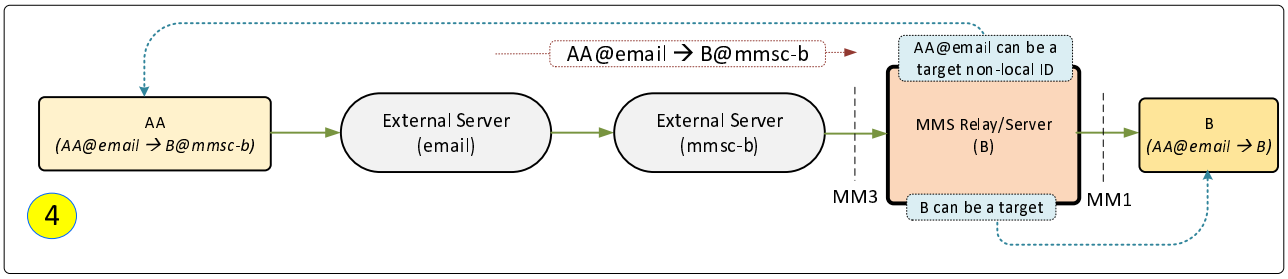


Figure 4.3.2.2.2.5-1: MM message from generic Email Address (AA) to MSISDN (B)

In this scenario, the phone number B can be a target in the MMS Relay/Server (B). The Email Address AA@email can be a target non-local ID in the MMS Relay/Server (B).

4.3.2.2.2.6 Scenario 5: Generic Email Address to generic Email Address

Email Client AA sends an email message to Email Client BB.

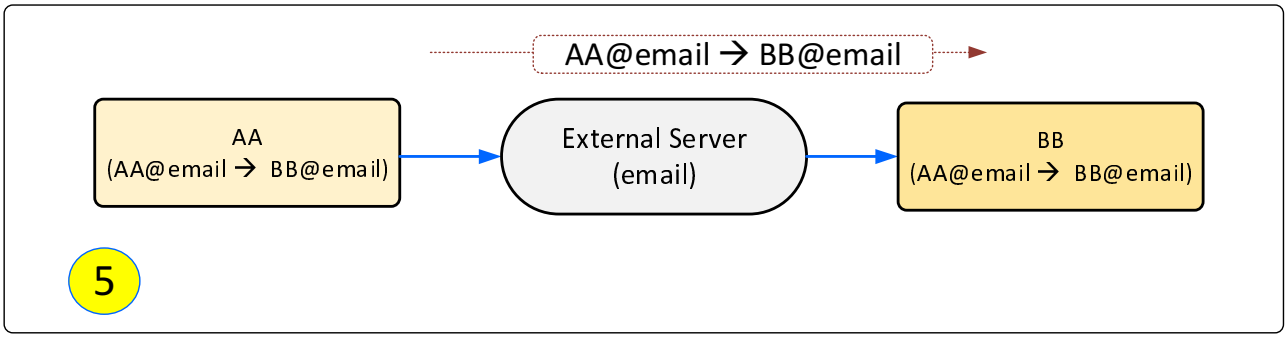


Figure 4.3.2.2.2.6-1: Email message from generic Email Address (AA) to generic Email Address (BB)

MMS LI does not apply to this scenario, since the email message from AA@email to BB@email does not hit the MMS/Relay Server.

4.3.2.3 One MMS Relay/Server

4.3.2.3.1 Overview

Figure 4.3.2.3.1-1 illustrates the cases of different identities that are applicable for MMS LI when there is no redirection and only MMS Relay Server is involved in the MMS flow.

In this case, it is presumed that MMS User Agent (A) and MMS User Agent (B) are served by the same MMS Relay/Server.

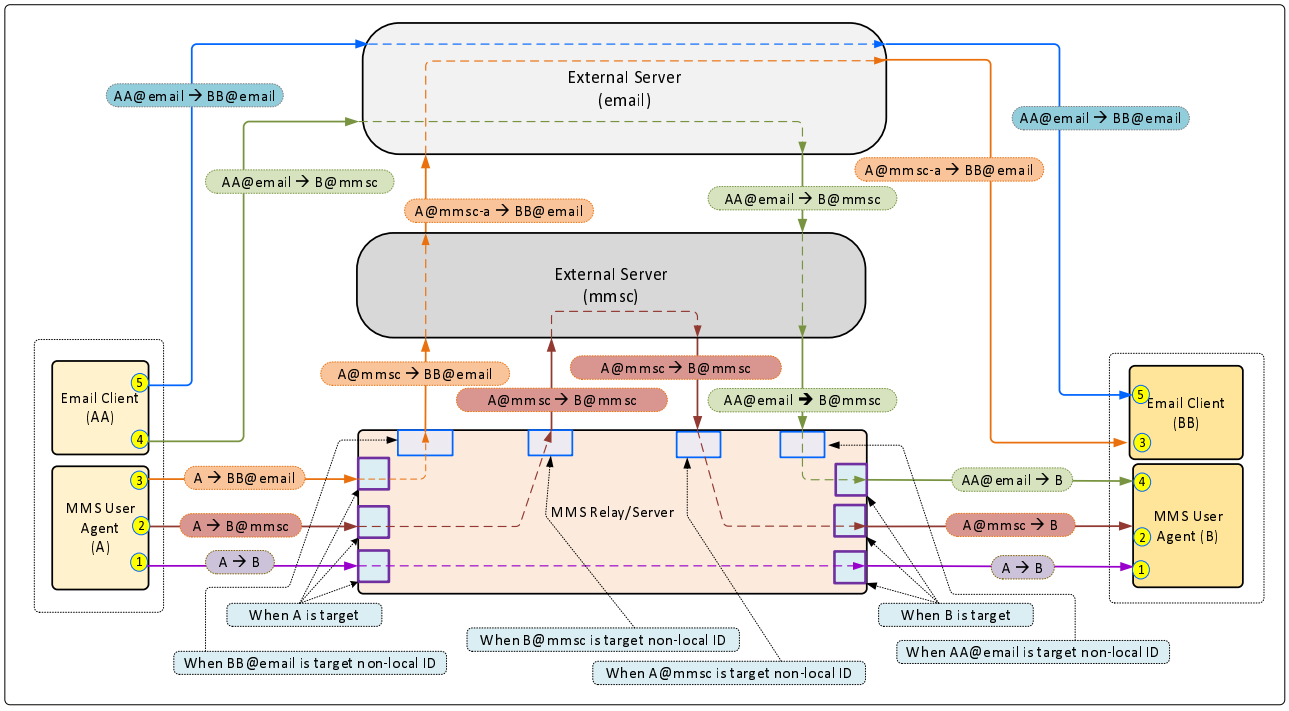


Figure 4.3.2.3.1-1 Scope of target identities without redirection (one Relay/Server)

As shown in figure 4.3.2.3.1-1, MM messages originated from an Email Client (i.e. generic Email Address) do not hit the MMS Relay/Server at the originating end. Likewise, the MM messages destined to an Email Client (i.e. generic Email Address) do not hit the MMS Relay Server at the terminating end. Therefore, the MM messages from a generic Email Address to a generic Email Address are outside the scope of MMS LI. However, the MM messages do hit the MMS Relay Server when at least one end of the MM message is a phone number (MSISDN, GPSI (MSISDN), E.164 number).

As illustrated in clause 4.3.2.2, the phone number may be in an email format within the MM message (i.e. MSISDN based Email Address). In this case, the local MMS Relay/Server (originating end or terminating end) adds or removes the domain name. The email domain name could identify the MMS Relay/Server. Even though the MMS User Agent (A) and MMS User Agent (B) are served by the same MMS Relay Server, when an MSISDN based Email Address is used for either of them, they can only be a target non-local ID. When an MMS User Agent sends or receives an MM message from a generic Email Address, that generic Email Address can be a target not local ID.

The five scenarios illustrated in figure 4.3.2.3.1-1 are further described in clause 4.3.2.3.2.

4.3.2.3.2 Scenario descriptions

4.3.2.3.2.1 General

The same five scenarios described in clause 4.3.2.2.2 are applicable to the illustrations shown in figure 4.3.2.3.1-1 as well.

4.3.2.3.2.2 Scenario 1: MSISDN to MSISDN - case 1

MMS User Agent (A) uses the own phone number (MSISDN, GPSI (MSISDN), E.164 number) and addresses the destination (MMS User Agent (B)) also with a phone number (MSISDN, GPSI (MSISDN), E.164 number).

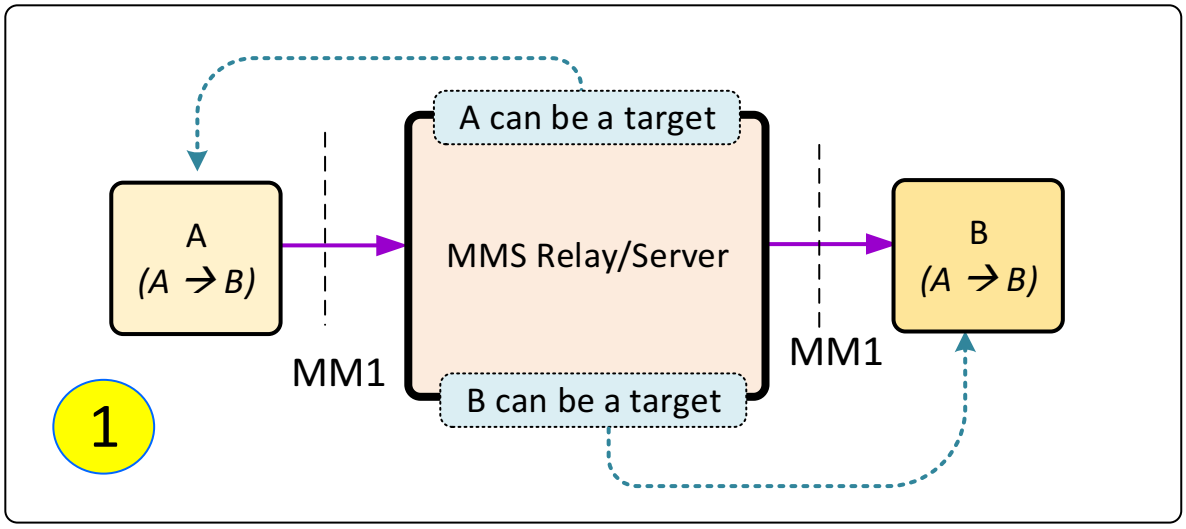


Figure 4.3.2.3.2.2-1: MM message from MSISDN (A) to MSISDN (B) - case 1

In this scenario, phone number A and the phone number B can be the targets in the MMS Relay/Server.

4.3.2.3.2.3 Scenario 2: MSISDN to MSISDN - case 2

MMS User Agent (A) uses the own phone number (MSISDN, GPSI (MSISDN), E.164 number) and addresses the destination (MMS User Agent (B)) with an MSISDN based Email Address (shown as B@mmsc).

The MMS Relay/Server (A) adds the domain name (shown as @mmsc) to the A's identity (shown as A@mmsc) before forwarding the MM message to the External Server serving the domain mmsc.

The MMS Relay/Server (B) would remove the domain name (shown as @mmsc) at the terminating end.

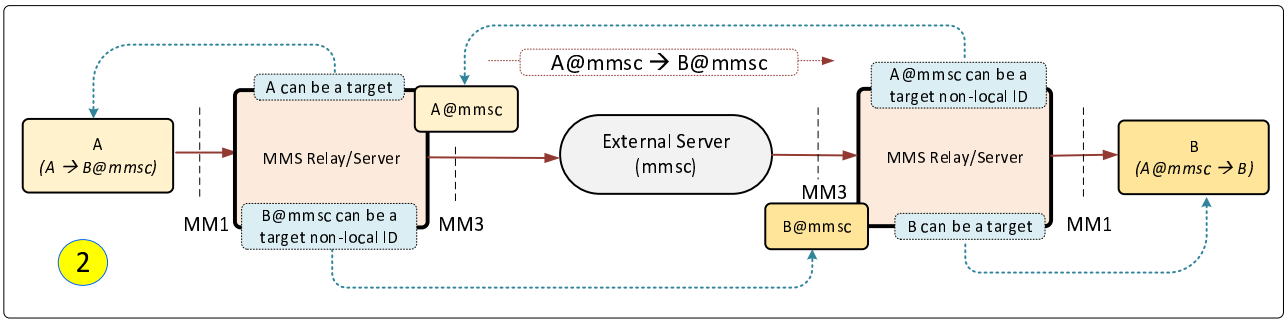


Figure 4.3.2.3.2.3-1: MM message from MSISDN (A) to MSISDN (B) – case 2

In this scenario, the phone number A and the phone number B can be the targets in the MMS Relay/Server and MSISDN base Email Address A@mmsc and the MSISDN based Email Address B@mmsc can be the target non-local IDs in the MMS Relay/Server.

Note that it is the same MMS Relay/Server at the two ends of the MM message path.

4.3.2.3.2.4 Scenario 3: MSISDN to generic Email Address

MMS User Agent (A) using the own phone number (MSISDN, GPSI (MSISDN), E.164 number) sends an MM message to a generic Email Address (shown as BB@email). This MM message goes from the external server (email) to the Email Client BB.

The MMS Relay/Server (A) adds the domain name (shown as @mmsc) to the A's identity (shown as A@mmsc) before forwarding the MM message to the External Server serving the domain mmsc.

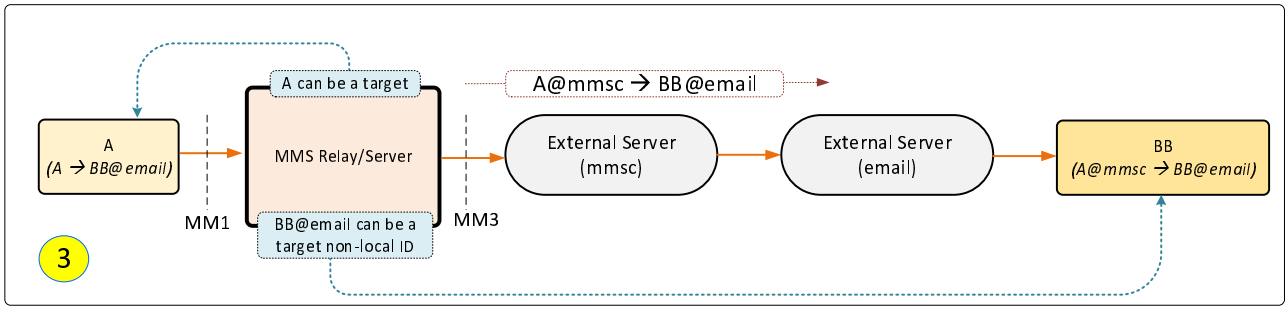


Figure 4.3.2.3.2.4-1: MM message from MSISDN (A) to generic Email Address

In this scenario, the phone number A can be a target in the MMS Relay/Server. The Email Address BB@email can be a target non-local ID in the MMS Relay/Server.

4.3.2.3.2.5 Scenario 4: Generic Email Address to MSISDN

Email Client (shown as AA) sends an MM message to an MSISDN based Email Address (shown as B@mmsc).

The MMS Relay/Server removes the domain name (shown as @mmsc) at the terminating end.

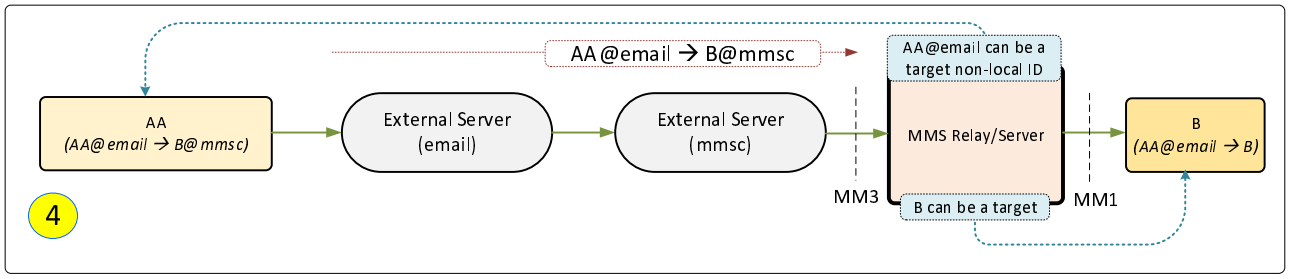


Figure 4.3.2.3.2.5-1: MM message from generic email Address to MSISDN

In this scenario, the phone number B can be a target in the MMS Relay/Server. The Email Address AA@email can be a target non-local ID in the MMS Relay/Server.

4.3.2.3.2.6 Scenario 5: Generic Email Address to generic Email Address

Email Client AA sends an email message to Email Client BB.

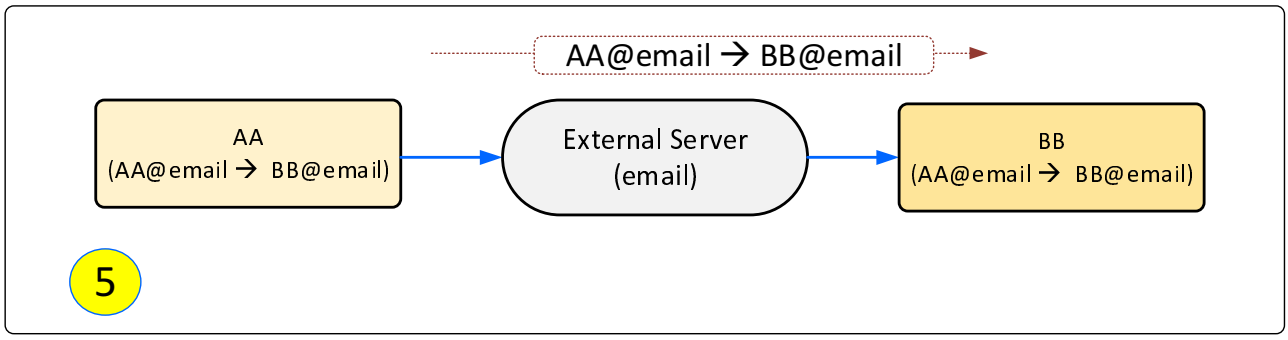


Figure 4.3.2.3.2.6-1: Email message from generic email address to generic email address

MMS LI does not apply to this scenario, since the email message from AA@email to BB@email does not hit the MMS/Relay Server. Note that this case same as the one shown in figure 4.3.2.2.6-1.

4.3.3 Topology views - with redirection

4.3.3.1 General

The redirection may be initiated by the MMS Relay/Server (i.e. redirecting the MM message) that serves the MMS User Agent or by the MMS User Agent (forwarding the MM message) upon receiving a notification of an incoming MM message (without a retrieval), in the former case, it may be through a prior setup at the MMS Relay/Server.

The MMS User Agent may also forward an incoming MM message after the retrieving the MM message, however, in this case, it would be same as if that MMS User Agent is sending the MM message, initially.

The Email Client may also forward the received MM message.

NOTE 1: The term "forward" is used for the actions performed by the MMS User Agent or the Email Client.

NOTE 2: The term "redirect" is used when the action is done by the MMS Relay/Server to avoid any confusion with the MMS Relay/Server's action of forwarding an MM message from input to output (e.g. MMS Relay/Server forwards the MM message from MM1 to MM3 or to MM4). The term "redirect" is also used to refer to the functional entity that performs the redirection/forwarding (e.g. redirecting MMS User Agent").

When the MMS User Agent or the Email Client forwards the MM message, the same steps as in the initial message transfer occur and hence, all the scenarios described in in clause 4.3.2 apply for a redirected MM message as well.

- The redirecting MMS User Agent can forward the MM message to an Email Address or to a phone number.
- The redirecting Email Client can forward the MM message to an Email Address or to a phone number.

All the three permutations as illustrated in clause 4.3.2.2.1 for MM message originations from a phone number (MSISDN, GPSI (MSISDN), E.164 number) are applicable when the redirection happens from a phone number (MSISDN, GPSI (MSISDN), E.164 number).

Both of the two permutations as illustrated in clause 4.3.2.2.1 for MM message originations from a generic Email Address are applicable when the redirection happens from a generic Email Address.

The MM messages redirected from the MMS Relay/Server are also handled in the same way except that in this case the initial MM1 message (between the redirecting MMS Relay/Server and the first terminating MMS User Agent) is not needed.

The subsequent clauses illustrate the user-forwarded cases of different identities that are applicable for MMS LI when separate MMS Relay Servers (when applicable) are involved in the MMS flow.

4.3.3.2 Multiple MMS Relay/Servers

4.3.3.2.1 General

The following cases are illustrated in the subsequent clauses (all are from the perspective of originating end):

- Initial MM message from MSISDN to MSISDN.
- Initial MM message from MSISDN to MSISDN based Email Address.
- Initial MM message from MSISDN to generic Email Address.
- Initial MM message from generic Email Address to MSISDN based Email Address.
- Initial MM message from generic Email Address to generic Email Address.

Multiple MMS Relay/Servers refers to the MMS Relay/Servers serving the MMS User Agents, when applicable, are different.

4.3.3.2.2 Initial MM message from MSISDN to MSISDN

4.3.3.2.2.1 Overview

MMS User Agent (A) initially sends an MM message using own phone number (MSISDN, GPSI (MSISDN), E.164 number) to MMS User Agent (B) (addressed with a phone number (MSISDN, GPSI (MSISDN), E.164 number) and the MMS User Agent (B) forwards the received MM message to destinations as illustrated.

In this illustration, it is presumed that MMS User Agent (A), MMS User Agent (B) and MMS User Agent (C) are served by different MMS Relay Servers.

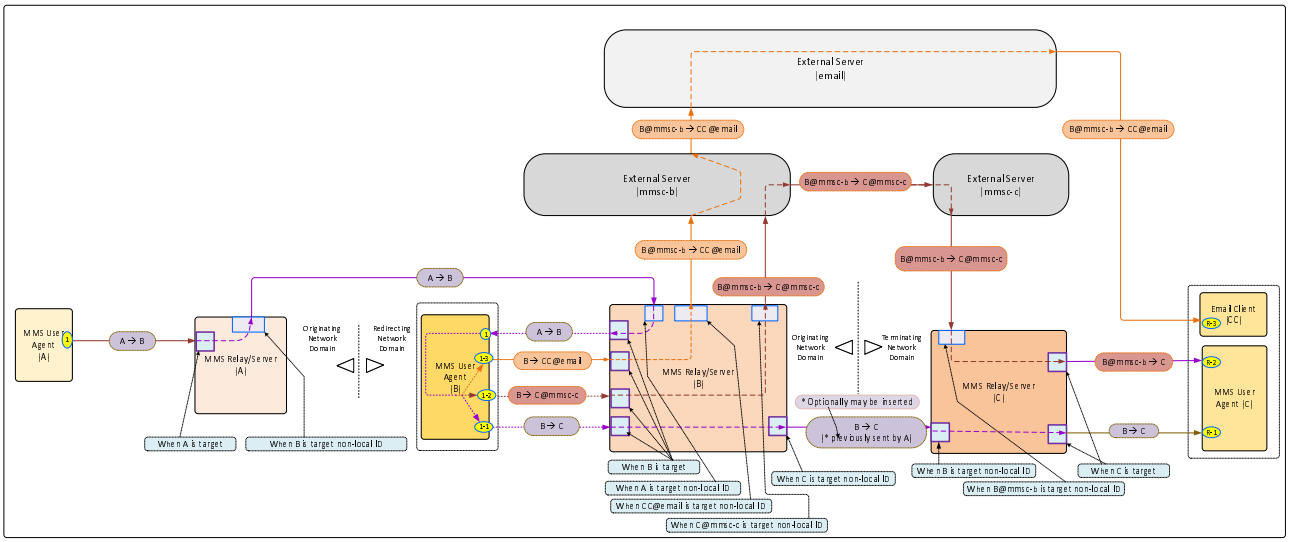


Figure 4.3.3.2.2.1-1: Scope of target identities with redirection (multiple Relay/Servers – case 1)

The path of the MM message flow from MMS User Agent (A) to MMS User Agent (B) is same as in the scenario 1 of the illustrations in clause 4.3.2.2 (figure 4.3.2.2.1-1 and 4.3.2.2.2.2-1). For this part, the phone number A can be a target in the MMS Relay/Server (A) and a target non-local ID in the MMS Relay/Server (B). The phone number B can be a target in the MMS Relay/Server (B) and a target non-local ID in the MMS Relay/Server (A).

The redirection of the MM message may happen to a phone number or to an Email Address.

When the MMS User Agent (B) forwards the MM message to an MSISDN based Email Address (shown as C@mmsc-c), the redirecting MMS Relay/Server (B) adds the domain name for B (shown as mmsc-b) and the redirected-to MMS Relay/Server (C) would remove the domain from for C (i.e. removes the mmsc-c from C@mmsc-c).

The three scenarios of redirected MMS flow shown as R-1, R-2, and R-3 in figure 4.3.3.2.2.1-1 are further illustrated in figure 4.3.3.2.2.1-2.

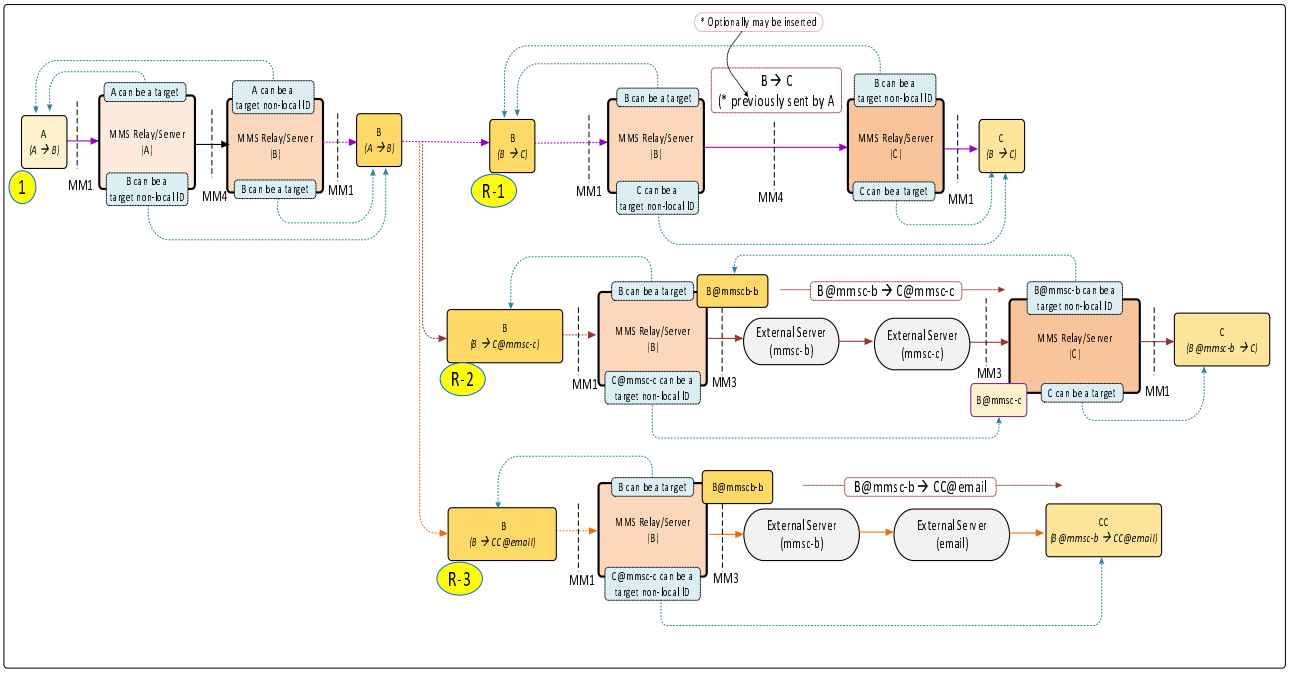


Figure 4.3.3.2.2.1-2: Three scenarios for the redirected MM message for case 1

The three redirected scenarios are described in clause 4.3.3.2.2.2.

4.3.3.2.2.2 Descriptions of MM message redirections from a phone number

4.3.3.2.2.2.1 Scenario R-1

In this scenario, the redirecting MMS User Agent (B) forwards the received MM message to a phone number (MSISDN, GPSI (MSISDN), E.164 number). The MMS User Agent (C) is the recipient of that redirected MM message.

The phone number B can be a target in the MMS Relay/Server (B) for the redirected MM message.

The phone number B can be a target non-local ID in the MMS Relay/Server (C).

The phone number C can be a target in the MMS Relay/Server (C) and a target non-local ID in the MMS Relay/Server (B).

4.3.3.2.2.2.2 Scenario R-2

In this scenario, the MMS User Agent (B) forwards the received MM message to a phone number (MSISDN, GPSI (MSISDN), E.164 number) by addressing the destination using an MSISDN based Email Address (shown as C@mmsc-c). The MMS User Agent (C) is the recipient of that redirected MM message.

The MMS Relay/Server (B) adds the domain name (shown as @mmsc-B) to the B's identity (shown as B@mmsc-b) before forwarding the MM message to the External Server serving the domain mmsc-c.

The MMS Relay/Server (C) would remove the domain name (shown as @mmsc-c) at the terminating end.

The phone number B can be target in the MMS Relay/Server (B) for the redirected MM message.

The MSISDN based Email Address B@mmsc-b can be a target non-local ID in the MMS Relay/Server (C).

The phone number C can be a target in the MMS Relay/Server (C) and C@mmsc-c can be a target non-local ID in the MMS Relay/Server (B).

4.3.3.2.2.2.3 Scenario R-3

In this scenario, MMS User Agent (B) forwards the received MM message to a generic Email Address (shown as CC@email). This MM message goes from the external server (email) to the Email Client (CC).

The MMS Relay/Server (B) adds the domain name (shown as @mmsc-b) to the B's identity (shown as A@mmsc-b) before forwarding the MM message to the External Server.

The phone number B can be target in the MMS Relay/Server (B) for the redirected MM message.

The Email Address CC@email can be a target non-local ID in the MMS Relay/Server (B).

4.3.3.2.3 Initial MM message from MSISDN to MSISDN based Email Address

MMS User Agent (A) initially sends an MM message using own phone number (MSISDN, GPSI (MSISDN), E.164 number) to MMS User Agent (B), addressed with an MSISDN based Email Address (shown as B@mmsc-b). The MMS User Agent (B) forwards the MM message to destinations as illustrated.

In this illustration, it is presumed that MMS User Agent (A), MMS User Agent (B) and MMS User Agent (C) are served by different MMS Relay Servers.

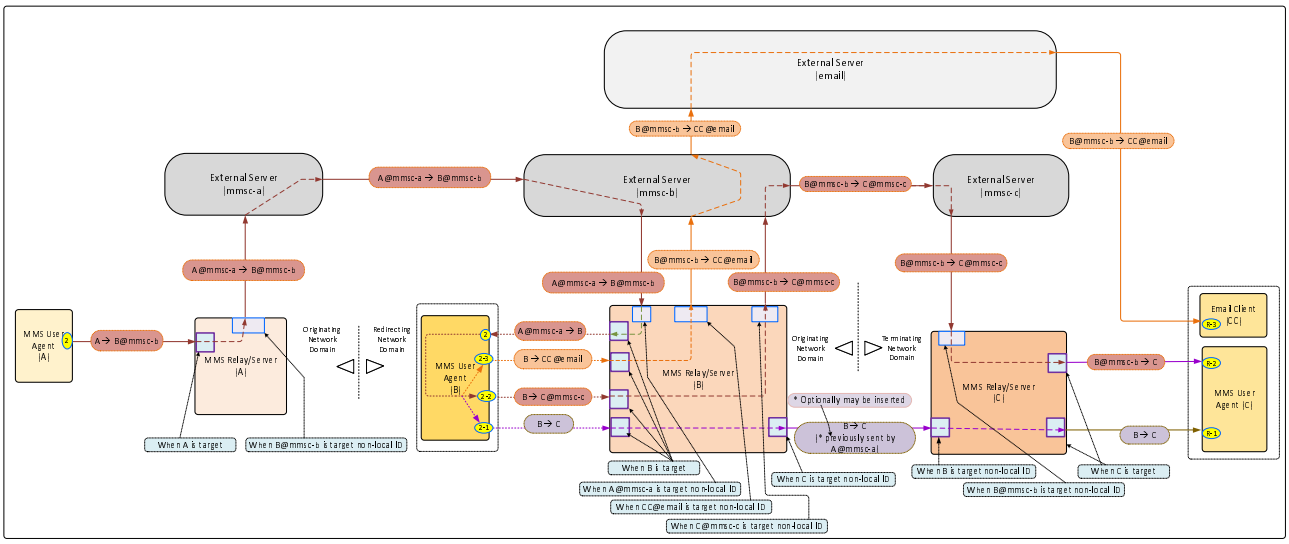


Figure 4.3.3.2.3-1: Scope of target identities with redirection (multiple Relay/Servers – case 2)

The path of the MM message from MMS User Agent (A) to MMS User Agent (B) is same as in the scenario 2 of the illustrations in clause 4.3.2.2 (figure 4.3.2.2.1-1 and 4.3.2.2.3-1). For this part, the phone number A can be a target in the MMS Relay/Server (A) and A@mmsc-a can be target non-local ID in the MMS Relay/Server (B). Likewise, The phone number B can be a target in the MMS Relay/Server (B) and B@mmsc-b can be a target non-local ID in the MMS Relay/Server (A).

The redirection of the MM message may happen to a phone number or to an Email Address.

When the MMS User Agent (B) forwards the MM message to an MSISDN based Email Address (shown as C@mmsc-c), the redirecting MMS Relay/Server (B) adds the domain name for B (shown as mmsc-b) and the redirected-to MMS Relay/Server (C) would remove the domain from for C (i.e. removes the mmsc-c from C@mmsc-c).

The three redirected scenarios of MMS flow, shown as R-1, R-2, R-3 in figure 4.3.3.2.3-1, are further illustrated in figure 4.3.3.2.3-2.

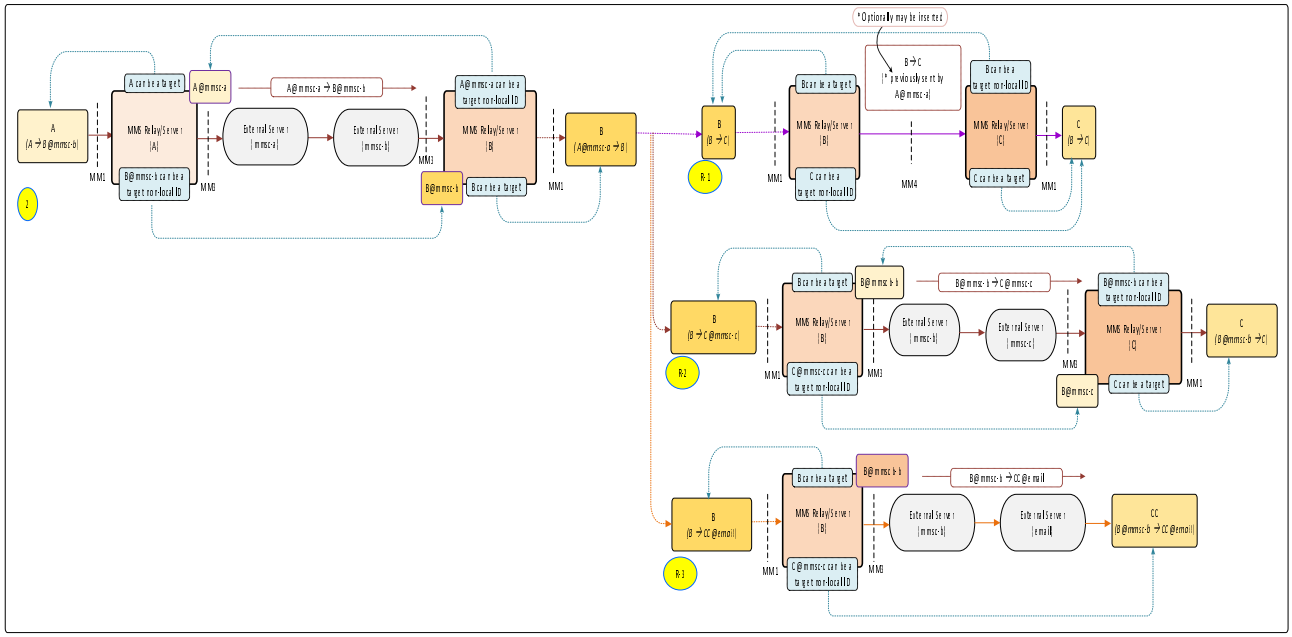


Figure 4.3.3.2.3-2: Three scenarios for the redirected MM message for case 2

The three redirected scenarios are described in clause 4.3.3.2.2.2.

4.3.3.2.4 Initial MM message from MSISDN to generic Email Address

4.3.3.2.4.1 Overview

MMS User Agent (A) initially sends an MM message using own phone number (MSISDN, GPSI (MSISDN), E.164 number) to a generic Email Address (shown as BB@email). The recipient of that MM message, Email Client (BB), forwards the MM message to destinations as illustrated.

In this illustration, it is presumed that MMS User Agent (A) and MMS User Agent (C) are served by different MMS Relay Servers.

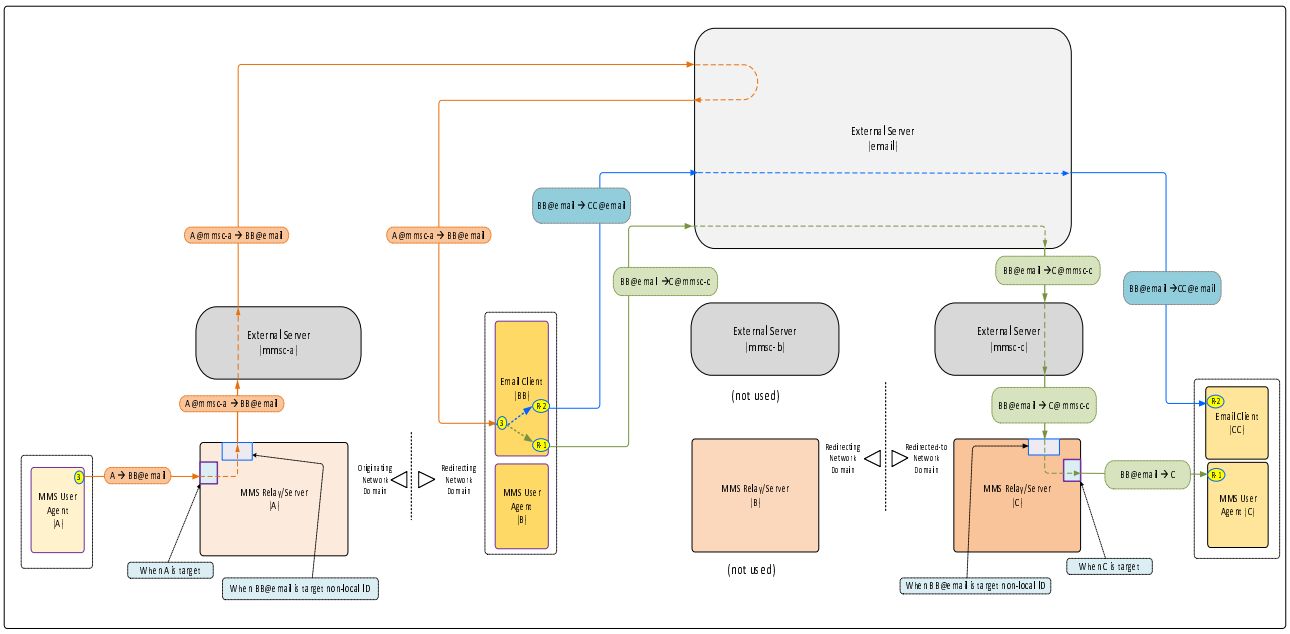


Figure 4.3.3.2.4.1-1: Scope of target identities – with redirection (multiple Relay/Servers – case 3)

The path of the MM message from MMS User Agent (A) to Email Client (BB) is same as in the scenario 3 of the illustrations in clause 4.3.2.2 (figure 4.3.2.2.1-1 and 4.3.2.2.2.4-1). For this part, the phone number A can be a target in the MMS Relay/Server (A) and the Email Address BB@email can be a non-local ID in the MMS Relay/Server (A).

The redirection of the MM message may happen to a phone number or to an Email Address.

The Email Client (BB) uses the phone number of the redirected-to MMS User Agent (C) in an email format (i.e. MSISDN based Email Address). In this case, the redirected-to MMS Relay/Server (C) would remove the domain from for C (i.e. removes the mmsc-c from C@mmsc-c).

The two redirected scenarios, shown as R-1, R-2 in figure 4.3.3.2.4.1-1, are further illustrated in figure 4.3.3.2.4.1-2.

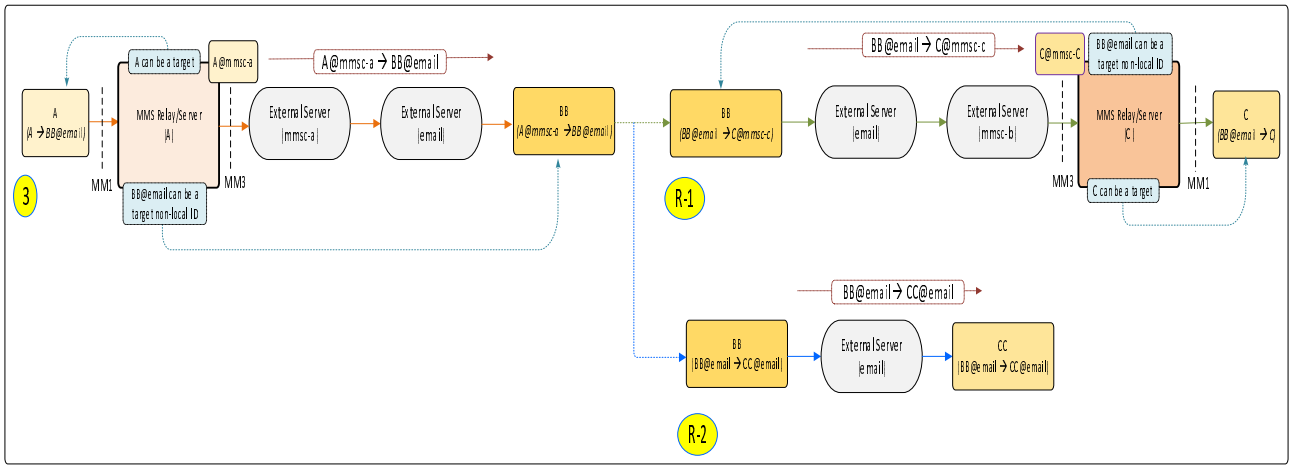


Figure 4.3.3.2.4.1-2: Two scenarios for the redirected MM message for case 3

The two redirected scenarios are described in clause 4.3.3.2.4.2.

4.3.3.2.4.2 Descriptions of MM message redirections from a phone number

4.3.3.2.4.2.1 Scenario R-1

In this scenario, the Email Client (BB) forwards the received MM message to a phone number (MSISDN, GPSI (MSISDN), E.164 number) by addressing the destination using an MSISDN based Email Address (shown as C@mmsc-c). The MMS User Agent (C) is the recipient of that redirected MM message.

The MMS Relay/Server (C) would remove the domain name (shown as @mmsc-c) at the terminating end.

The generic Email Address BB@email can be a target non-local ID in the MMS Relay/Server (C).

The phone number C can be a target in the MMS Relay/Server (C).

4.3.3.2.4.2.2 Scenario R-2

In this scenario, the Email Client (BB) forwards the received MM message to a generic Email Address (shown as CC@email). This MM message goes from the External Server (email) to the Email Client (CC).

MMS LI does not apply to this scenario, since the email message redirected from BB@email to CC@email does not hit the MMS/Relay Server.

4.3.3.2.5 Initial MM message from generic Email Address to MSISDN based Email Address

Email Client (shown as AA) sends an MM message to MMS User Agent (B), addressed with an MSISDN based Email Address (shown as B@mmsc-b). The MMS User Agent (B) forwards the MM message to destinations as illustrated.

In this illustration, it is presumed that MMS User Agent (B) and MMS User Agent (C) are served by different MMS Relay/Servers.

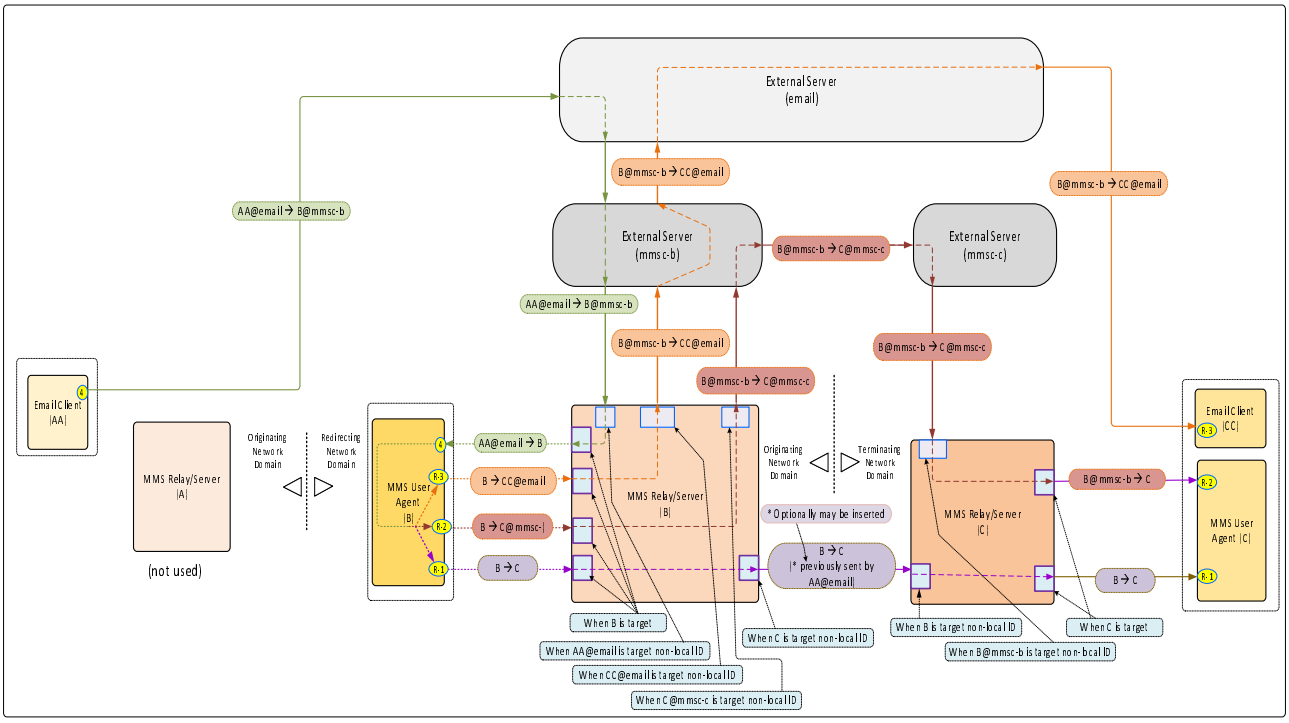


Figure 4.3.3.2.5-1: Scope of target identities with redirection (multiple Relay/Servers – case 4)

The path of the MM message from Email Client (AA) to MMS User Agent (B) is same as in the scenario 4 of the illustrations in clause 4.3.2.2 (figure 4.3.2.2.1-1 and 4.3.2.2.2.5-1). For this part, the phone number B can be a target in the MMS Relay/Server (B) and the Email Address AA@email can be a target non-local ID in the MMS Relay/Server (B).

The redirection of the MM message may happen to a phone number or to an Email Address.

When the MMS User Agent (B) forwards the MM message to an MSISDN based Email Address (shown as C@mmsc-c), the redirecting MMS Relay/Server (B) adds the domain name for B (shown as mmsc-b) and the redirected-to MMS Relay/Server (C) would remove the domain from for C (i.e. removes the mmsc-c from C@mmsc-c).

The three redirected scenarios, shown as R-1, R-2, R-3 in figure 4.3.3.2.5-1, are further illustrated in 4.3.3.2.5-2.

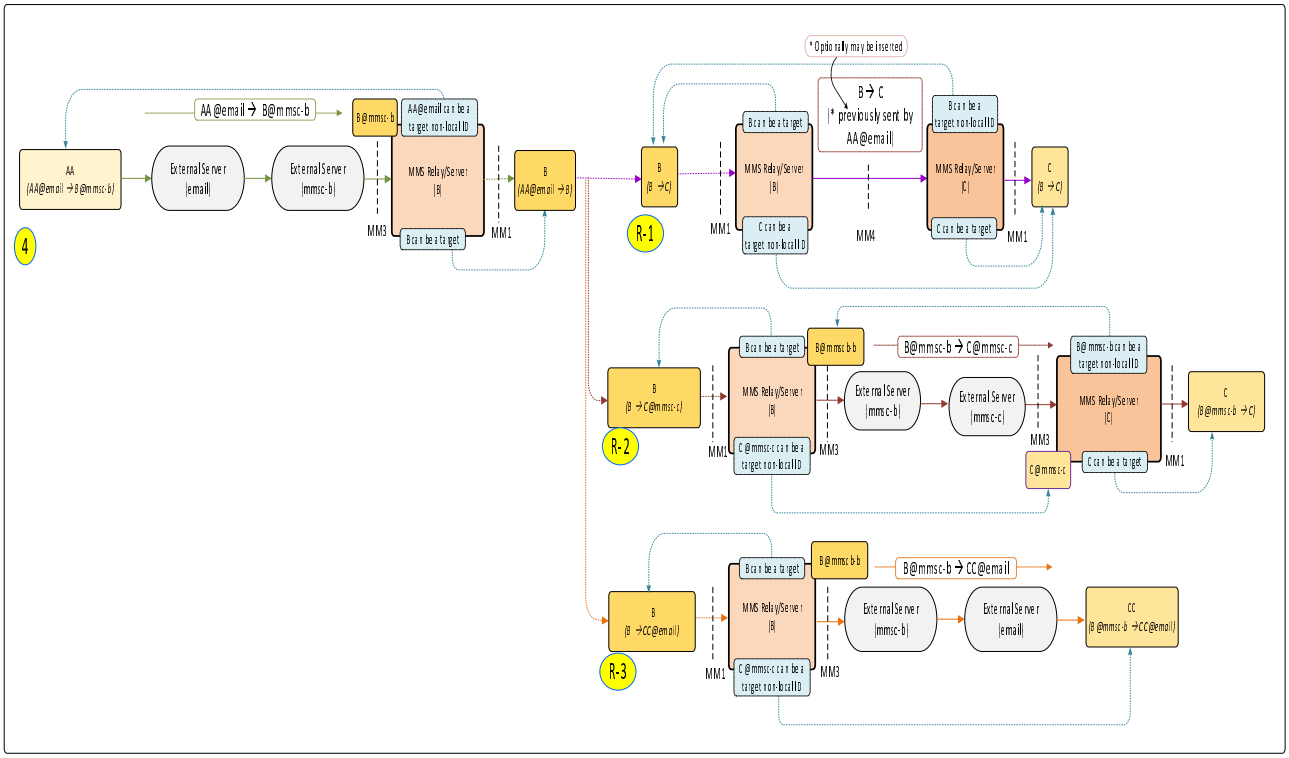


Figure 4.3.3.2.5-2: Three scenarios for the redirected MM message for case 4

The three redirected scenarios are described in clause 4.3.3.2.2.2.

4.3.3.2.6 Initial MM message from generic Email Address to generic Email Address

Email Client (shown as AA) sends an MM message to a generic Email Address (shown as BB@email). The Email Client (BB) who receives that MM message forwards the same to destinations as illustrated.

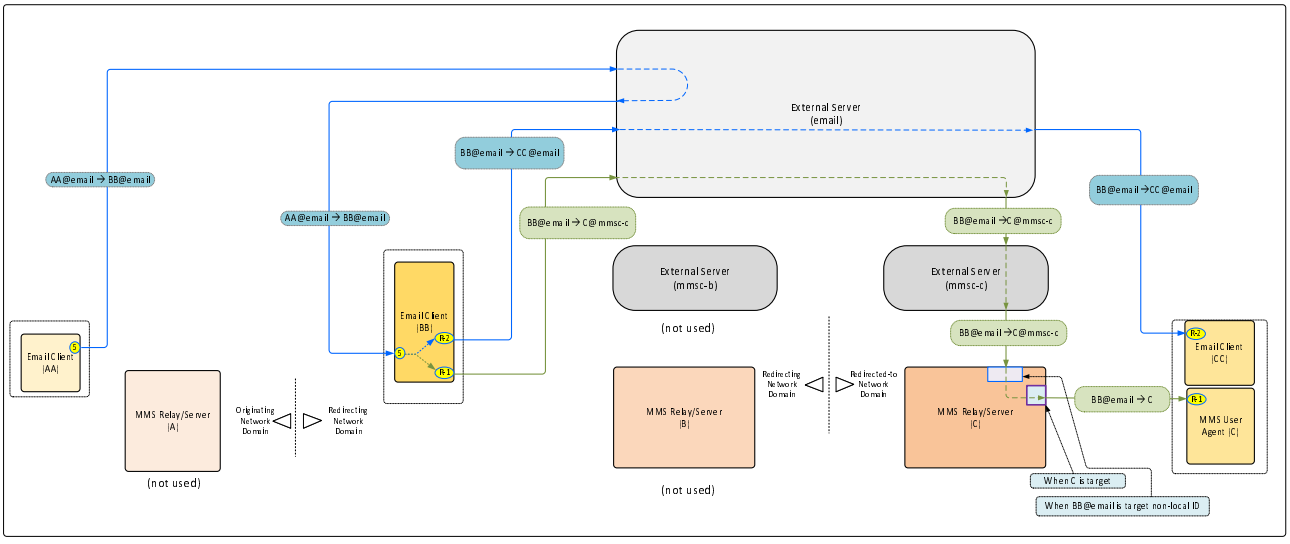


Figure 4.3.3.2.6-1: Scope of target identities with redirection (multiple Relay/Servers – case 5)

The path of the MM message from Email Client (AA) to Email Client (BB) is same as in the scenario 5 of the illustration in clause 4.3.2.2 (figure 4.3.2.2.1-1 and 4.3.2.2.2.6-1). For this part, MMS LI does not apply since the initial email message from AA@email to BB@email does not hit the MMS/Relay Server.

The redirection of the MM message by the Email Client (BB) may happen to an MSISDN based Email Address or to generic Email Address.

The two redirected scenarios, shown as R-1, R-2 in figure 4.3.3.2.6-1 are further illustrated in 4.3.3.2.6-2.

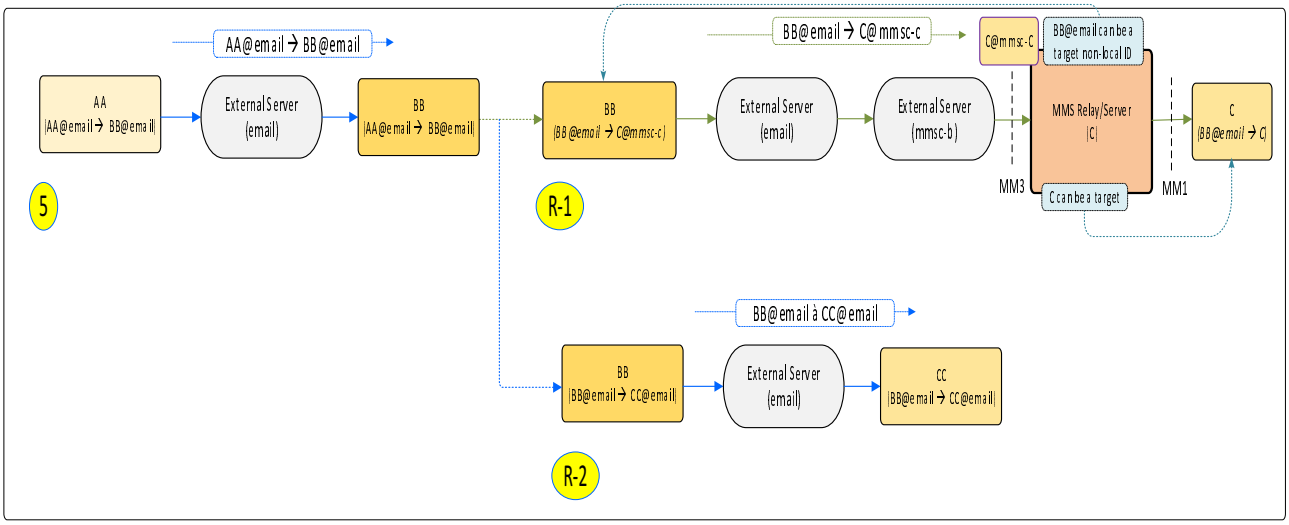


Figure 4.3.3.2.6-2: Two scenarios for the redirected MM message (for case 5)

The two redirected scenarios are described as in clause 4.3.3.4.2.

4.3.3.3 One MMS Relay Server

4.3.3.3.1 General

When all the MMS User Agents are served by the same MMS Relay/Server, the identity of the originating MMS User Agent is a phone number (MSISDN, GPSI (MSISDN), or E,164 Number).

In the initial MM message, the redirecting MMS User Agent may be addressed with a phone number (MSISDN, GPSI (MSISDN), or E,164 Number) or an MSISDN based Email Address.

The redirected MMS User Agent may be address using a with a phone number (MSISDN, GPSI (MSISDN), or E,164 Number) or an MSISDN based Email Address.

The generic Email Address based identity does not apply to this scenario.

4.3.3.3.2 Initial MM message from MSISDN to MSISDN

4.3.3.3.2.1 Overview

MMS User Agent (A) initially sends an MM message using own phone number (MSISDN, GPSI (MSISDN), E.164 number) to MMS User Agent (B) (addressed with a phone number (MSISDN, GPSI (MSISDN), E.164 number) and the MMS User Agent (B) forwards the received MM message to destinations as illustrated.

In this illustration, it is presumed that MMS User Agent (A), MMS User Agent (B) and MMS User Agent (C) are served by the same MMS Relay Server.

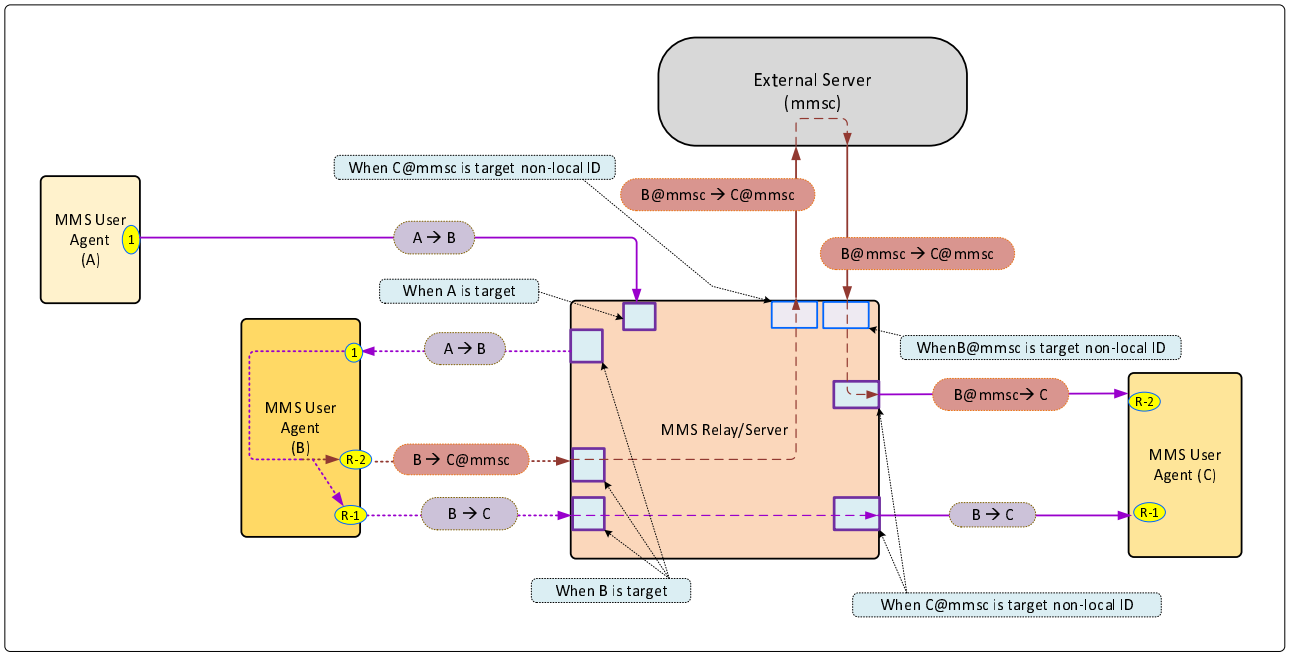


Figure 4.3.3.3.2.1-1: Scope of target identities with redirection (one Relay/Server) – case 1

The path of the MM message flow from MMS User Agent (A) to MMS User Agent (B) is same as in the scenario 1 of the illustrations in clause 4.3.2.3 (figure 4.3.2.3.1-1 and 4.3.2.3.2.2-1). The MMS User Agent (A) and MMS User Agent (B) can be the targets in the MMS Relay/Server. The redirection of the MM message may happen to a phone number or to an MSDDN based Email Address.

When the MMS User Agent (B) forwards the MM message to an MSISDN based Email Address (shown as C@mmsc), the MMS Relay/Server adds the domain name for B (shown as mmsc).

The two scenarios of redirected MMS flow shown as R-1, R-2 in figure 4.3.3.3.2.1-1, are further illustrated in figure 4.3.3.3.2.1-2.

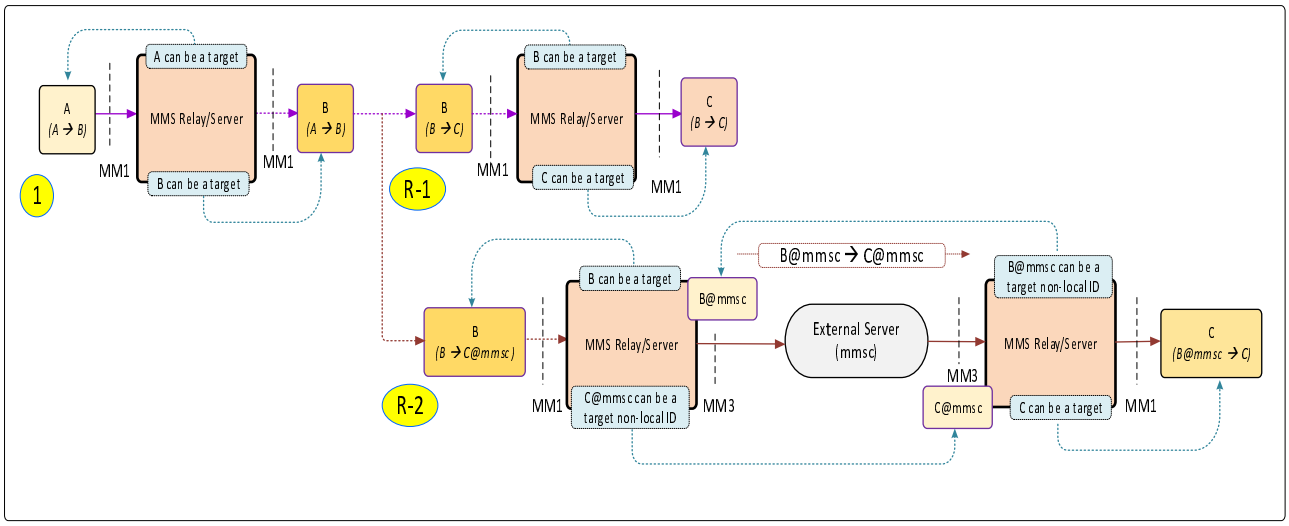


Figure 4.3.3.3.2.1-2: Two scenarios for the redirected MM message for case 1

The two redirected scenarios are described in clause 4.3.3.3.2.2.

4.3.3.3.2.2 Descriptions of MM message redirected MM messages

4.3.3.3.2.2.1 Scenario R-1

In this scenario, the redirecting MMS User Agent (B) forwards the received MM message to a phone number (MSISDN, GPSI (MSISDN), E.164 number). The MMS User Agent (C) is the recipient of that redirected MM message.

The phone number B can be a target in the MMS Relay/Server for the redirected MM message. The phone number C can be a target in the MMS Relay/Server.

4.3.3.3.2.2.2 Scenario R-2

In this scenario, the MMS User Agent (B) forwards the received MM message to a phone number (MSISDN, GPSI (MSISDN), E.164 number) by addressing the destination using an MSISDN based Email Address (shown as C@mmsc). The MMS User Agent (C) is the recipient of that redirected MM message.

The MMS Relay/Server adds the domain name (shown as @mmsc) to the B's identity (shown as B@mmsc) before forwarding the MM message to the External Server serving the domain mmsc.

The MMS Relay/Server would remove the domain name (shown as @mmsc) from C@mmsc from C@mmsc, at the terminating end of the flow.

The phone number B can be target in the MMS Relay/Server (B) for the redirected MM message.

The MSISDN based Email Address B@mmsc can be a target non-local ID in the MMS Relay/Server.

The phone number C can be a target in the MMS Relay/Server. The MSISDN based Email Address C@mmsc- can be a target non-local ID in the MMS Relay/Server.

4.3.3.3.3 Initial MM message from MSISDN to MSISDN based Email Address

MMS User Agent (A) initially sends an MM message using own phone number (MSISDN, GPSI (MSISDN), E.164 number) to MMS User Agent (B), addressed with an MSISDN based Email Address (shown as B@mmsc). The MMS User Agent (B) forwards the MM message to destinations as illustrated.

In this illustration, it is presumed that MMS User Agent (A), MMS User Agent (B) and MMS User Agent (C) are served by the same MMS Relay Server.

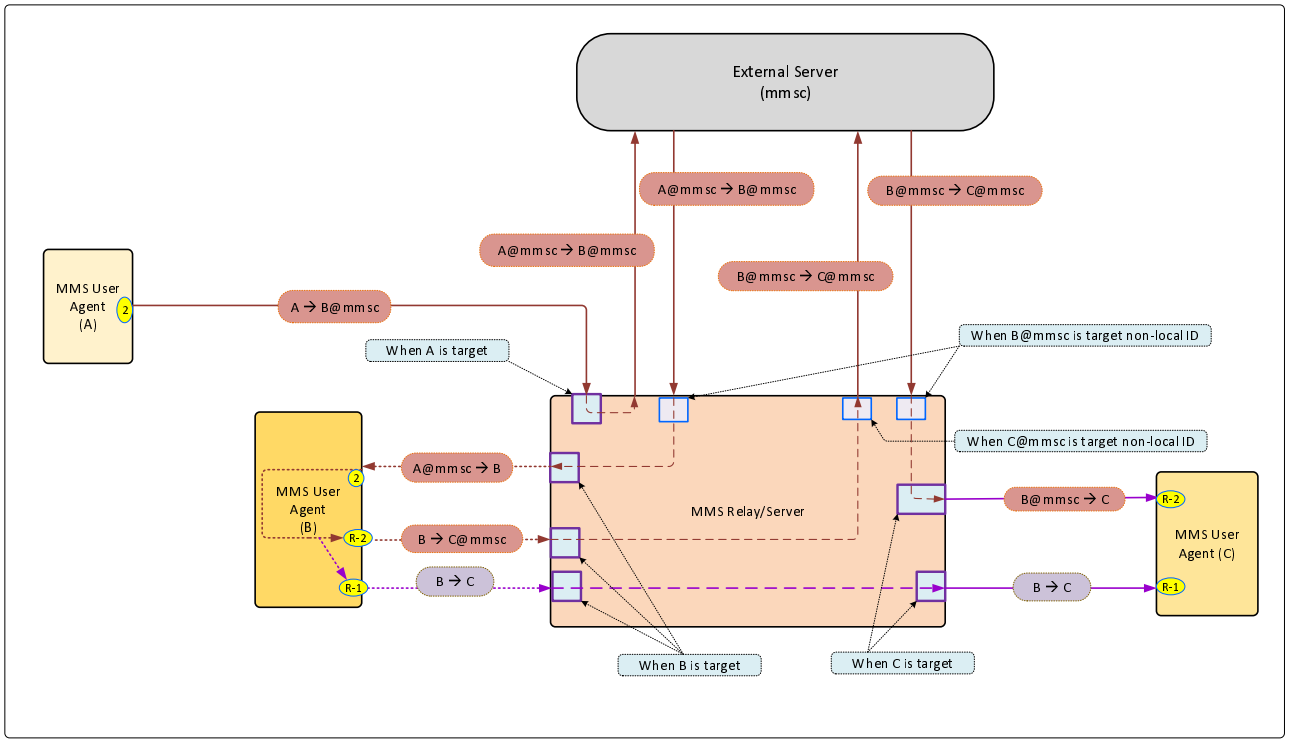


Figure 4.3.3.3.3-1: Scope of target identities with redirection (one Relay/Server) – case 4

The path of the MM message flow from MMS User Agent (A) to MMS User Agent (B) is same as in the scenario 2 of the illustrations in clause 4.3.2.3 (figure 4.3.2.3.1-1 and 4.3.2.3.2.3-1). The MMS User Agent (A) and MMS User Agent (B) can be the targets in the MMS Relay/Server. The MSISDN based Email Address of B (i.e. B@mmsc) and the MSISDN based Email Address of B (i.e. B@mmsc) can be the target non-local identities in the MMS Relay/Server.

The redirection of the MM message may happen to a phone number or to an MSDDN based Email Address.

When the MMS User Agent (B) forwards the MM message to an MSISDN based Email Address (shown as C@mmsc), the MMS Relay/Server adds the domain name for B (shown as mmsc).

The two scenarios of redirected MMS flow shown as R-1, R-2 in figure 4.3.3.3.3-1 are further illustrated in figure 4.3.3.3.3-2.

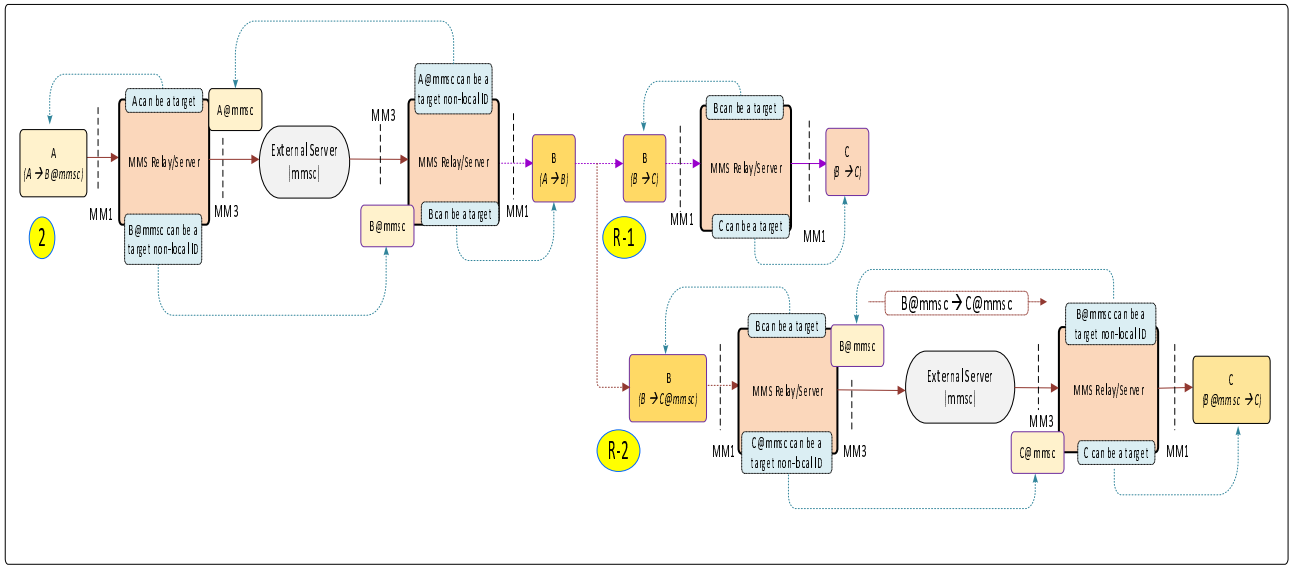


Figure 4.3.3.3.3-2: Two scenarios for the redirected MM message for case 2

The two redirected scenarios are described in clause 4.3.3.3.2.2.

4.4 Scope of MMS messages reported

4.4.1 General

As illustrated in clause 4.3, when the MMS User Agent, local to an MMS Relay/Server, is the target, the xIRIs generated by the IRI-POI in the MMS Relay/Server are based off of the MM1 messages as defined in the OMA specification [19].

For target non-local ID represented by a phone number (E.164 number, GPSI (MSISDN), MSISDN), the xIRIs generated by the IRI-POI in the MMS Relay/Server are based off of the MM4 messages as defined in TS 23.140 [18].

For target non-local ID represented by an Email Address, the xIRIs generated by the IRI-POI in the MMS Relay/Server are based off of the MM1 messages as defined in the OMA specification [19] transfer over or from the MM3 interface. MM3 messages by themselves are not referenced.

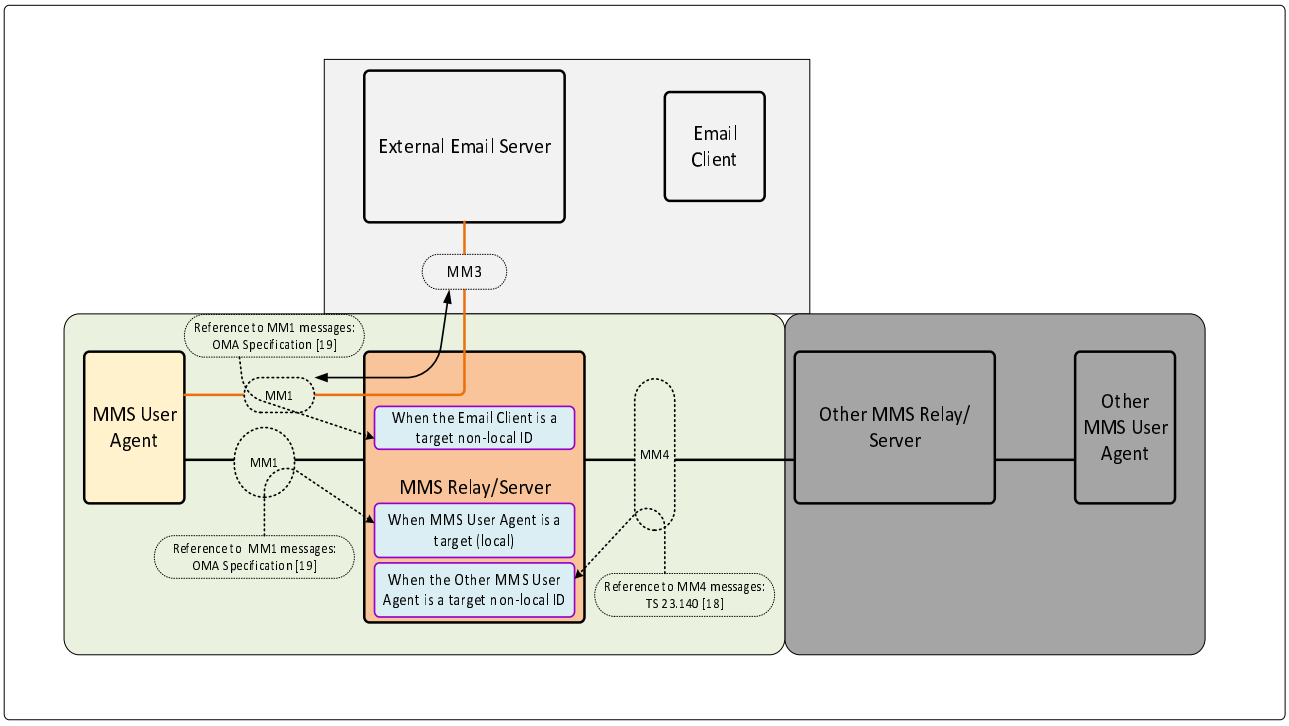


Figure 4.4.1-1: Referenced source specifications for MMS LI

The TS 33.127 [3] and TS 33.128 [4] use the MMS architecture defined in TS 23.140 [18] for MMS LI. However, the TS 23.140 [18] does not define the stage 3 details for the MM1 messages. For that reason, the TS 33.128 [4] uses the OMA spec [19] for the message details.

4.4.2 3GPP, OMA and MMS LI messages

4.4.2.1 General

A comparison of the message-names used in the TS 23.140 [18] and the message names used in the OMA specification [19] along with associated xIRI messages defined in TS 33.128 [4] are given in this clause.

The MMBox related operations result in XIRIs only when the target that is local to the MMS Relay/Server.

4.4.2.2 MM1 to xIRIs (originating side)

When the target is local to the MMS Relay/Server or when the target non-local ID is represented by an Email Address, the MM1 messages are used to generate the xIRIs.

Table 4.4.2.2-1: MMS LI from (MM1, MMS_M) messages (target is local MMS Relay/Server)

MM1 [18]	MMS_M [19]	MM message direction	MMS LI (xIRI [4])
MM1_submit.REQ	m-send-req	UA → Relay/Server	MMSSend (NOTE 1)
MM1_submit.RES	m-send-conf	Relay/Server → UA	MMSSend (NOTE 1)
MM1_delivery_report.REQ	m-delivery-ind	Relay/Server → UA	MMSDeliveryReport
MM1_read_reply_originator.REQ	m-read-orig-ind	Relay/Server → UA	MMSReadReport
MM1_mmbox_view.REQ	m-mbox-view-req	UA → Relay/Server	MMSMBoxViewRequest
MM1_mmbox_view.RES	m-mbox-view-conf	Relay/Server → UA	MMSMBoxViewResponse
MM1_mmbox_upload.REQ	m-mbox-upload-req	UA → Relay/Server	MMSMBoxUpload (NOTE 2)
MM1_mmbox_upload_RES	m-mbox-upload-conf	Relay/Server → UA	MMSMBoxUpload (NOTE 2)
MM1_mmbox_store_REQ	m-mbox-store-req	UA → Relay/Server	MMSMBoxStore (NOTE 3)
MM1_mmbox_store.RES	m-mbox-store-conf	Relay/Server → UA	MMSMBoxStore (NOTE 3)
MM1_mmbox_delete.REQ	m-mbox-delete-req	UA → Relay/Server	
MM1_mmbox_delete.RES	m-mbox-delete-conf	Relay/Server → UA	MMSMBoxDelete
NOTE 1: The xIRI is constructed from the m-send-req, but the trigger for the xIRI is m-send-conf. NOTE 2: The xIRI is constructed from the m-mbox-upload-req but the trigger for the xIRI is m-mbox-upload-conf. NOTE 3: The xIRI is constructed from the m-mbox-store-req but the trigger for the xIRI is m-mbox-store-conf. Furthermore, at the originating end, this procedure is only used to update the state of an already stored MM message.			

As shown in table 4.4.2.2-1, most of the xIRIs are generated when a confirmation is sent to the MMS User Agent. An illustration of the message mapping shown in table 4.4.2.2-1 along with the directional aspects are illustrated in clause 4.4.3.

Table 4.4.2.2-2: MMS LI from (MM1, MMS_M) messages (target is Email Address, non-local ID)

MM1 [18]	MMSM [19]	MM message direction	MMS LI (xIRI [4])
MM1_submit.REQ	m-send-req	UA → Relay/Server	MMSConvertedToEmail (NOTE)
MM1_submit.RES	m-send-conf	Relay/Server → UA	MMSConvertedToEmail (NOTE)
MM1_delivery_report.REQ	m-delivery-ind	Relay/Server → UA	-
MM1_read_reply_originator.REQ	m-read-orig-ind	Relay/Server → UA	-
NOTE: The xIRI is constructed from the m-send-req, but the trigger for the xIRI is m-send-conf.			

The MMBox specific messages are not included in table 4.4.2.2-2 as such operations do not apply to target non-local ID.

An illustration of the message mapping shown in table 4.4.2.2-2 along with the directional aspects are illustrated in clause 4.4.3.

4.4.2.3 MM1 to xIRIs (terminating side)

When the target is local to the MMS Relay/Server or when the target non-local ID is represented by an Email Address, the MM1 messages are used to generate the xIRIs.

Table 4.4.2.3-1: MMS LI from (MM1, MMS_M) messages (target is local MMS Relay/Server)

MM1 [18]	MMS_M [19]	MM message direction	MMS LI (xIRI [4])
MM1_notification.REQ	m-notification-ind	Relay/Server → UA	MMSNotification
MM1_notification.RES	m-notifyresp-ind	UA → Relay/Server	MMSNotificationResponse
MM1_retrieve.REQ	WEP/HTTP GET	UA → Relay/Server	
MM1_retrieve.RES	m-retrieve-conf	Relay/Server → UA	MMSRetrieval
MM1_acknowledge.REQ	m-notifyresp-ind (NOTE 1)	UA → Relay/Server	MMSDeliveryAck
MM1_acknowledge.REQ	m-acknowledge-ind (NOTE 2)	UA → Relay/Server	MMSDeliveryAck
MM1_read_reply_receipient.REQ	m-read-rec-ind	UA → Relay/Server	MMSReadReport
MM1_forward.REQ	m-forward-req	UA → Relay/Server	MMSForward (NOTE 3)
MM1_forward.RES	m-forward-conf	Relay/Server → UA	MMSForward (NOTE 3)
MM1_delete.REQ	m-delete-req	UA → Relay/Server	
MM1_delete.RES	m-delete-conf	Relay/Server → UA	MMSDeleteFromRelay
MM1_cancel.REQ	m-cancel-req	Relay/Server → UA	MMSCancel
MM1_cancel.RES	m-cancel-conf	UA → Relay/Server	
MM1_mmbox_store.REQ	m-mbox-store-req	UA → Relay/Server	MMSMboxStore (NOTE 4)
MM1_mmbox_store.RES	m-mbox-store-conf	Relay/Server → UA	MMSMboxStore (NOTE 4)
MM1_mmbox_view.REQ	m-mbox-view-req	UA → Relay/Server	MMSMBoxViewRequest
MM1_mmbox_view.RES	m-mbox-view-conf	Relay/Server → UA	MMSMBoxViewResponse
MM1_mmbox_delete.REQ	m-mbox-delete-req	UA → Relay/Server	
MM1_mmbox_delete.RES	m-mbox-delete-conf	Relay/Server → UA	MMSMBoxDelete
NOTE 1: Immediate retrieval NOTE 2: Delayed retrieval NOTE 3: The xIRI is constructed from the m-forward-req, but the trigger for the xIRI is m-forward-conf. NOTE 4: The xIRI is constructed from the m-mbox-store, but the trigger for the xIRI is m-mbox-store-conf.			

An illustration of the message mapping shown in table 4.4.2.3-1 along with the directional aspects are illustrated in clause 4.4.3.

Table 4.4.2.3-1: MMS LI from (MM1, MMS_M) messages (target is Email Address, non-local ID)

MM1 [18]	MMS_M [19]	MM message direction	MMS LI (xIRI [4])
MM1_notification.REQ	m-notification-ind	Relay/Server → UA	-
MM1_notification.RES	m-notifyresp-ind	UA → Relay/Server	-
MM1_retrieve.REQ	WEP/HTTP GET	UA → Relay/Server	
MM1_retrieve.RES	m-retrieve-conf	Relay/Server → UA	MMSConvertedFromEmail
MM1_acknowledge.REQ	m-notifyresp-ind (NOTE 1)	UA → Relay/Server	
MM1_acknowledge.REQ	m-acknowledge-ind (NOTE 2)	UA → Relay/Server	-
MM1_read_reply_receipient.REQ	m-read-rec-ind	UA → Relay/Server	-
MM1_forward.REQ	m-forward-req	UA → Relay/Server	-
MM1_forward.RES	m-forward-conf	Relay/Server → UA	-
MM1_delete.REQ	m-delete-req	UA → Relay/Server	-
MM1_delete.RES	m-delete-conf	Relay/Server → UA	-
MM1_cancel.REQ	m-cancel-req	Relay/Server → UA	-
MM1_cancel.RES	m-cancel-conf	UA → Relay/Server	-

The MMBox specific messages are not included in table 4.4.2.2-2 as such operations do not apply to target non-local ID.

An illustration of the message mapping shown in table 4.4.2.2-2 along with the directional aspects are illustrated in clause 4.4.3.

4.4.2.4 MM4 to XIRIs

When the target non-local ID is represented by a phone number, the MM4 messages as defined in TS 33.128 [4] are used for the xIRI message generation. The following tables map the MM4 messages [18] to the xIRI messages [4].

Table 4.4.2.4-1: MMS LI from MM4 messages (originating side)

MM4 [18]	MM message direction	MMS LI (xIRI [4])
MM4_forward.REQ	Relay/Server (A) to Relay Server (B)	MMSToNonLocalTarget
MM4_forward.RES	Relay/Server (B) to Relay Server (A)	
MM4_delivery_report.REQ	Relay/Server (B) to Relay Server (A)	MMSDeliveryReportNonLocalTarget
MM4_delivery_report.RES	Relay/Server (A) to Relay Server (B)	
MM4_read_reply_report.REQ	Relay/Server (B) to Relay Server (A)	MMSReadReportNonLocalTarget
MM4_read_reply_report.RES	Relay/Server (A) to Relay Server (B)	

An illustration of the message mapping shown in table 4.4.2.4-1 along with the directional aspects are illustrated in clause 4.4.3.

Table 4.4.2.4-2: MMS LI from MM4 messages (terminating side)

MM4 [18]	MM message direction	MMS LI (xIRI [4])
MM4_forward.REQ	Relay/Server (A) to Relay Server (B)	MMSSendByNonLocalTarget
MM4_forward.RES	Relay/Server (B) to Relay Server (A)	
MM4_delivery_report.REQ	Relay/Server (B) to Relay Server (A)	MMSDeliveryReportNonLocalTarget
MM4_delivery_report.RES	Relay/Server (A) to Relay Server (B)	
MM4_read_reply_report.REQ	Relay/Server (B) to Relay Server (A)	MMSReadReportNonLocalTarget
MM4_read_reply_report.RES	Relay/Server (A) to Relay Server (B)	

An illustration of the message mapping shown in table 4.4.2.4-2 along with the directional aspects are illustrated in clause 4.4.3.

4.4.3 Topology views

4.4.3.1 General

4.4.3.1.1 Scope

The illustrations show the scope of MMS LI messages which are based on the OMA specification [19] for MM1 messages and TS 23.140 [18] for the MM4 messages because that is how the most MMS implementations are, and that is how the TS 33.128 [4] defines the stage 3 MMS LI.

A comparison of MM1 messages as defined in TS 23.140 [18] and OMA spec [19] as they apply to MMS LI is given in clause 4.4.2.

The illustrations include the following different cases:

- Target local to the MMS Relay/Server.
- Target non-local ID at the MMS Relay/Server.
- MMS origination from an MMS User Agent. The destination can be another MMS User Agent or an Email Client.
- MMS termination to an MMS User Agent. The origination can be from another MMS User Agent or an Email Client:
  - Immediate retrieval.
  - Deferred retrieval.
  - Deletion without retrieval.
  - Redirection without retrieval (same as MMS Redirection from an MMS User Agent, listed below).
- MMS redirection from an MMS User Agent. The redirection can be to an MMS User Agent or to an Email Client. Here, the initial origination can be from an MMS User Agent or an Email Client.

- When an MM message transfer involves two or more MMS User Agents, those MMS User Agents may be served by different MMS Relay/Servers or may also be served by the same MMS Relay/Server.

As illustrated in clause 4.3 (e.g. scenario 2, clause 4.3.2.2.2.3) a MM message addressed to an MSISDN based Email Address is routed to the External Server over the MM3 interface, even though the final destination is an MMS User Agent. Likewise, an MM message originated from an MSISDN based Email Address comes to the terminating MMS Relay/Server from the External Server via the MM3 interface.

In other words, depending on the addressing scheme used, the MM messages, even if routed via the External Server, may have the MMS User Agents as the origination and the destination. However, from an MMS Relay/Server perspective, the MM messages are handled in the same way as Emails (i.e. MM1 to MM3 or vice-versa).

4.4.3.1.2 MMBox

The MMBox related steps are shown as and when an MM message transfer happens. However, the use of MMBox is optional. Therefore, all the steps along with the associated xIRIs are present if and only if MMBox is used by the target MMS User Agent.

When MMBox is not used, it is presumed that the MM messages are stored locally at the MMS/Relay Server.

When the MMBox is used, the MMS User Agent may also perform a few MMBox specific operations as illustrated in clause 4.4.3.2.2.

LI does not apply to such MMBox specific operations for target non-local ID.

4.4.3.1.3 Delivery reports

When an MM message is retrieved by the terminating MM User Agent, a delivery report (with an indication of retrieval) may be sent to the originating MMS User Agent only when both of the following are true:

- The originating MMS User Agent requests for a delivery report.
- The terminating MMS User Agent permits such a delivery report.

The delivery reports are initiated by the MMS Relay/Server that serves the terminating MMS User Agent.

An intermediate delivery report may be sent by the originating MMS Relay/Server to the originating MMS User Agent as and when the MM message is forwarded to the destination, if the first condition listed above is true.

When MM message is redirected, a delivery report (with an indication of forwarding) may be sent to the originating MMS User Agent, if the above listed two conditions are met (here, the permission is from the redirecting MMS User Agent).

When the redirected-to MMS User Agent retrieves the redirected MM message, a delivery report (with an indication of retrieval) may be sent to the redirecting MMS User Agent if the above listed two conditions are met (here, the request for the delivery report is from the redirecting MM User Agent, and the permission is from the redirected-to MMS User Agent).

An illustration of the above conditions in a non-redirecting case is in figure 4.4.3.1.3-1.

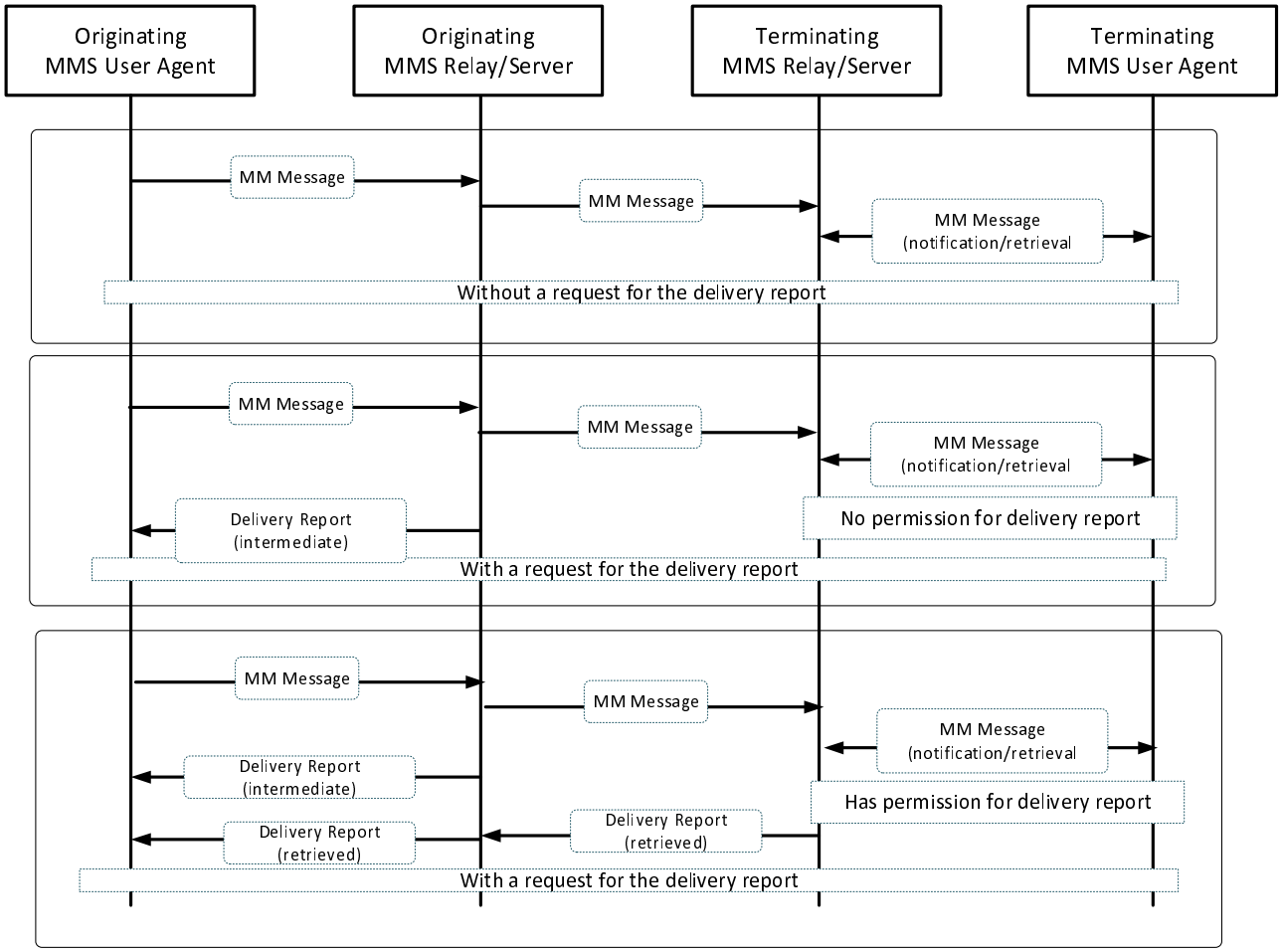


Figure 4.4.3.1.3-1: Examples of delivery report conditions

The delivery report is not received if the destination is an Email Address. At the terminating end, a request for the delivery report is presumed to be not received, if the origination is an Email Address.

4.4.3.1.4 Read reports

A read report may be sent to the originating MMS User Agent when both of the following are true:

- The originating MMS User Agent requests for a read report.
- The terminating MMS User Agent sends read report.

The read reports are generated by the terminating MMS User Agent.

When MM message is redirected, a read report may be sent to the redirecting MMS User Agent, if the above listed two conditions are met (here, the request is from the redirecting MMS User Agent, and read report is generated from the redirected-to MMS User Agent). When an MM message is redircted, the read-report, if generated, is sent only to the redirecting MMS User Agent.

An illustration of the above conditions in a non-redirecting case is in figure 4.4.3.1.4-1.

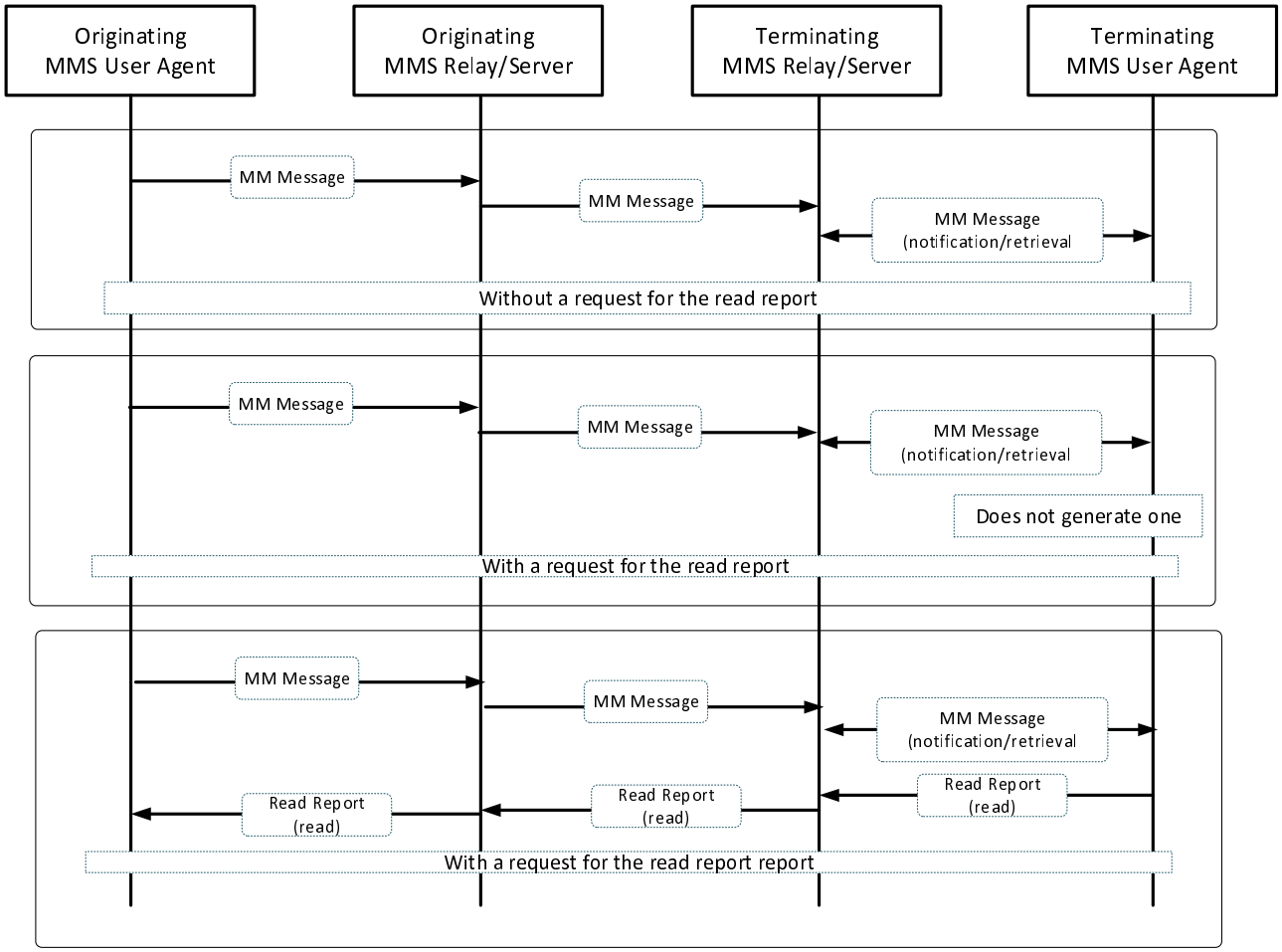


Figure 4.4.3.1.4-1: Examples of read report conditions

The read report is not received if the destination is an Email Address. At the terminating end, a request for report report is presumed to be not received, if the origination is from an Email Address.

4.4.3.2 MMBox specific operations and LI

4.4.3.2.1 Overview

The MMS User Agents may locally upload, view, retrieve, and delete the MM messages at the MMBox located within the local MMS Relay/Server. The MMS User Agents may also store an incoming MM message, update the status of a stored/uploaded MM message, in the MMBox.

This clause refers to these steps which may be taken by an MMS User Agent as MMBox specific operations.

Figure 4.4.3.2.1-1 illustrates the MMS LI messages as they relate to the MMSBox specific operations performed by a target MMS User Agent.

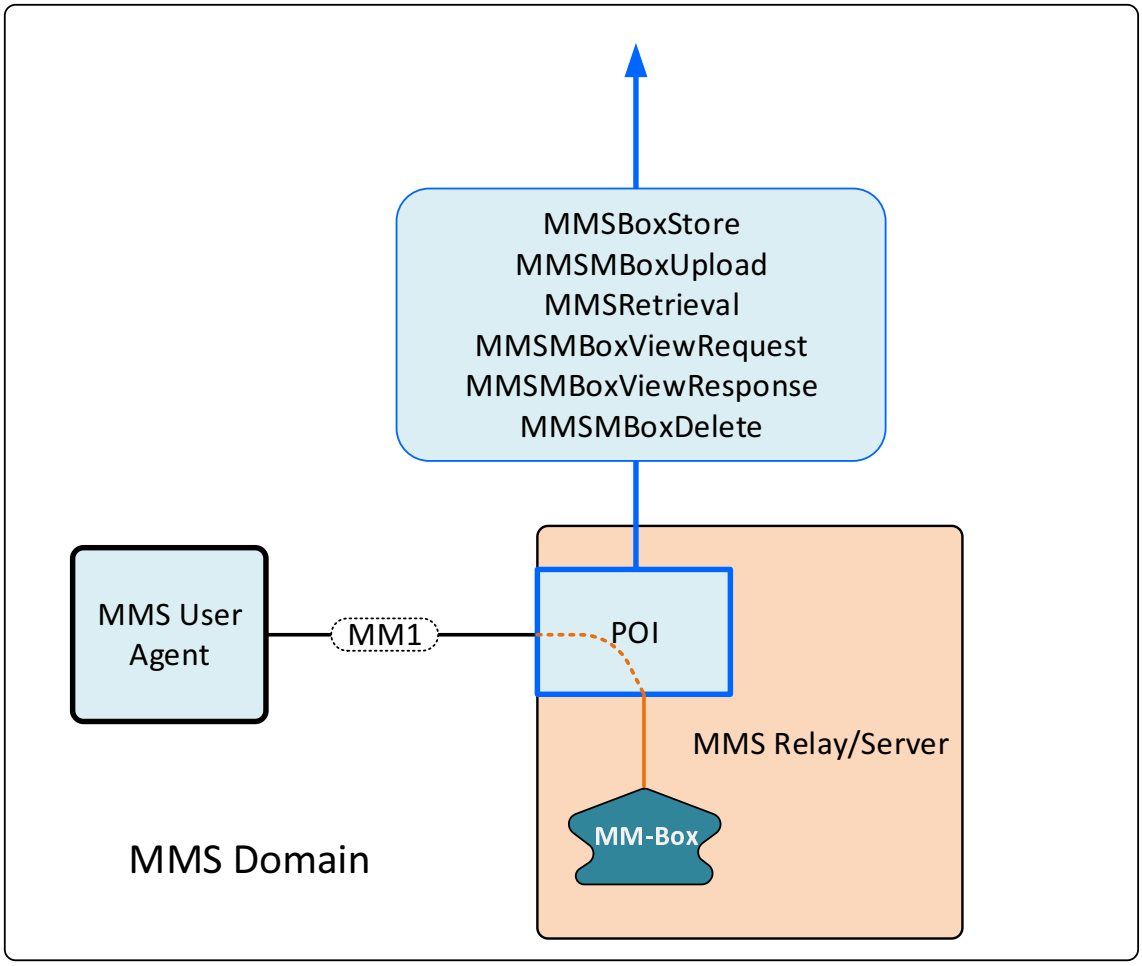


Figure 4.4.3.2.1-1: MMBox specific operations

From an LI perspective, the xIRIs listed are only applicable to the targets that are local to the MMS Relay/Server.

4.4.3.2.2 MMBox messages and MMS LI

Figure 4.4.3.2.2-1 illustrates the generation of xIRIs in reference to the MMBox specific operations performed by the MMS User Agent as a target.

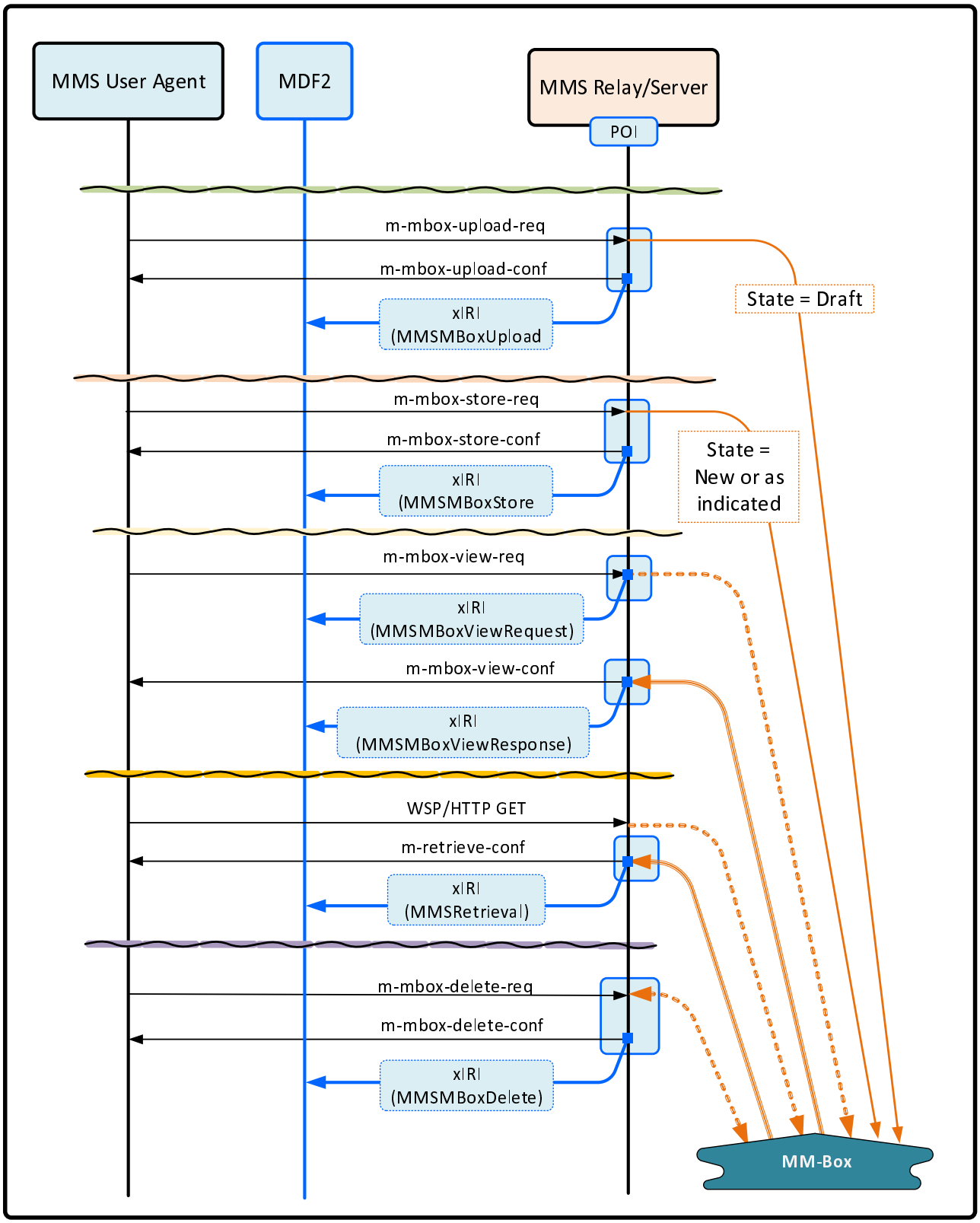


Figure 4.4.3.2.2-1: MMBox specific operations

The m-mbox-store-req is normally sent when the terminating MMS User Agent receives an incoming MM message to request the MMS Relay/Server to store the incoming MM message, if it is not stored automatically. However, m-mbox-store-req may also be used to update the state of the MM message stored in the MMBox. From that point of view, this is applicable to the originating MMS User Agent as well.

The WSP/HTTP GET is used to retrieve an MM message irrespective of it being stored in an MMBox. Accordingly, the xIRI MMSRetrieval is used to report when the target retrieves an MM message, irrespective of whether such a retrieval happens for internally stored MM message or for an MM message stored at the MMBox.

View, retrieval and deletion of MM messages with respect to the MMBox may happen for one or more MM messages.

The MMBox related operations that may also happen as and when the MM message is sent, received, or forwarded are illustrated in the subsequent clauses.

LI specifics

The xIRI MMSMBoxUpload is generated when the MMS Relay/Server sends the m-mbox-upload-conf message to the target MMS User Agent. The xIRI also includes the information received the m-mbox-upload.req message.

The xIRI MMSMBoxStore is generated when the MMS Relay/Server sends the m-mbox-store-conf message to the target MMS User Agent. The xIRI also includes the information received the m-mbox-store.req message.

The xIRI MMSViewRequest is generated when the MMS Relay/Server receives the m-mbox-view-req message from the target MMS User Agent. The xIRI MMSViewResponse is generated when the MMS Relay/Server sends the m-mbox-view-conf message to the target MMS User Agent.

The xIRI MMSMBoxDelete is generated when the MMS Relay/Server sends the m-mbox-delete-conf message to the target MMS User Agent.

The xIRI MMSRetrieval is generated when the MMS Relay/Server sends the m-retrieve-conf to the target MMS User Agent.

4.4.3.3 Originating MMS domain

4.4.3.3.1 General

4.4.3.3.1.1 Scope

Three cases are illustrated in the subsequent clauses:

- Target is local to the MMS Relay/Server, destination is an MMS User Agent is served by a different MMS Relay/Server.
- Target is local to the MMS Relay/Server, destination is an Email Client or an MMS User Agent with an MSISDN based Email Address.
- Target non-local ID.

In the case of target non-local ID identified with a phone number, two MMS Relay/Servers are always used. The target non-local ID is the destination of the MM message.

4.4.3.3.1.2 MMBox

When MM message is sent by the originating MMS User Agent, if requested so, a copy of the MM message is stored in the MMBox with the status showing "sent", if an MMBox is supported for MMS User Agent (A). Therefore, the storing of the MM message at the MMBox does not happen if the store indication is not sent in the m-send-req message, even if the MMBox is supported for the MMS User Agent (A).

When MMBox is not used, it is presumed that the MM messages are stored locally at the MMS/Relay Server.

When the MMBox is used, the MMS User Agent may also perform a few MMBox specific operations as illustrated in clause 4.4.3.2.2.

LI does not apply to such MMBox specific operations for target non-local ID.

4.4.3.3.1.3 Delivery reports

The receipt of delivery report from the terminating MMS domain is conditional. The delivery report is received only if the terminating MMS Relay/Server sends such a report. The terminating MMS Relay/Server sends such a report if and

only if the originating MMS User Agent has requested for such a report and the terminating MMS User Agent permits the sending of such a report.

The delivery report is not received if the destination is an Email Address.

4.4.3.3.1.4 Read reports

The receipt of read report from the terminating MMS User Agent is conditional. The read report is received if the originating MMS User Agent has requested for such a report and the terminating MMS User Agent sends such a report.

The read report is not received, if the destination is an Email Address.

4.4.3.3.2 Target is local to MMS Relay Server

4.4.3.3.2.1 Overview

Figure 4.4.3.3.2.1-1 gives an overview of the MMS LI messages applicable to the originating MMS domain when the target is local to the MMS Relay/Server and the destination can be an Email Client or an MMS User Agent served by a different MMS Relay/Server with a phone number or an MSISDN based Email Address.

When the destination MMS User Agent uses an MSISDN based Email Address, from the MMS Relay/Server (A)'s perspective, the handling is similar to when the destination is an Email Client since the outbound MM messages are transferred over the MM3 interface in an email message format.

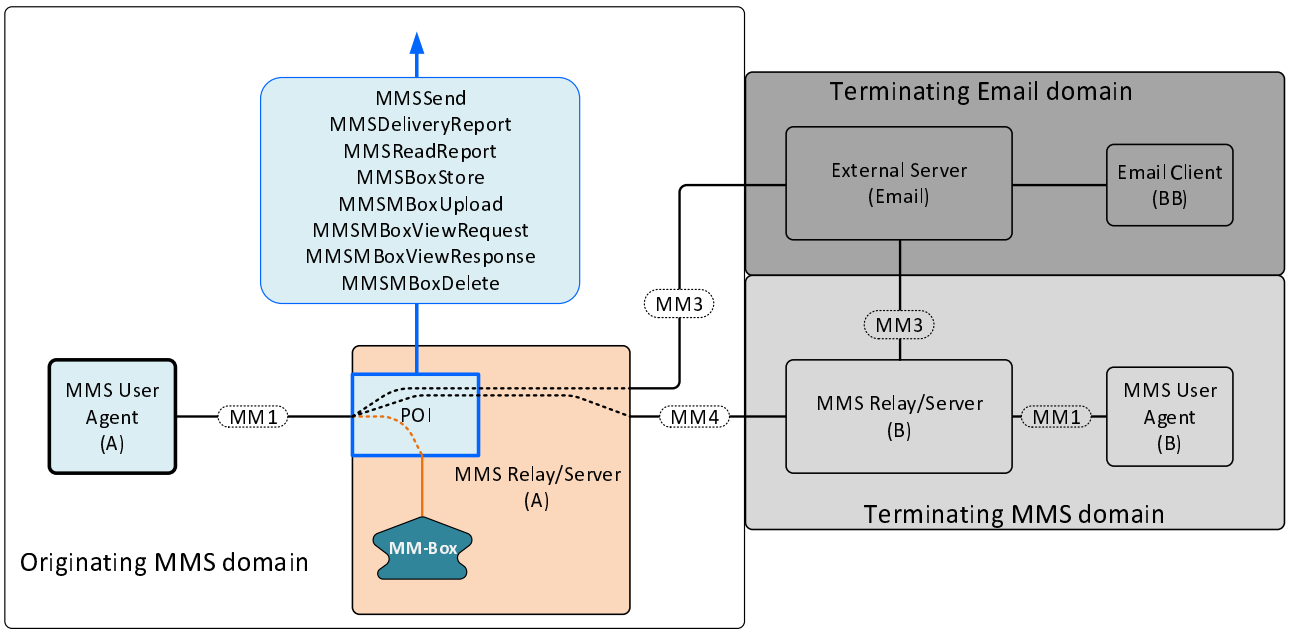


Figure 4.4.3.3.2.1-1: MMS LI messages – originating MMS domain (target is local)

The MM1 messages between the MMS User Agent (A), the target and the MMS Relay/Server (A) are used to generate the xIRIs.

The type of destination (i.e. Email Client Vs MMS User Agent) has no impact on the MMS LI when the originating MMS User Agent is a target and local to the MMS Relay/Server, except that the conditional delivery reports and the read reports are never received when the destination is an Email Client.

4.4.3.3.2.2 MM messages to MMS LI messages

Figure 4.4.3.3.2.2-1 gives an overview of the MMS LI messages applicable to the originating MMS domain when the target is local to the MMS Relay/Server.

In this illustration, the destination is an MMS User Agent served (identifier by a phone number) by a different MMS Relay/Server (the other MMS Relay/Server is not shown in figure 4.4.3.3.2.2-1).

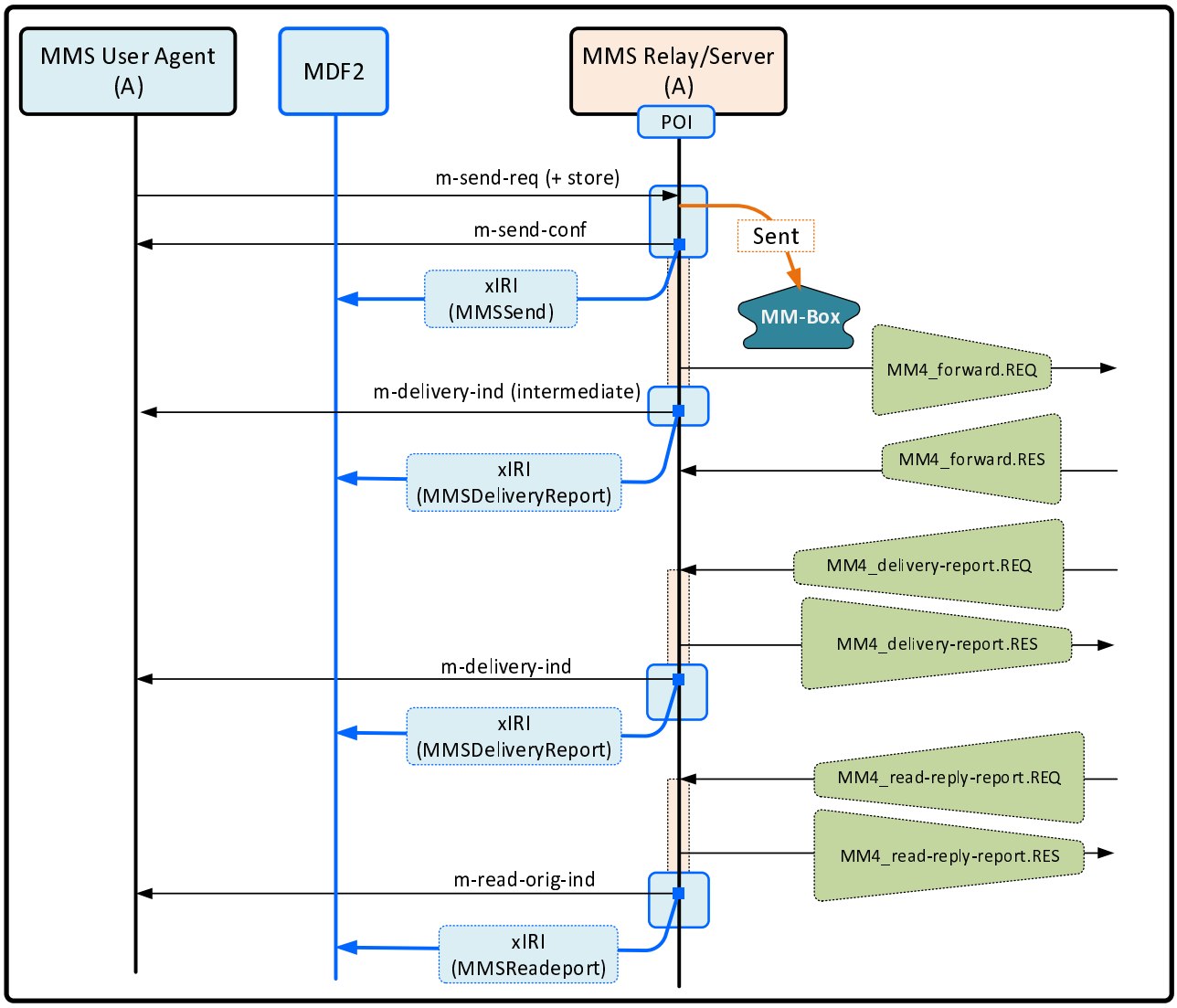


Figure 4.4.3.3.2.2-1: MMS LI messages from MM messages – MMS Relay/Server (A) – case 1

When MM message is sent by the originating MMS User Agent, if requested so, a copy of the MM message is stored in the MMBox. Therefore, the storing of the MM message at the MMBox does not happen if the store indication is not sent in the m-send-req message.

Even though the trigger for the generation of xIRI MMSSend is the m-send-conf message that the MMS-Relay/Server (A) sends to the MMS User Agent (A), the xIRI includes information from the m-send-req as well.

As shown, there are two instances where the m-delivery-ind is sent to the MMS User Agent:

- When the MM message is forwarded to the destination (using MM4_forward,REQ).
- When a delivery report is received from the destination (using MM4_delivery-report.REQ).

Since the receipt of delivery report from the destination (MM4_delivery-report-REQ) is conditional, the sending of associated m-delivery-ind is also conditional. Therefore, the generation of the xIRI MMSDeliveryReport is conditional.

Since the receipt of the read-reply report from the destination (MM4_read-reply-report.REQ) is conditional, the sending of m-read-orig-ind is also conditional. Therefore, the generation of the xIRI MMSReadReport is conditional.

4.4.3.3.2.3 Destination is an email address

Figure 4.4.3.3.2.3-1 gives an overview of the MMS LI messages applicable to the originating MMS domain when the target is local to the MMS Relay/Server and the destination is an Email Client or an MMS User Agent with an MSISDN based email address.

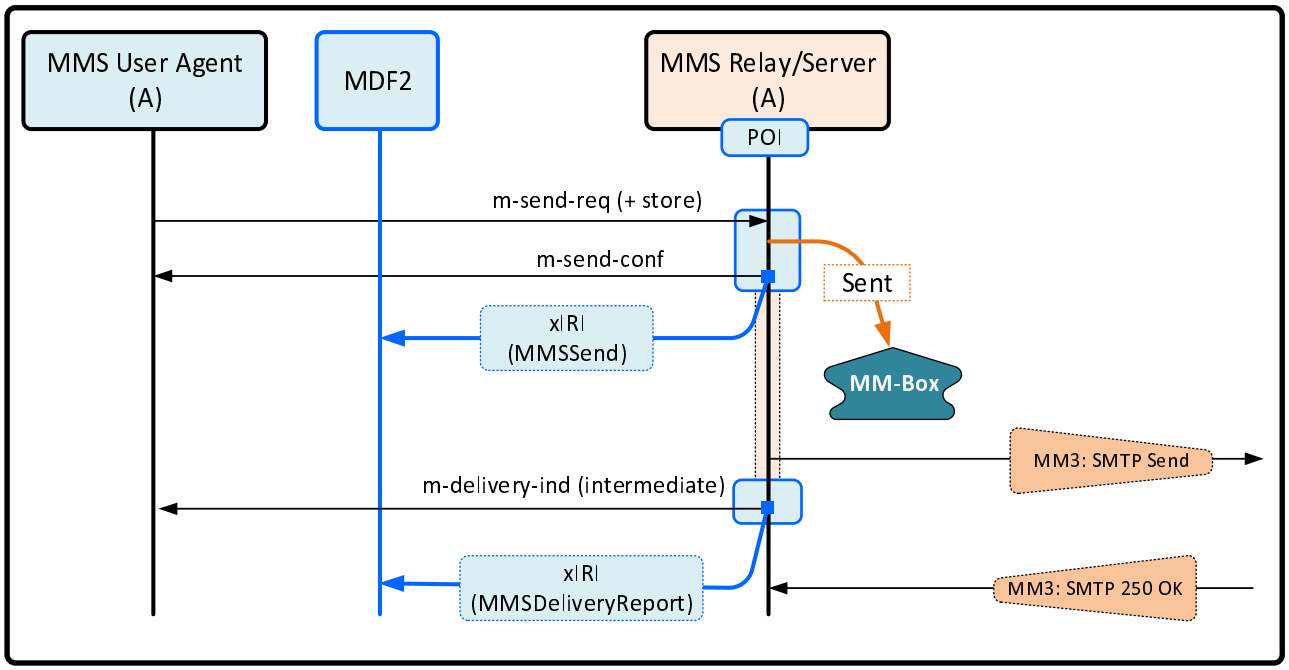


Figure 4.4.3.3.2.3-1: MMS LI messages from MM messages – MMS Relay/Server (A) - case 2

When MM message is sent by the originating MMS User Agent, if requested so, a copy of the MM message is stored in the MMBox. Therefore, the storing of the MM message at the MMBox does not happen if the store indication is not sent in the m-send-req message.

Even though the trigger for the generation of xIRI MMSSend is the m-send-conf message that the MMS-Relay/Server (A) sends to the MMS User Agent (A), the xIRI includes information from the m-send-req as well.

As shown, a m-delivery-ind is sent to the MMS User Agent when the MM message (in the email format) is forwarded to the destination (using MM3L SMTP Send).

As can be inferred, there are subsequent delivery reports or a read report from the destination end.

4.4.3.3.3 Target non-local ID

4.4.3.3.3.1 Overview

Figure 4.4.3.3.3.1-1 gives an overview of the MMS LI messages applicable to the originating MMS domain for the target non-local ID. Here, the identity of the target non-local ID is either an E.164 number or an MSISDN (i.e. another MMS user Agent).

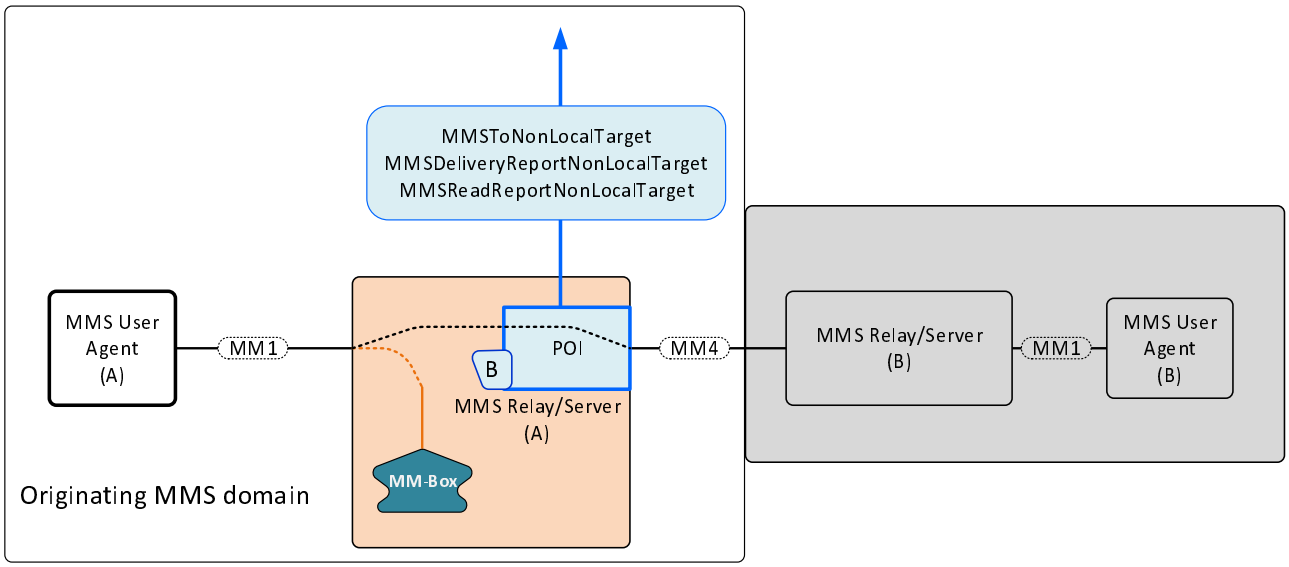


Figure 4.4.3.3.3.1-1: MMS LI messages – originating MMS domain (target non-local ID)

As shown in figure 4.4.3.3.2.1-1, the MM4 messages are referenced for the MMS LI when the destination is a target non-local ID and MMS User Agent.

4.4.3.3.3.2 MM messages to MMS LI messages

Figure 4.4.3.3.3.2-1 illustrates the generation of xIRIs from the MM4 messages as the MMS User Agent (A) sends an MM message to another MMS User Agent (target non-local ID) in a different network domain.

Note that the non-LI aspect of this flow is same as in figure 4.4.3.3.2.2-1.

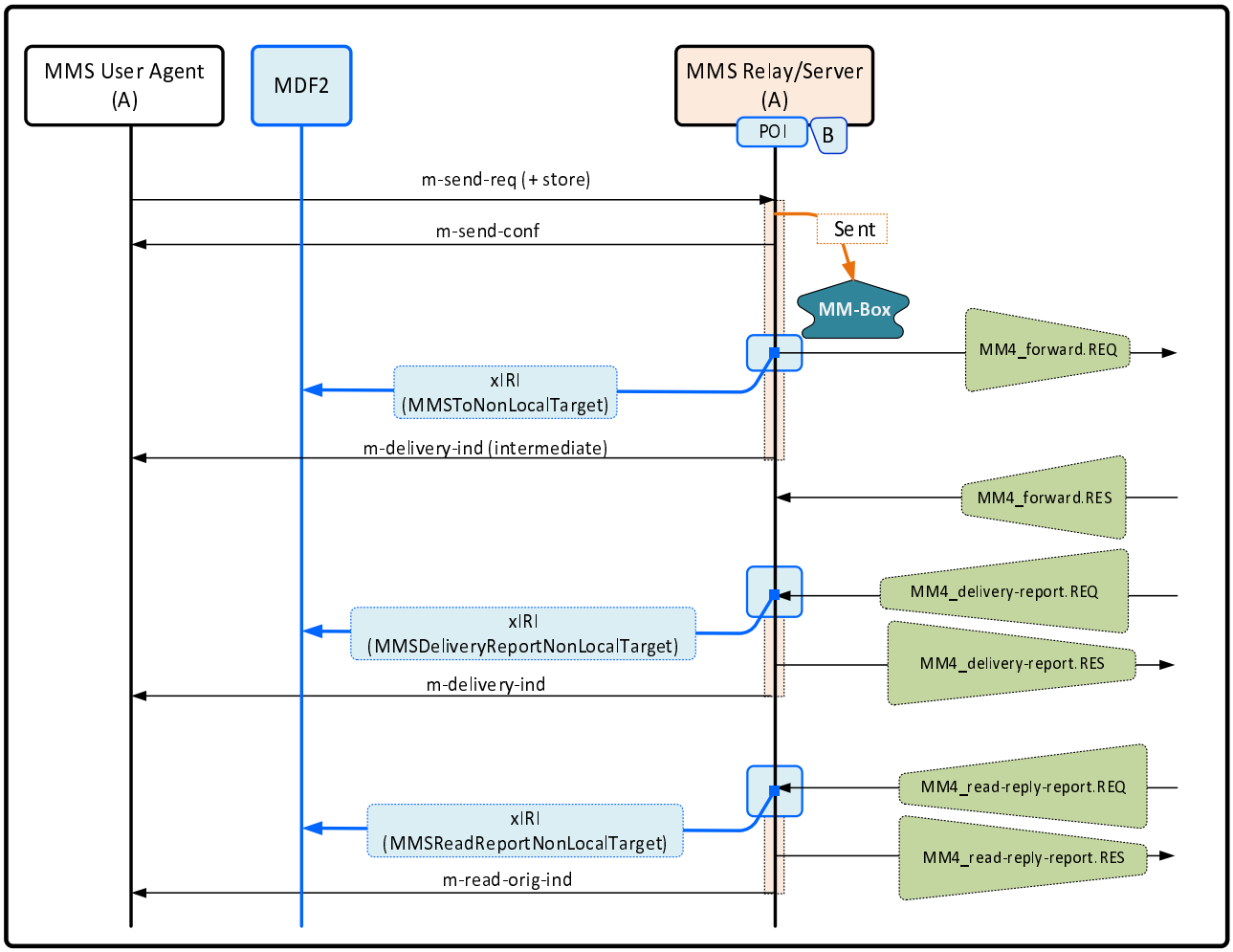


Figure 4.4.3.3.3.2-1: MMS LI messages from MM messages – MMS Relay/Server (A)

When MM message is sent by the originating MMS User Agent, if requested so, a copy of the MM message is stored in the MMBox. Therefore, the storing of the MM message at the MMBox does not happen if the store indication is not sent in the m-send-req message.

The triggers for the xIRIs are sending or receiving the MM4 messages as shown in figure 4.4.3.3.3.2-1.

Since the receipt of MM4_delivery-report.REQ is conditional, the generation of xIRI MMSDeliveryReportNonLocalTarget is also conditional.

Since the receipt of MM4_ read-reply-report.REQ is conditional, the generation of xIRI MMSReadReportNonLocalTarget is also conditional.

4.4.3.4 Terminating MMS domain

4.4.3.4.1 General

4.4.3.4.1.1 Scope

Three cases are illustrated in the subsequent clauses:

- Target is local to the MMS Relay/Server, origination is an MMS User Agent is served by a different MMS Relay/Server.
- Target is local to the MMS Relay/Server, origination is an Email Client or an MMS User Agent with an MSISDN based Email Address.
- Target non-local ID.

In the case of target non-local ID identified with a phone number, two MMS Relay/Servers are always used. The target non-local ID is the origination end of the MM message.

4.4.3.4.1.2 MMBox

When the MMBox is supported, the incoming MM message is automatically stored in the MMBox , if setup and configured so, by the terminating MMS User Agent. The status of thus stored MM message would be "New". In that case, MMS User Agent would point to the MM message stored in the MMBox for any subsequent actions (e.g. retrieval, deletion, forwarding). The status of the MM message changes according to the action taken (e.g. "Retrieved", "Forwarded").

If the MMBox is supported, but not configured to store the incoming MM message automatically, then the MMS User Agent (B) may request the MMS Relay/Server (B) to store the MM message in the MMBox using the m-mbox-store-req message as illustrated in clause 4.4.3.2.2.

If the MMBox is not supported or not setup by the MMS User Agent, then the MMBox related aspects shown in subsequent clauses are not applicable, and in that case, the MMS Relay/Server (B) stores the message locally and reference then would be to the internally stored message for any subsequent actions (e.g. Retrieve, Forward, Delete).

The MMS User Agent may also perform a few MMBox specific operations as illustrated in clause 4.4.3.2.2.

The LI does not apply to such MMBox specific operations for target non-local ID.

4.4.3.4.1.3 Delivery reports

The sending of delivery report to the originating MMS domain is conditional. The delivery report is sent only if the originating MMS User Agent has requested such a report and the terminating MMS User Agent permits the sending of such a report.

The delivery report is not sent if the origination from an Email Address.

4.4.3.4.1.4 Read reports

The sending of the read report to the originating MMS domain is conditional. The read report is sent only if the originating MMS User Agent has requested for such a report and the terminating MMS User Agent sends such a report.

The read report is not sent, if the origination is from an Email Address.

4.4.3.4.1.5 Retrieval of MM message

The terminating MMS User Agent upon receiving a notification for an incoming MM message may retrieve the MM message either automatically (referred to as immediate retrieval) or manually (referred to as deferred retrieval).

4.4.3.4.1.6 Deletion of MM message

The terminating MM user upon receiving a notification for an incoming MM message may delete the MM message without retrieval.

4.4.3.4.2 Target is local to MMS Relay Server

4.4.3.4.2.1 Overview

Figure 4.4.3.4.2.1-1 gives an overview of the MMS LI messages applicable to the terminating MMS domain when the target is local to the MMS Relay/Server. The MM message may have been originated from an MMS User Agent or an Email Client or can be an MMS User Agent with an MSISDN based Email Address.

When the origination MMS User Agent uses an MSISDN based Email Address, from the MMS Relay/Server (B)'s perspective, the handling is similar to when the origination is from an Email Client since the inbound MM messages are transferred over the MM3 interface in an email message format.

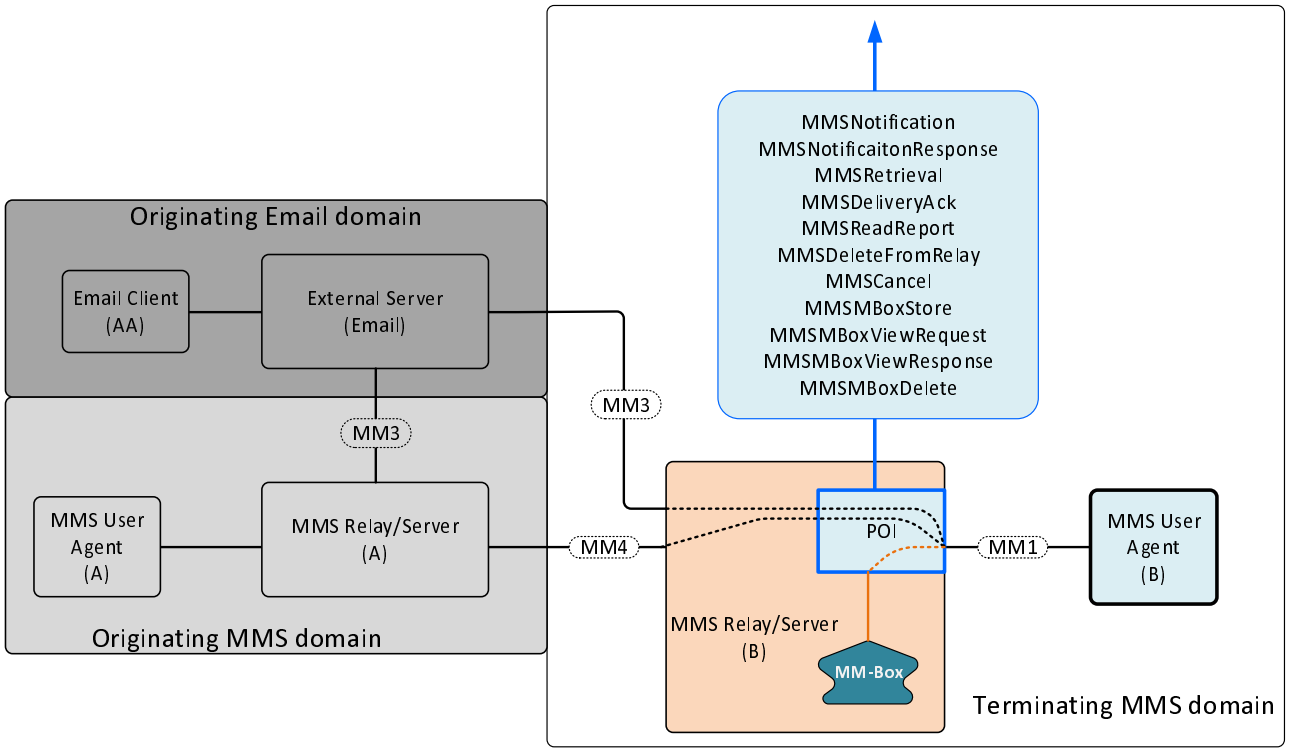


Figure 4.4.3.4.2.1-1: MMS LI messages – terminating MMS domain (target is local)

The MM1 messages between the MMS Relay Server (B) and the MMS User Agent (B), the target, are used to generate the xIRIs.

The type of MMS origination (i.e. Email Client Vs MMS User Agent) has no impact on the MMS LI when the terminating MMS User Agent is a target and local to the MMS Relay/Server.

4.4.3.4.2.2 MM messages to MMS LI messages

4.4.3.4.2.2.1 Immediate retrieval

Figure 4.4.3.4.2.2.1-1 illustrates the generation of xIRIs from the MMS messages as the MMS User Agent (B) is notified and the MMS User Agent (B) automatically retrieves the MM message, referred to as immediate retrieval.

In this illustration, the origination is an MMS User Agent (identified with a phone number) served by a different MMS Relay/Server (the other MMS Relay/Server is not shown in figure 4.4.3.4.2.2.1-1).

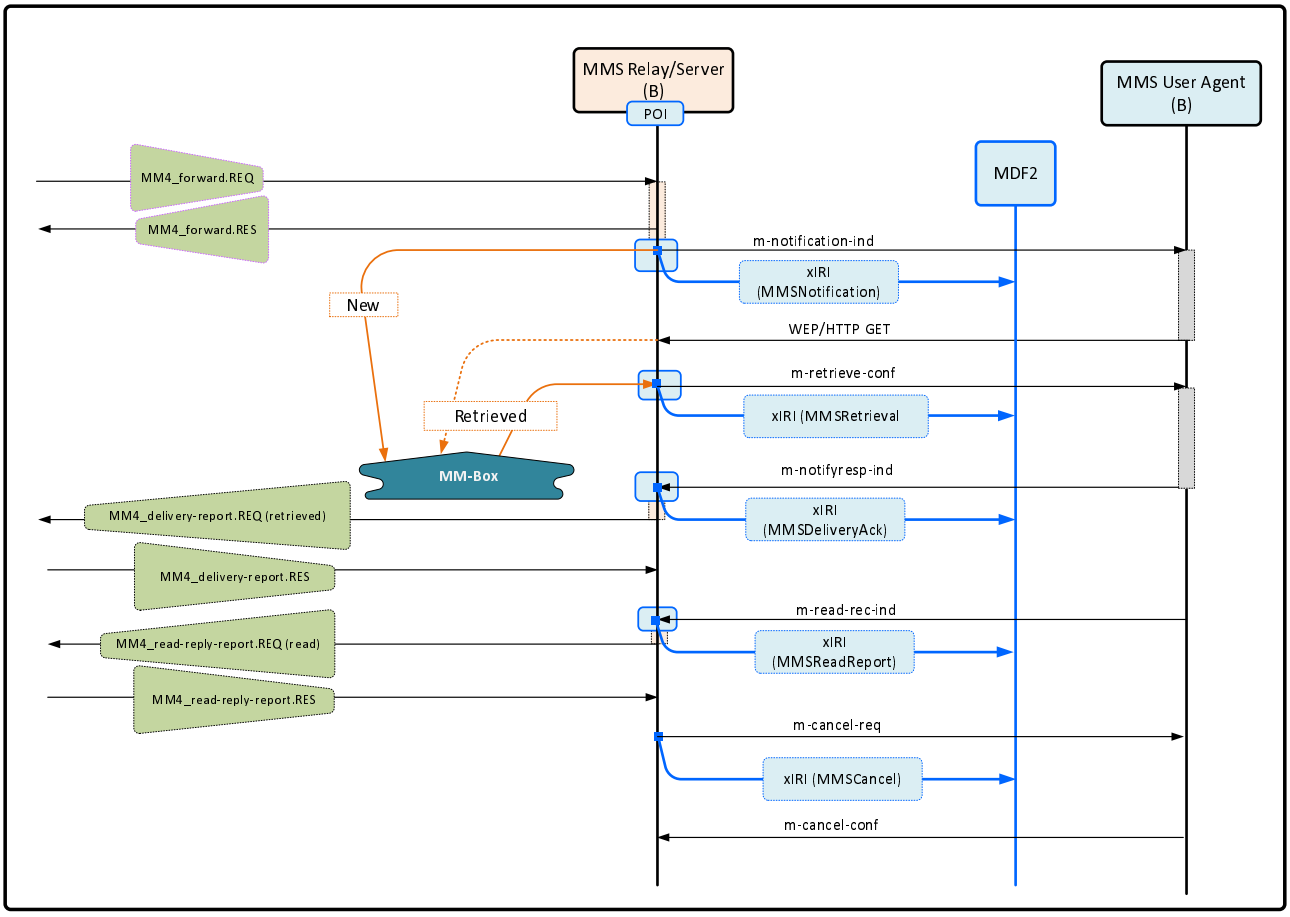


Figure 4.4.3.4.2.2.1-1: MMS LI messages from MM messages (immediate) – MMS Relay/Server (B)

The trigger for xIRI MMSDeliveryAck is MMS Relay/Server (B) receiving the m-notifyresp.ind message.

The trigger for the xIRIMMSReadReport is MMS Relay Server (B) receiving the m-read-rec-ind from the target MMS User Agent (B). However, MMS User Agent (B) sending of the m-read-rec-ind is conditional, and therefore, the generation of xIRI MMSReadReport is conditional.

The MMS Relay/Server (B) may cancel (shown in figure 4.4.3.4.2.2.1-1) the MM message retrieved by the MMS User Agent (B). This step is not part of the normal flow. However, if the MMS Relay/Server (B) cancels the retrieved MM message at the MMS User Agent (B), then the xIRI MMCancel is generated as shown.

4.4.3.4.2.2.2 Deferred retrieval

Figure 4.4.3.4.2.2.2-1 illustrates the generation of xIRIs from the MMS messages as the MMS User Agent (B) is notified and subsequently retrieves the MM message, referred to as deferred retrieval.

In this illustration, the origination is an MMS User Agent (identified with a phone number) served by a different MMS Relay/Server (the other MMS Relay/Server is not shown in figure 4.4.3.4.2.2.2-1).

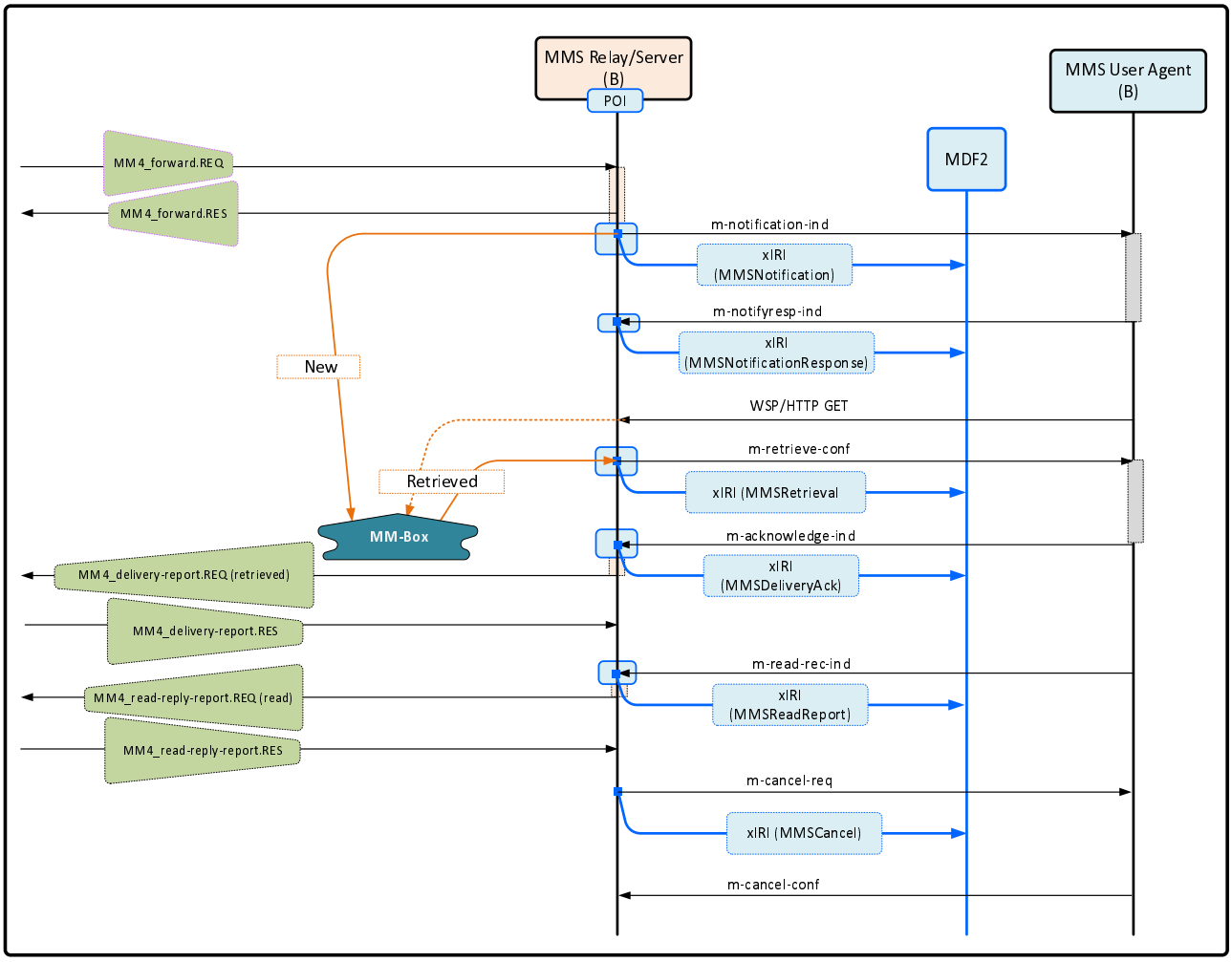


Figure 4.4.3.4.2.2.2-1: MMS LI messages from MM messages (deferred) – MMS Relay/Server (B)

The trigger for xIRI MMSDeliveryAck is MMS Relay/Server (B) receiving the m-acknowledge.ind message.

The trigger for the xIRIMMSReadReport is MMS Relay Server (B) receiving the m-read-rec-ind from the target MMS User Agent (B). However, MMS User Agent (B) sending of the m-read-rec-ind is conditional, and therefore, the generation of xIRI MMSReadReport is conditional.

The MMS Relay/Server (B) may cancel (shown in figure 4.4.3.4.2.2.2-1) the MM message retrieved by the MMS User Agent (B). This step is not part of the normal flow. However, if the MMS Relay/Server (B) cancels the retrieved MM message at the MMS User Agent (B), then the xIRI MMCancel is generated as shown.

4.4.3.4.2.2.3 Deletion of MM message

The MMS User Agent (B) may delete the MM message from the MMS Relay/Server instead of the retrieval.

Figure 4.4.3.4.2.2.3-1 illustrates the generation of xIRIs from the MMS messages as the MMS User Agent (B) is notified and the MMS User Agent (B) deletes the MM message from the MMS Relay/Server instead of retrieving. This may happen only with the deferred retrieval case.

In this illustration, the origination is an MMS User Agent (identified with a phone number) served by a different MMS Relay/Server (the other MMS Relay/Server is not shown in figure 4.4.3.4.2.2.3-1).

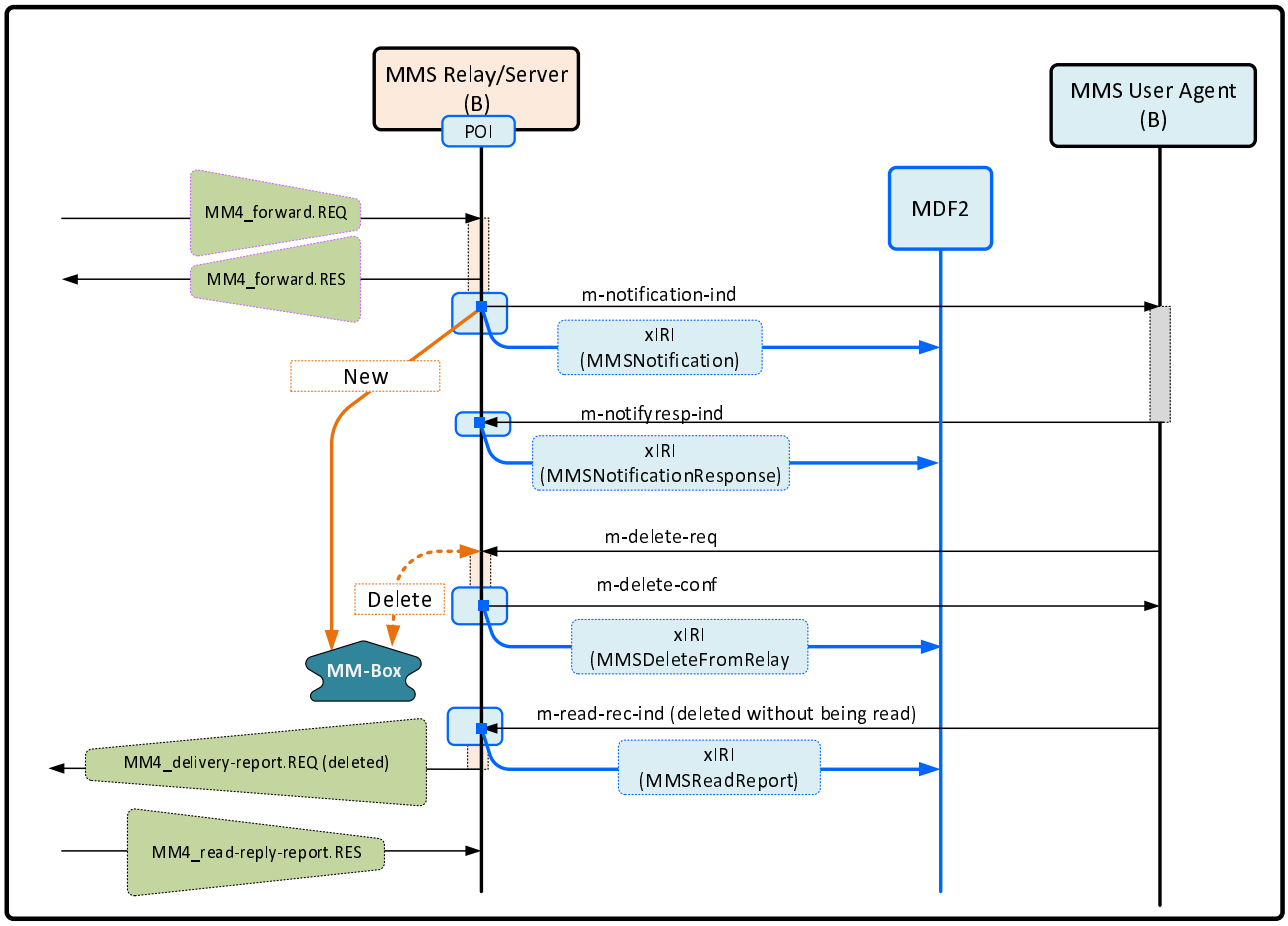


Figure 4.4.3.4.2.2.3-1: MMS LI messages from MM messages (deletion) – MMS Relay/Server (B)

If a read report is requested by the originating MMS User Agent, then the MMS User Agent (B) may send a m-read-rec-ind with "deleted without being read". The xIRI MMSReadReport is generated when the MMS Relay/Server (B) receives the m-read-rec-ind message.

4.4.3.4.2.2.4 Origination is an email address

Figure 4.4.3.4.2.2.4-1 gives an overview of the MMS LI messages applicable to the terminating MMS domain when the target is local to the MMS Relay/Server and the origination is an Email Client or an MMS User Agent with an MSISDN based email address. The deferred retrieval case is illustrated below.

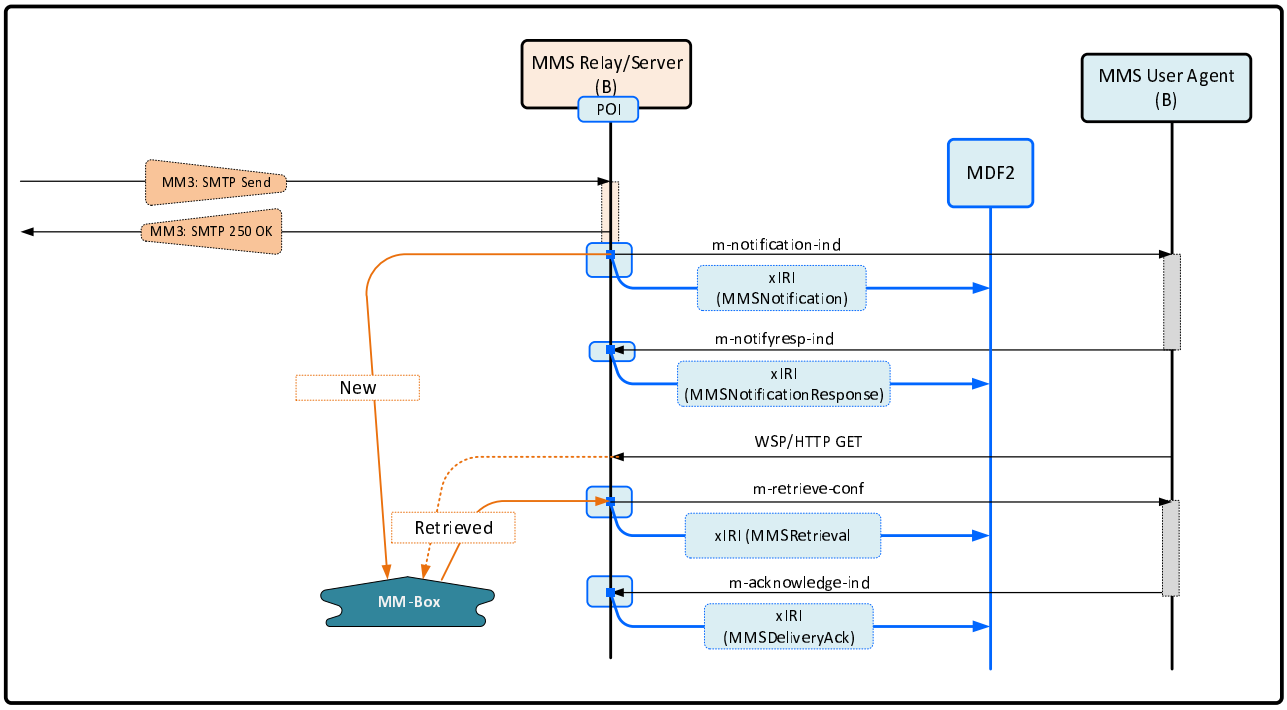


Figure 4.4.3.3.2.4-1: MMS LI messages from MM messages – MMS Relay/Server (from email address)

As can be inferred, there are no delivery report or a read report since the originating end is an Email Address.

4.4.3.4.3 Target non-local ID

4.4.3.4.3.1 Overview

Figure 4.4.3.4.3.1-1 gives an overview of the MMS LI messages applicable to the terminating MMS domain for the target non-local ID. Here, the identity of the target non-local ID is either an E.164 number or an MSISDN (i.e. another MMS user Agent).

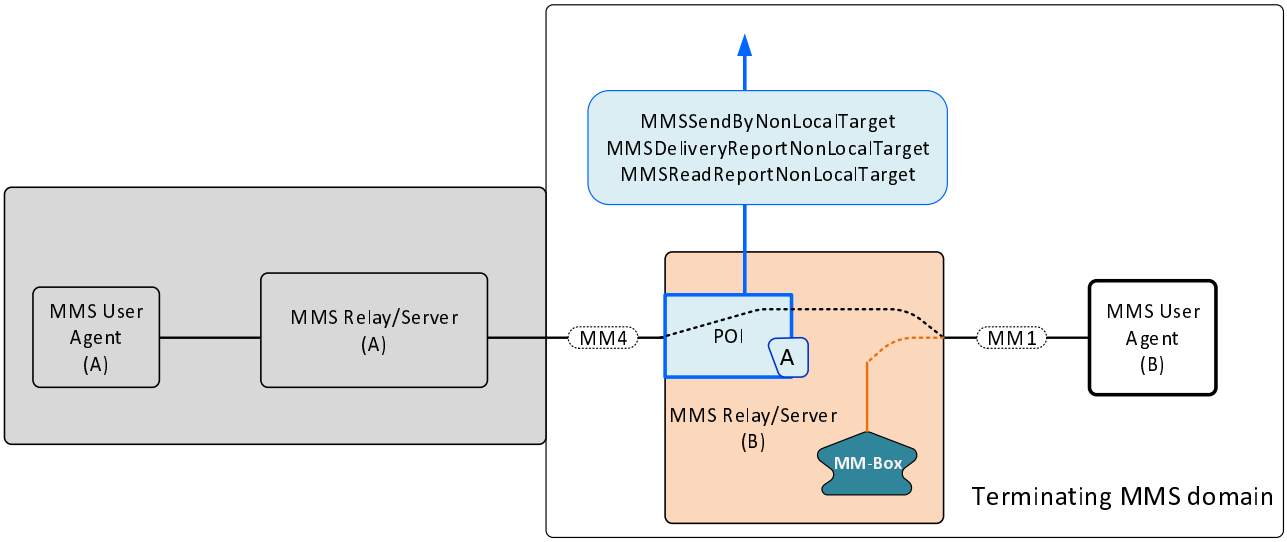


Figure 4.4.3.4.3.1-1: MMS LI messages – originating MMS domain (target non-local ID)

As shown in figure 4.4.3.4.3.1-1, the MM4 messages are referenced for the MMS LI when the origination target non-local ID is an MMS User Agent.

The MMS LI for target non-local ID in the terminating MMS Relay/Server (B) is same for immediate retrieval and deferred retrieval cases. Only the immediate retrieval case and deletion before retrieval cases are illustrated.

4.4.3.4.3.2 MM messages to MMS LI messages

4.4.3.4.3.2.1 Immediate retrieval

Figure 4.4.3.4.3.2.1-1 illustrates the generation of xIRIs from the MMS messages as the MMS User Agent (B) is notified and automatically retrieves the MM message sent from MMS User Agent (target non-local ID) present in a different network domain.

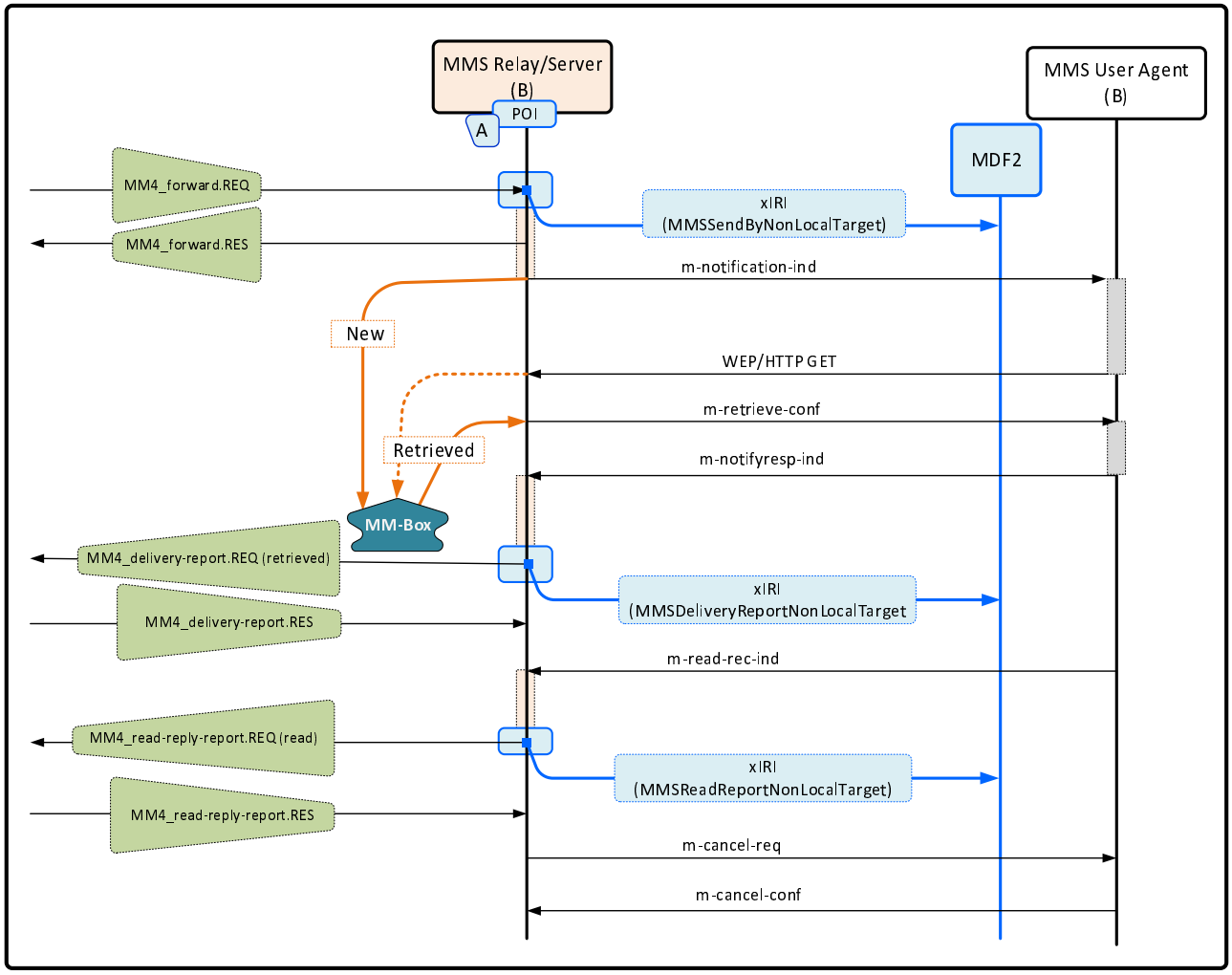


Figure 4.4.3.4.3.2.1-1: MMS LI messages from MM messages (immediate) – MMS Relay/Server (B)

The triggers for the xIRIs are sending or receiving the MM4 messages as shown in figure 4.4.3.4.3.2-1.

Since the sending of MM4_delivery-report.REQ is conditional, the generation of xIRI MMSDeliveryReportNonLocalTarget is also conditional.

Since the sending of MM4_ read-reply-report.REQ is conditional, the generation of xIRI MMSReadReportNonLocalTarget is also conditional.

4.4.3.4.3.2.2 Deferred retrieval

Figure 4.4.3.4.3.2.2-1 illustrates the generation of xIRIs from the MMS messages as the MMS User Agent (B) is notified and subsequently retrieves the MM message sent from MMS User Agent (target non-local ID) present in a different network domain.

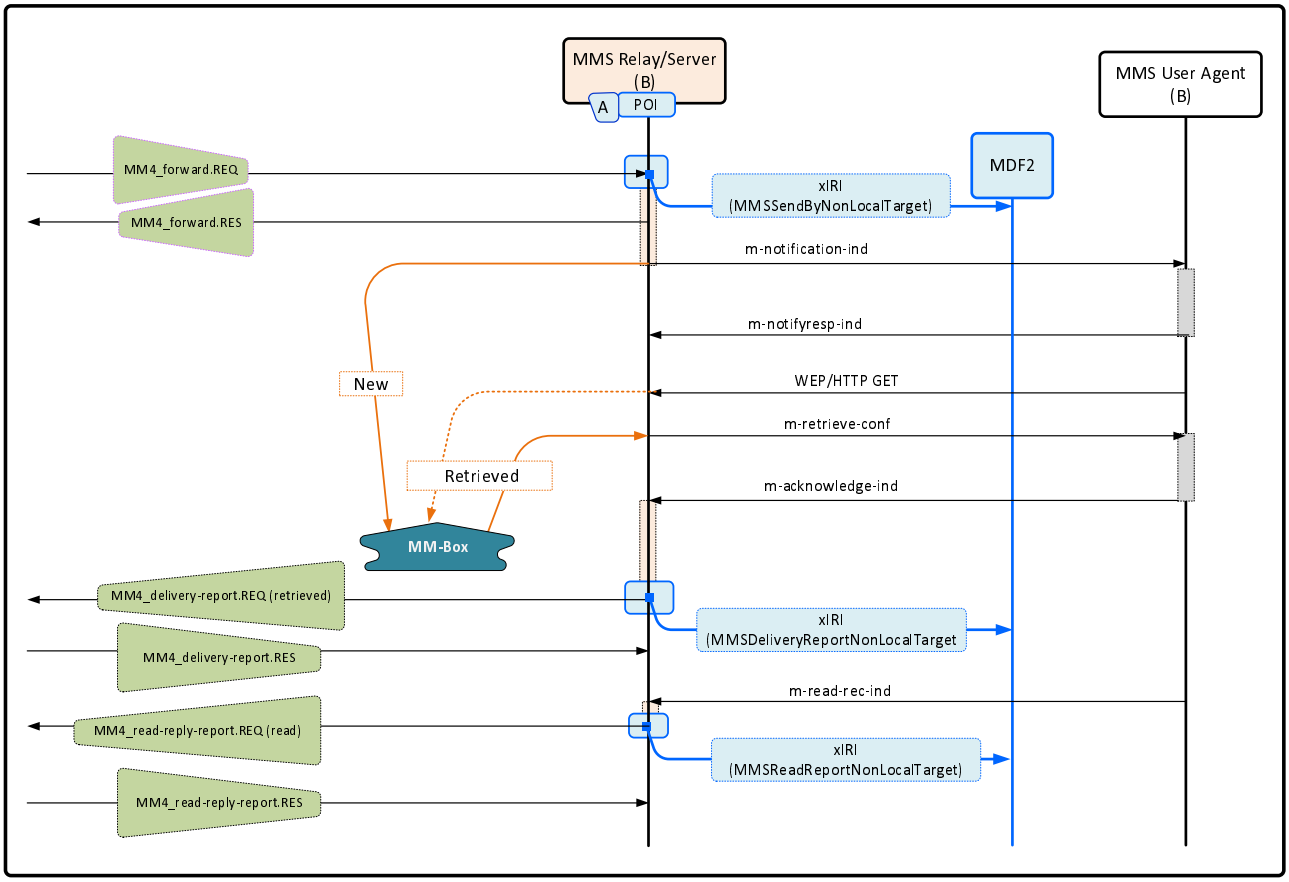


Figure 4.4.3.4.3.2.2-1: MMS LI messages from MM messages (deferred) – MMS Relay/Server (B)

The triggers for the xIRIs are sending or receiving the MM4 messages as shown in figure 4.4.3.4.3.2-1.

Since the sending of MM4_delivery-report.REQ is conditional, the generation of xIRI MMSDeliveryReportNonLocalTarget is also conditional.

Since the sending of MM4_ read-reply-report.REQ is conditional, the generation of xIRI MMSReadReportNonLocalTarget is also conditional.

4.4.3.4.3.2.3 Deletion of MM message

The MMS User Agent (B) may delete the MM message from the MMS Relay/Server instead of the retrieval.

Figure 4.4.3.4.3.2.3-1 illustrates the generation of xIRIs from the MMS messages as the MMS User Agent (B) is notified and the MMS User Agent (B) deletes the MM message from the MMS Relay/Server instead of retrieving.

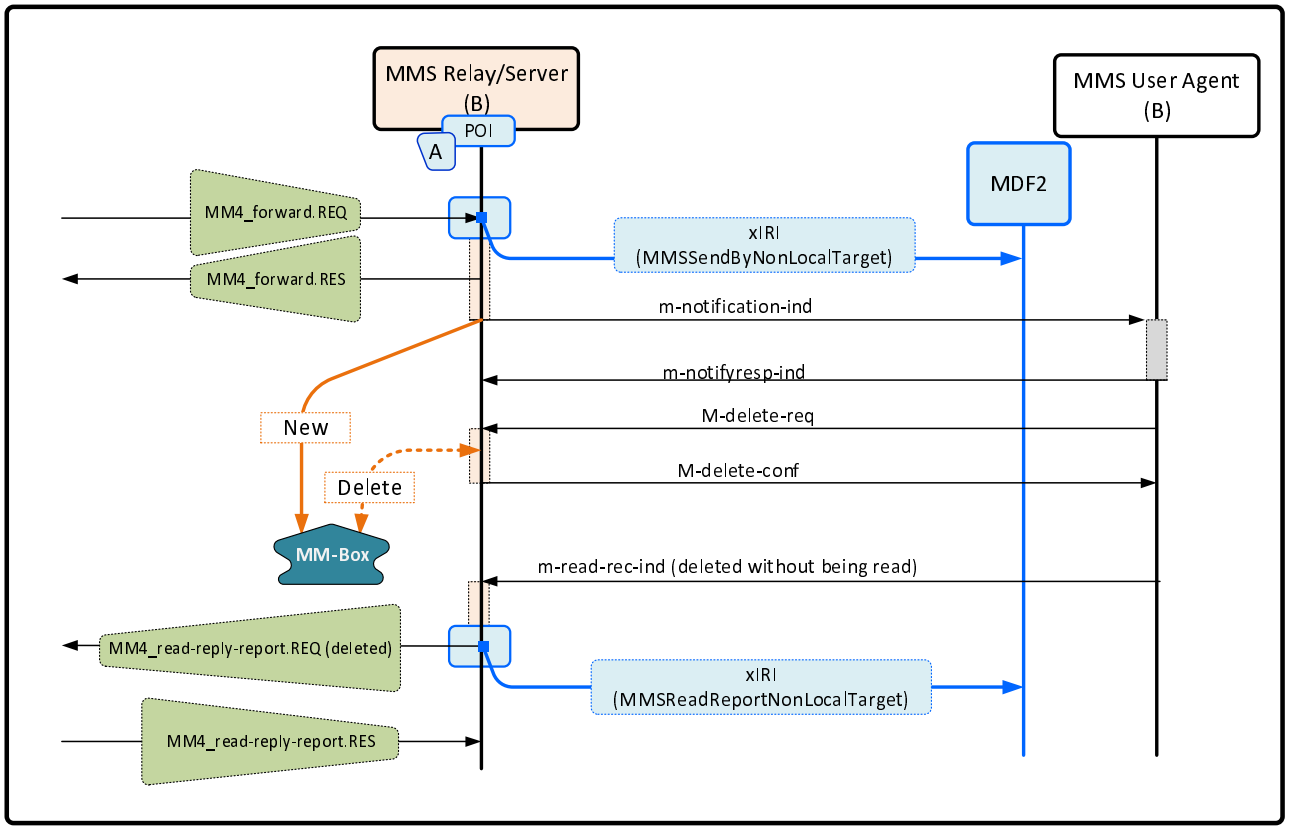


Figure 4.4.3.4.3.2.3-1: MMS LI messages from MM messages (deletion) – MMS Relay/Server (B)

If a read report is requested by the originating MMS User Agent, then the MMS User Agent (B) may send a m-read-rec-ind with "deleted without being read". Upon receiving such a m-read-rec-ind from the MMS User Agent (B), the MMS Relay/Server sends the MM4_read-reply-report.REQ to the originating MMS domain. The sending of MM4_read-reply-report.REQ triggers the sending of xIRI MMSReadReportNonLocalTarget.

The MMS User Agent (B) may also separately delete the MM message stored in the MMBox using the m-mbox-delete-req message as illustrated in clause 4.4.3.2.2. However, this has no impact on the MMS LI since the target is non-local ID.

4.4.3.5 Redirecting MMS domain

4.4.3.5.1 General

4.4.3.5.1.1 Scope

The MMS User Agent that is notified of an incoming MM message may forward the message to different destination (Email Client or MMS User Agent). The forwarding happens instead of the retrieval. Note that the forwarding of the MM message applies only for the deferred retrieval case.

In the subsequent descriptions, originating end refers to the originator the initial MM message and the destination refers to the destination after the forwarding (i.e. redirected-to end). The MMS User Agent who does the act of forwarding is referred to as redirecting MMS User Agent.

As far as the MMS LI is concerned, two aspects have to be addressed:

- LI for the incoming leg MM messages.
- LI for forwarded leg MM messages.

The LI for the incoming leg of MM message is similar to the MMS LI in a terminating MMS domain and the LI for the forwarded leg (i.e. redirected leg) is similar to the MMS LI in an originating MMS domain, except the messages used on the MM1 interface.

The following three cases are illustrated in the subsequent clauses:

- Redirecting MMS User Agent is a target (local to the MMS Relay/Server).
- Originating MMS User Agent is a target non-local ID at the redirecting MMS Relay/Server.
- The redirected-to (i.e. destination) MMS User Agent is a target non-local ID at the redirecting MMS/Relay/Server.

In the case of target non-local ID, the illustrations show the use of three separate MMS Relay/Servers.

Two alternatives of forwarding

The MMS User Agent may first retrieve the MM message and then forward the retrieved MM message using the MMS origination procedures (i.e. using the m-send-req message). In this case, the MMS origination procedures as described in clause 4.4.3.3 are used.

Yet as another alternative, through a prior setup, the MMS Relay/Server could automatically, forward the incoming MM message to an MMS User Agent to a pre-setup destination. In this a user-action, at the time, when an MM message is not required.

These are not described as alternate methods of forwarding in TS 23.140 [18] or the OMA spec [19], but mentioned here, as some deployments may have such implementations.

4.4.3.5.1.2 MMBox

When the MMBox is supported, the incoming MM message is automatically stored in the MMBox, if setup and configured so, by the redirecting MMS User Agent. The status of thus stored MM message would be "New". In that case, redirecting MMS User Agent would point to the MM message stored in the MMBox in the forwarding request. Once forwarded, the status of the MM message in the MMBox is changed to "forwarded".

If the MMBox is supported, but not configured to store the incoming MM message automatically, then the MMS User Agent may request the MMS Relay/Server to store the MM message in the MMBox, when a forwarding request is sent. The status of the thus stored MM message would be "forwarded".

If the MMBox is not supported or not setup by the MMS User Agent, then MMBox related steps shown in the subsequent clauses are not applicable, and in that case, the MMS Relay/Server (B) stores the MM message locally and a reference then would be to the internally stored MM message.

4.4.3.5.1.3 Delivery reports

Delivery report to the originating MMS domain

The sending of delivery report to the originating MMS domain is conditional. The delivery report is sent only if the originating MMS User Agent has requested such a report and the redirecting MMS User Agent permits the sending of such a report.

The delivery report is not sent if the origination from an Email Address.

Delivery report from the redirected-to MMS domain

A delivery report may be received from a redirected-to MMS domain if the requests for such a report and the redirected-to MMS User Agent permits the sending of the same. When the redirection happens, an intermediate delivery report is sent to the redirecting MMS User Agent, if the same is requested by that redirecting MMS User Agent.

4.4.3.5.1.4 Read reports

A read report is not sent to the originating MMS domain.

A read report may be received from the redirected-to MMS domain to the redirecting MMS domain, if the redirecting MMS User Agent requests for such a report and the redirected-to MMS User Agent sends such a report.

The read report is not received, if the MMS is forwarded to an Email Address.

4.4.3.5.2 Target is local to MMS Relay Server

4.4.3.5.2.1 Overview

Figure 4.4.3.5.2.1-1 gives an overview of the MMS LI messages applicable to the redirecting MMS domain when the target is local to the MMS Relay/Server. The redirecting MMS domain is nothing but the terminating MMS domain except that here the terminating MMS User Agent forwards the MM message instead of retrieval or deletion.

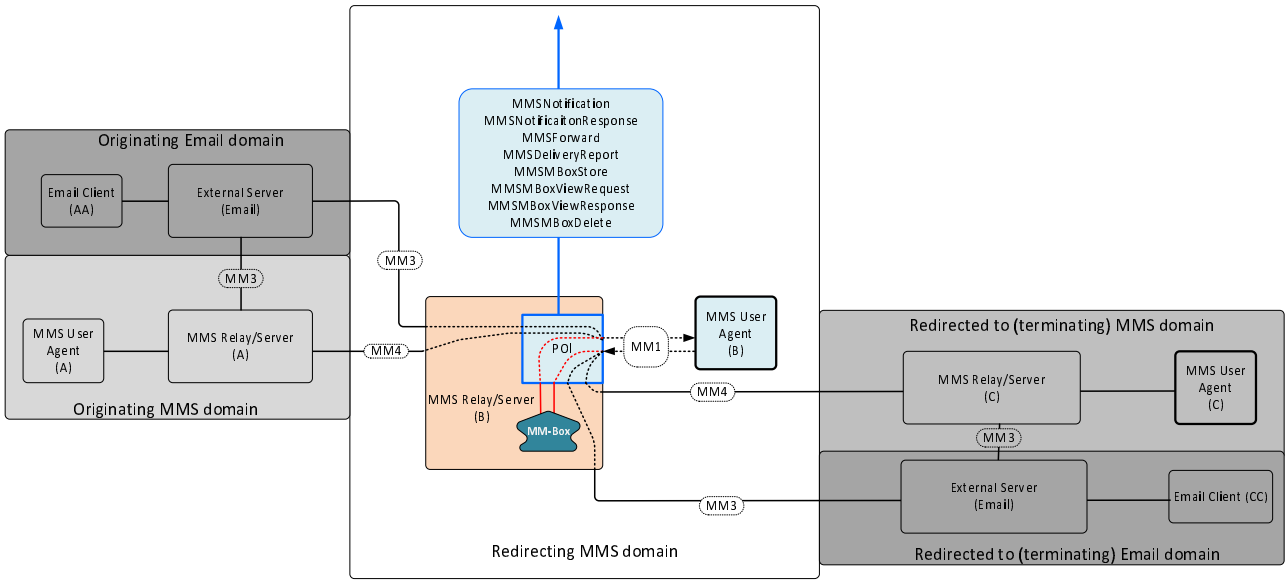


Figure 4.4.3.5.2.1-1: MMS LI messages – redirecting MMS domain (target is local)

As shown in figure 4.4.3.5.2.1-1, the MM message may have been originated from an MMS User Agent or an Email Client. The MM message may be forwarded to an MMS User Agent or to an Email Client.

The type of MMS origination (i.e. Email Client Vs MMS User Agent) or the destination end (Email Client Vs MMS User Agent), i.e. redirected to end, has no impact on the MMS LI when the redirecting MMS User Agent is a target and local to the MMS Relay/Server.

4.4.3.5.2.2 MM messages to MMS LI messages

Figure 4.4.3.5.2.2-1 illustrates the case where the MMS User Agent (B), the target, is notified and subsequently forwards the MM message.

In this illustration, the origination and the destination (i.e. redirected-to) is an MMS User Agent (identified with a phone number) served by a different MMS Relay/Server (the other MMS Relay/Server is not shown in figure .4.3.5.2.2-1).

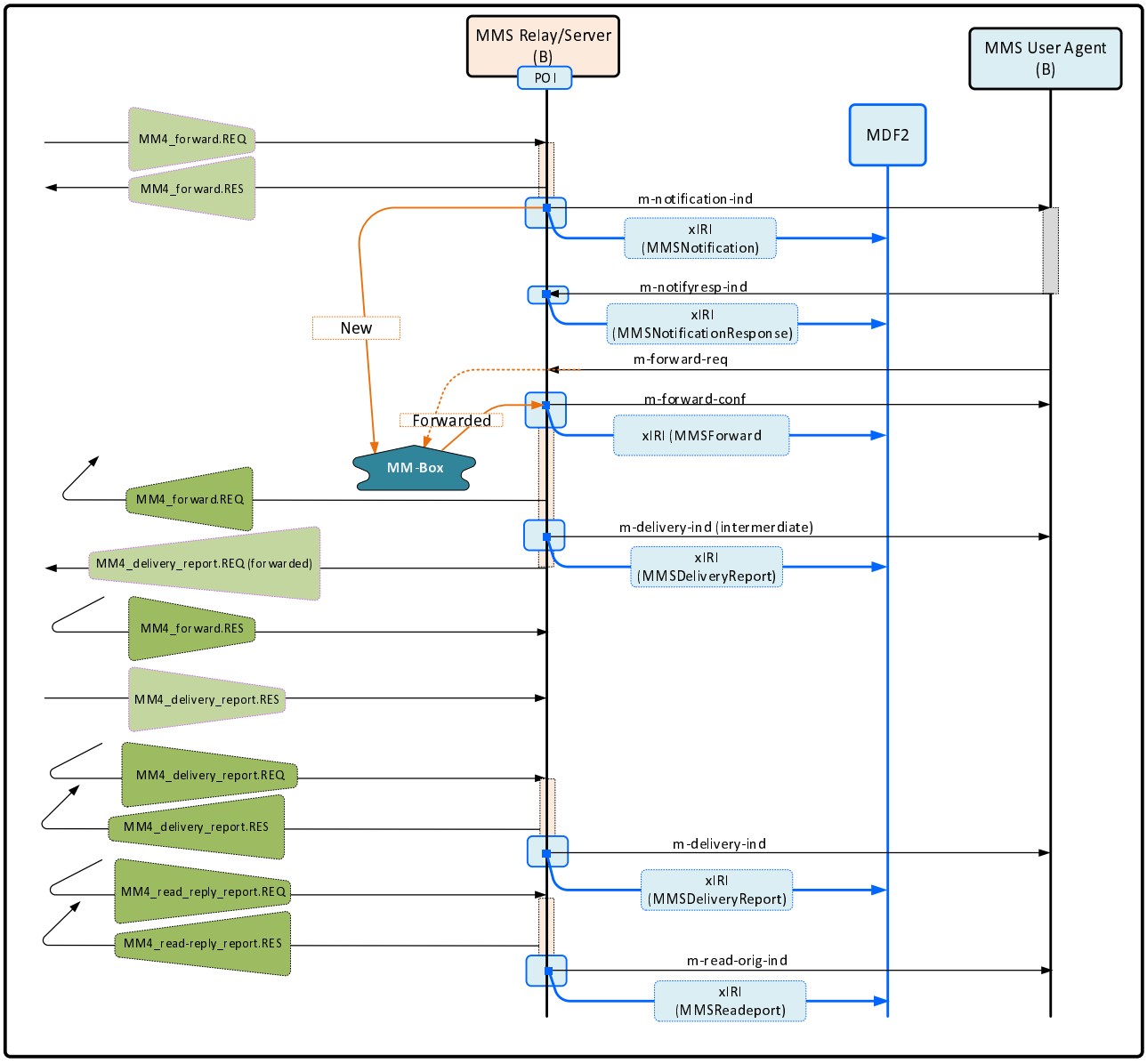


Figure 4.4.3.5.2.2-1: MMS LI messages from MM messages (forward)

As illustrated, the xIRIs generated a mixture of the xIRIs generated for the terminating MMS domain and originating MMS domain cases.

When the MMS User Agent (B) forwards the message, the xIRI MMSForward is sent when the m-forward-conf is sent to the MMS User Agent (B). The information from the MM message retrieved from the MMBox is used for the xIRI MMSForward message.

As shown, there are two instances where the m-delivery-ind is sent to the MMS User Agent (B):

- When the MM message is forwarded to the destination (using MM4_forward.REQ).
- When a delivery report is received from the destination (using MM4_delivery-report.REQ).

Since the receipt of delivery report from the destination (MM4_delivery-report-REQ) is conditional, the sending of associated m-delivery-ind is also conditional. Therefore, the generation of the xIRI MMSDeliveryReport is conditional.

Since the receipt of the read-reply report from the destination (MM4_read-reply-report.REQ) is conditional, the sending of m-read-orig-ind is also conditional. Therefore, the generation of the xIRI MMSReadReport is conditional.

4.4.3.5.2.3 Destination is an email address

Figure 4.4.3.5.2.3-1 gives an overview of the MMS LI messages applicable to the Redirecting MMS domain when the target is local to the MMS Relay/Server and the destination (i.e. redirected-to) is an Email Client or an MMS User Agent with an MSISDN based email address.

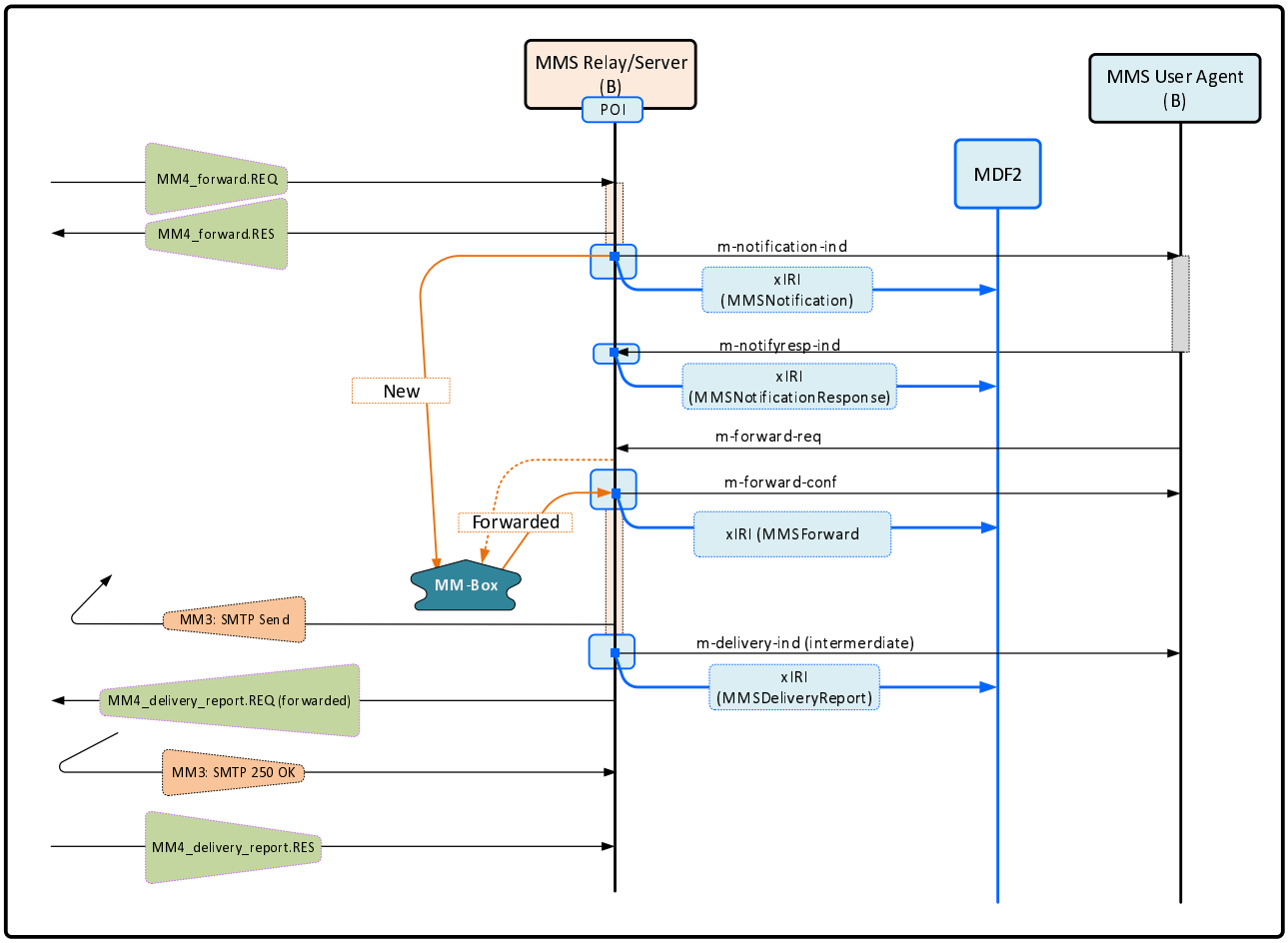


Figure 4.4.3.5.2.3-1: MMS LI messages from MM messages – MMS Relay/Server (B)

When the MMS User Agent (B) forwards the message, the xIRI MMSForward is sent when the m-forward-conf is sent to the MMS User Agent (B). The information from the MM message retrieved from the MMBox is used for the xIRI MMSForward message.

As shown, m-delivery-ind is sent to the MMS User Agent when the MM message (in the email format) is forwarded to the destination (using MM3: SMTP Send).

As can be inferred, there are subsequent delivery reports or a read report from the destination end.

4.4.3.5.2.4 Origination is an email address

Figure 4.4.3.5.2.4-1 gives an overview of the MMS LI messages applicable to the Redirecting MMS domain when the target is local to the MMS Relay/Server and the origination is an Email Client or an MMS User Agent with an MSISDN based email address.

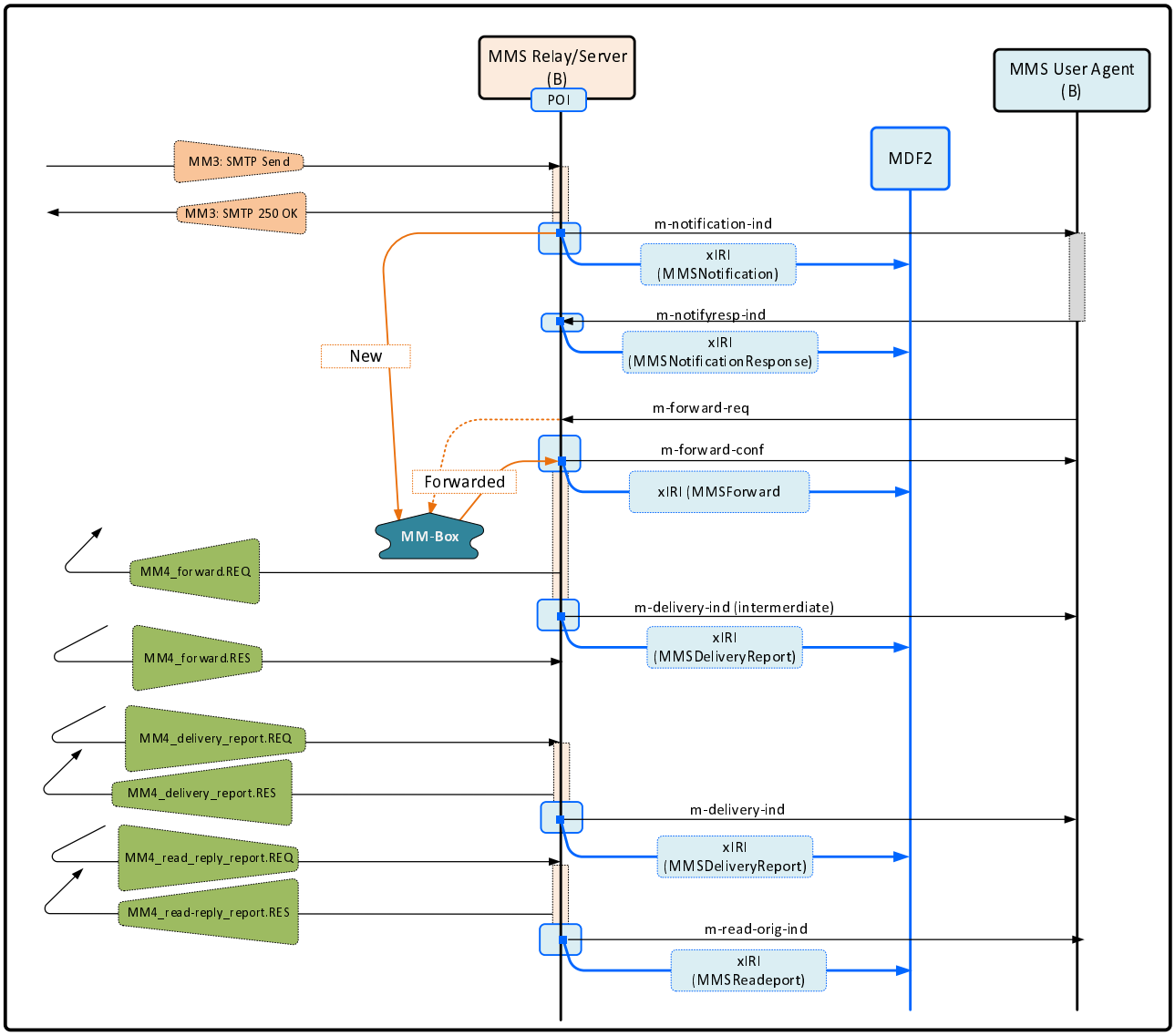


Figure 4.4.3.5.2.4-1: MMS LI messages from MM messages – MMS Relay/Server (B)

When the MMS User Agent (B) forwards the message, the xIRI MMSForward is sent when the m-forward-conf is sent to the MMS User Agent (B). The information from the MM message retrieved from the MMBox is used for the xIRI MMSForward message.

As shown, there are two instances where the m-delivery-ind is sent to the MMS User Agent (B):

- When the MM message is forwarded to the destination (using MM4_forward.REQ).
- When a delivery report is received from the destination (using MM4_delivery-report.REQ).

Since the receipt of delivery report from the destination (MM4_delivery-report-REQ) is conditional, the sending of associated m-delivery-ind is also conditional. Therefore, the generation of the xIRI MMSDeliveryReport is conditional.

Since the receipt of the read-reply report from the destination (MM4_read-reply-report.REQ) is conditional, the sending of m-read-orig-ind is also conditional. Therefore, the generation of the xIRI MMSReadReport is conditional.

4.4.3.5.2.5 Origination and destination are email addresses

Figure 4.4.3.5.2.5-1 gives an overview of the MMS LI messages applicable to the Redirecting MMS domain when the target is local to the MMS Relay/Server and the origination and destination are Email Clients or an MMS User Agents with an MSISDN based email addresses.

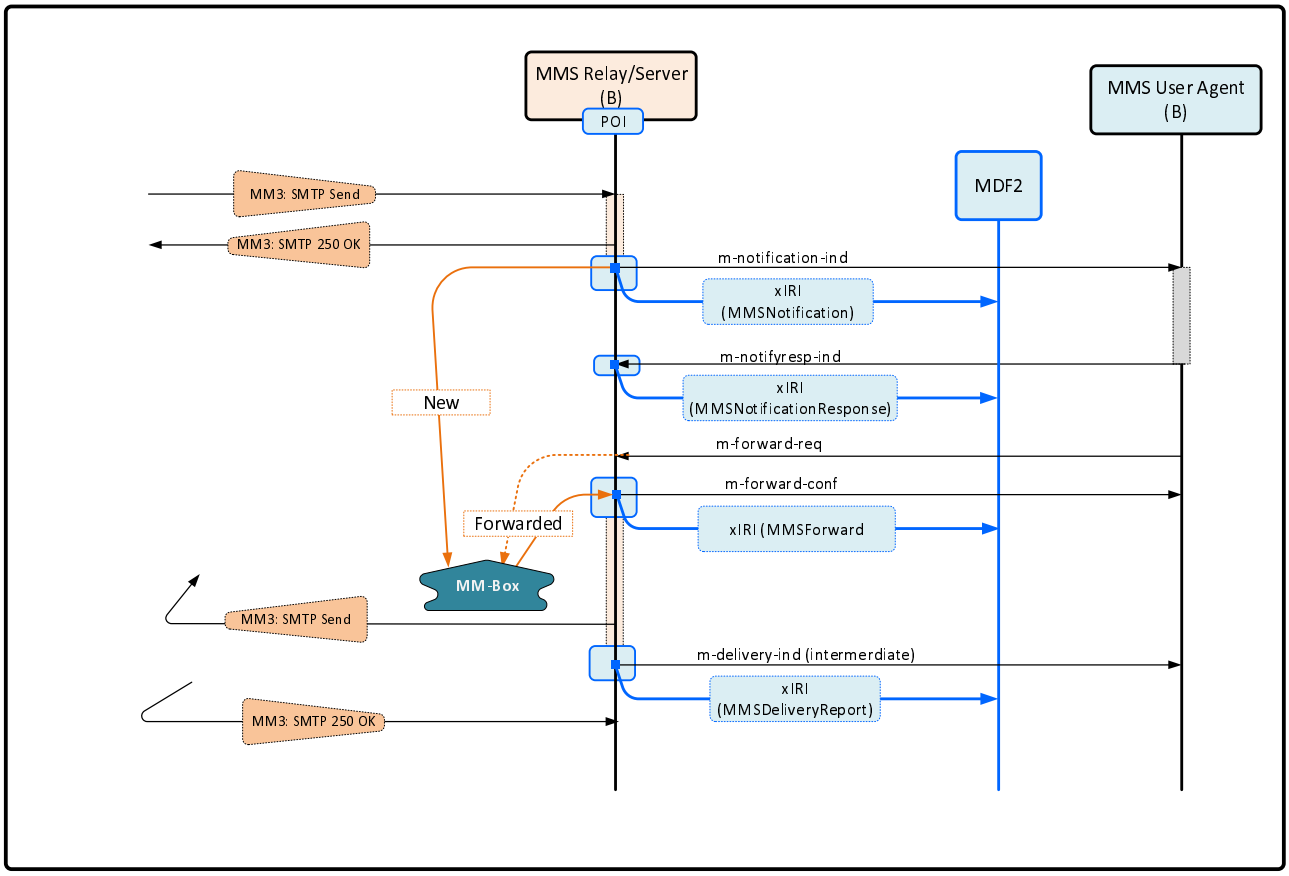


Figure 4.4.3.5.2.5-1: MMS LI messages from MM messages – MMS Relay/Server (B)

When the MMS User Agent (B) forwards the message, the xIRI MMSForward is sent when the m-forward-conf is sent to the MMS User Agent (B). The information from the MM message retrieved from the MMBox is used for the xIRI MMSForward message.

As shown, m-delivery-ind is sent to the MMS User Agent when the MM message (in the email format) is forwarded to the destination (using MM3: SMTP Send).

As can be inferred, there are subsequent delivery reports or a read report from the destination end.

4.4.3.5.3 Target non-local ID

4.4.3.5.3.1 Overview

Figure 4.4.3.5.3.1-1 gives an overview of the MMS LI messages applicable to the redirecting MMS domain when both originating and redirected-to MMS User Agents are target non-local IDs at the redirecting MMS Relay/Server. Note that either both (as shown) or one of them or none of them can be the target non-local IDs.

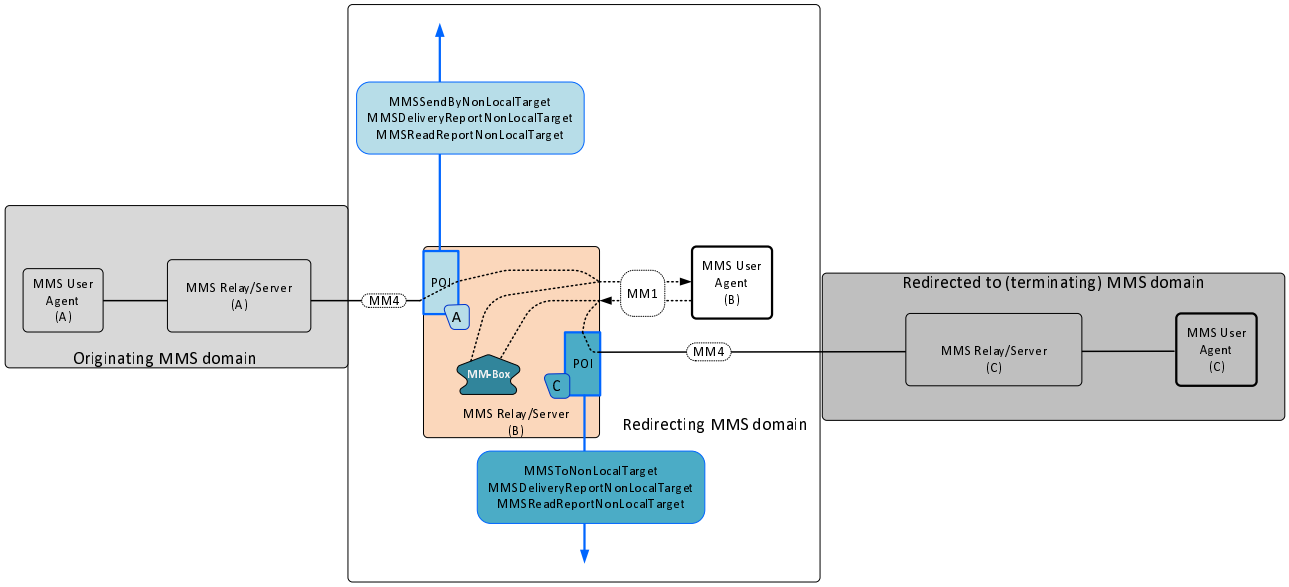


Figure 4.4.3.5.3.1-1: MMS LI messages – redirecting MMS domain (target non-local IDs)

As shown in figure 4.4.3.5.3.1-1, the incoming MM4 messages are referenced for the MMS LI for the originating target non-local ID, i.e. MMS User Agent (A) and outgoing MM4 messages are used for the MMS LI when the destination of the redirected MM message is a target non-local ID, i.e. MMS User Agent (C).

4.4.3.5.3.2 MM messages to MMS LI messages – origination is target non-local ID – case 1

Figure 4.4.3.5.3.2-1 illustrates the scenario for the generation of xIRIs from the MMS messages as the MMS User Agent (B) is notified and subsequently forwards the MM message received from MMS User Agent A (target non-local ID) to MMS User Agent (C). Both MMS User Agent (A) and MMS User Agent (C) are different network domains.

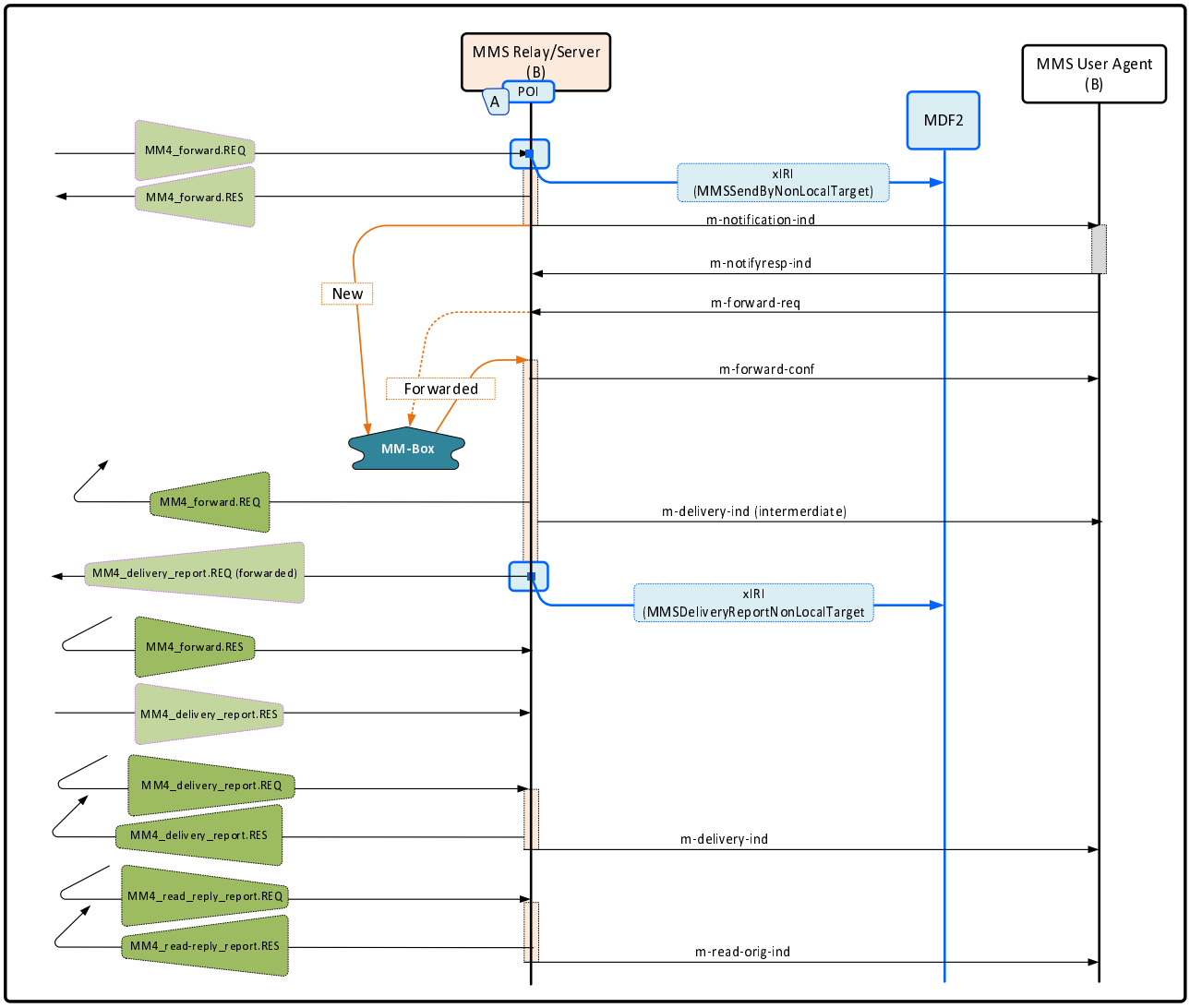


Figure 4.4.3.5.3.2-1: MMS LI messages from MM messages (origin – target non-local ID) - case 1

The triggers for xIRIs are the MM4 messages sent to, or received from, the originating MMS domain.

The MMSSendByNonLocalTarget is generated MMS Relay Server (B) receives the MM4_forward.REQ message.

The trigger for MMSDeliveryReportNonLocalTarget is MMS Relay Server (B) sending of the MM4_delivery-report.REQ message. However, sending of MM4_delivery-report.REQ is conditional, and therefore, the generation of MMSDeliveryReportNonLocalTarget is also conditional.

4.4.3.5.3.3 MM messages to MMS LI messages – origination is target non-local ID – case 2

Figure 4.4.3.5.3.3-1 illustrates the scenario for the generation of xIRIs from the MMS messages as the MMS User Agent (B) is notified and subsequently forwards the MM message received from MMS User Agent A (target non-local ID) to Email Client or to an MMS User Agent with MSISDN based Email Address.

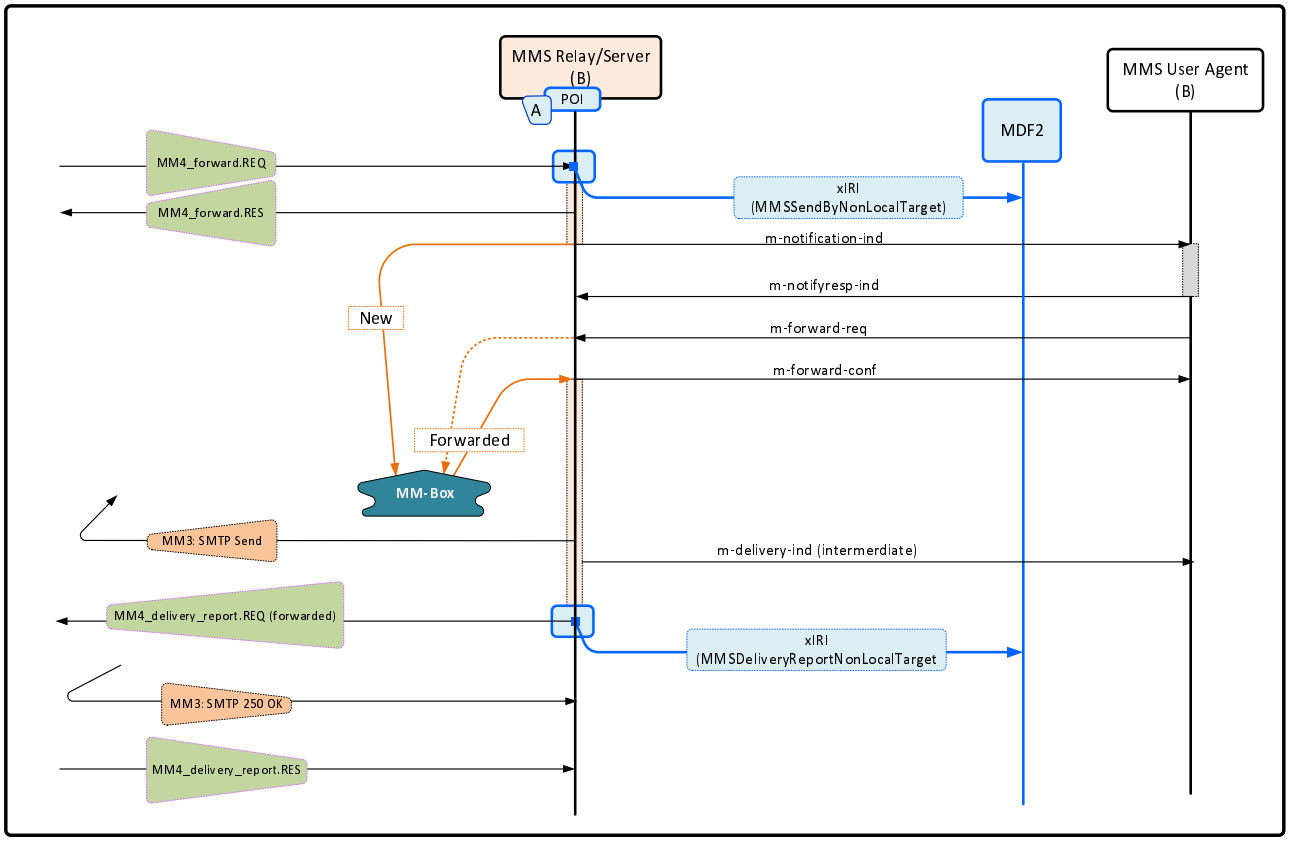


Figure 4.4.3.5.3.3-1: MMS LI messages from MM messages (origin – target non-local ID) - case 2

The triggers for xIRIs are the MM4 messages sent to, or received from, the originating MMS domain.

The MMSSendByNonLocalTarget is generated MMS Relay Server (B) receives the MM4_forward.REQ message.

The trigger for MMSDeliveryReportNonLocalTarget is MMS Relay Server (B) sending of the MM4_delivery-report.REQ message. However, sending of MM4_delivery-report.REQ is conditional, and therefore, the generation of MMSDeliveryReportNonLocalTarget is also conditional.

4.4.3.5.3.4 MM messages to MMS LI messages – destination is target non-local ID – case 1

Figure 4.4.3.5.3.4-1 illustrates the scenario for the generation of xIRIs from the MMS messages as the MMS User Agent (B) is notified and subsequently forwards the MM message received from MMS User Agent A to MMS User Agent C (target non-local ID). Both MMS User Agent (A) and MMS User Agent (C) are different network domains.

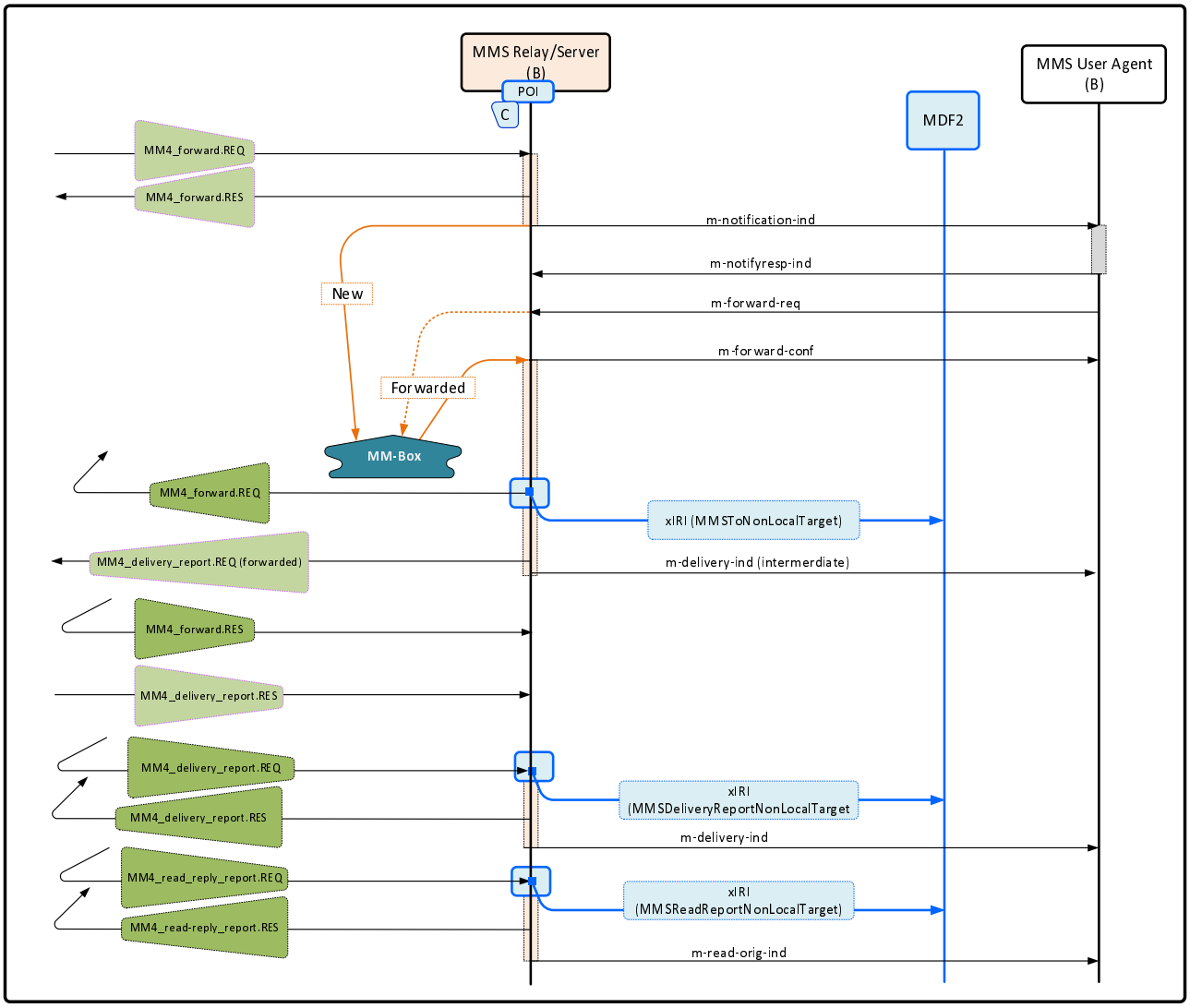


Figure 4.4.3.5.3.4-1: MMS LI messages from MM messages (destination – target non-local ID) - case 1

The triggers for xIRIs are the MM4 messages sent to, or received from, the destination MMS domain.

The MMSToNonLocalTarget is generated MMS Relay Server (B) sends the MM4_forward.REQ message toward the destination MMS domain.

The MM4_delivery_report.REQ received from the redirected-to end triggers the sending of xIRI MMSDeliveryReportNonLocalTarget. Since the receipt of MM4_delivery-report.REQ is conditional, the generation of xIRI MMSDeliveryReportNonLocalTarget is also conditional.

The MM4_read-reply-report.REQ received from the redirected-to end triggers the sending of xIRI MMSReadReportNonLocalTarget. Since the receipt of MM4_ read-reply-report.REQ is conditional, the generation of xIRI MMSReadReportNonLocalTarget is also conditional.

4.4.3.5.3.5 MM messages to MMS LI messages – destination is target non-local ID – case 2

Figure 4.4.3.5.3.5-1 illustrates the scenario for the generation of xIRIs from the MMS messages as the MMS User Agent (B) is notified and subsequently forwards the MM message received from Email Client or MMS User Agent with an MSISDN based Email Address to MMS User Agent C (target non-local ID).

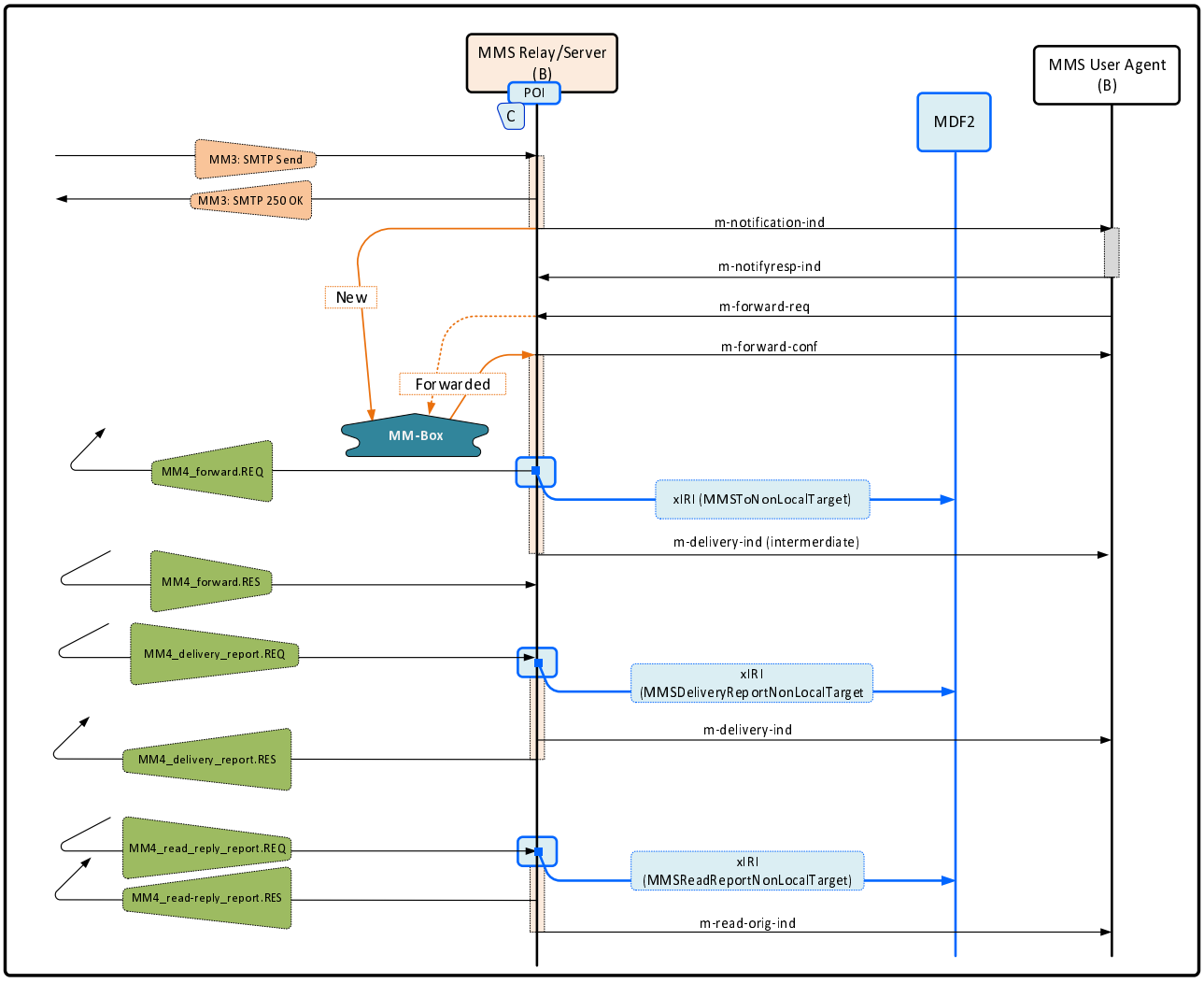


Figure 4.4.3.5.3.5-1: MMS LI messages from MM messages (destination – target non-local ID) - case 2

The triggers for xIRIs are the MM4 messages sent to, or received from, the destination MMS domain.

The MMSToNonLocalTarget is generated MMS Relay Server (B) sends the MM4_forward.REQ message toward the destination MMS domain.

The MM4_delivery_report.REQ received from the redirected-to end triggers the sending of xIRI MMSDeliveryReportNonLocalTarget. Since the receipt of MM4_delivery-report.REQ is conditional, the generation of xIRI MMSDeliveryReportNonLocalTarget is also conditional.

The MM4_read-reply-report.REQ received from the redirected-to end triggers the sending of xIRI MMSReadReportNonLocalTarget. Since the receipt of MM4_ read-reply-report.REQ is conditional, the generation of xIRI MMSReadReportNonLocalTarget is also conditional.

4.4.3.6 Single MMS Relay/Server – non-redirection case

4.4.3.6.1 General

4.4.3.6.1.1 Scope

In the illustrations, both originating and terminating MMS User Agents are served by the same MMS Relay/Server. Six cases are illustrated in the subsequent clauses:

- Immediate retrieval case, originating MMS User Agent is the target.
- Immediate retrieval case, terminating MMS User Agent is the target.

- Deferred retrieval case, originating MMS User Agent is the target.
- Deferred retrieval case, terminating MMS User Agent is the target.
- Deletion case, originating MMS User Agent is the target.
- Deletion case, terminating MMS User Agent is the target.

The case of Email Client or MSISDN based Email Address or target non-local ID do not come into play in these illustrations, since both users are local to the MMS Relay Server.

4.4.3.6.1.2 MMBox

When MM message is sent by the originating MMS User Agent, if requested so, a copy of the MM message is stored in the MMBox with the status showing "sent", if an MMBox is supported for the originating MMS User Agent. Therefore, the storing of the MM message at the MMBox does not happen if the store indication is not sent in the m-send-req message, even if the MMBox is supported for the originating MMS User Agent.

Likewise, when the MMBox is supported, the incoming MM message is automatically stored in the MMBox , if setup and configured so, by the terminating MMS User Agent. The status of thus stored MM message would be "New". In that case, the terminating MMS User Agent would point to the MM message stored in the MMBox for any subsequent actions (e.g. retrieval, deletion, forwarding). The status of the MM message changes according to the action taken (e.g. "Retrieved", "Forwarded").

When MMBox is not used, it is presumed that the MM messages are stored locally at the MMS/Relay Server.

When the MMBox is used, the MMS User Agent may also perform a few MMBox specific operations as illustrated in clause 4.4.3.2.2.

Each MMS User Agent has own inboxes in the MMBox. In the illustrations, separate MMBox shown for each MMS User Agent to infer that individuality. Either both, or one, or none may have the MMBox.

4.4.3.6.1.3 Delivery reports

The delivery report is conditional. The MMS Relay/Server sends a delivery report to the originating MMS User Agent if and only if the originating MMS User Agent has requested for such a report and the terminating MMS User Agent permits the delivery of such a report.

4.4.3.6.1.4 Read reports

The read report from the terminating MMS User Agent is conditional. The terminating MMS User Agent may send a read-report if the originating MMS User Agent requests for such a report and the terminating MMS User chooses to send such a report.

4.4.3.6.2 Overview

Figure 4.4.3.6.2-1 gives an overview where both originating and terminating MMS User Agents are served by the same MMS Relay/Server. In the illustration, both originating and terminating MMS User Agents are the targets.

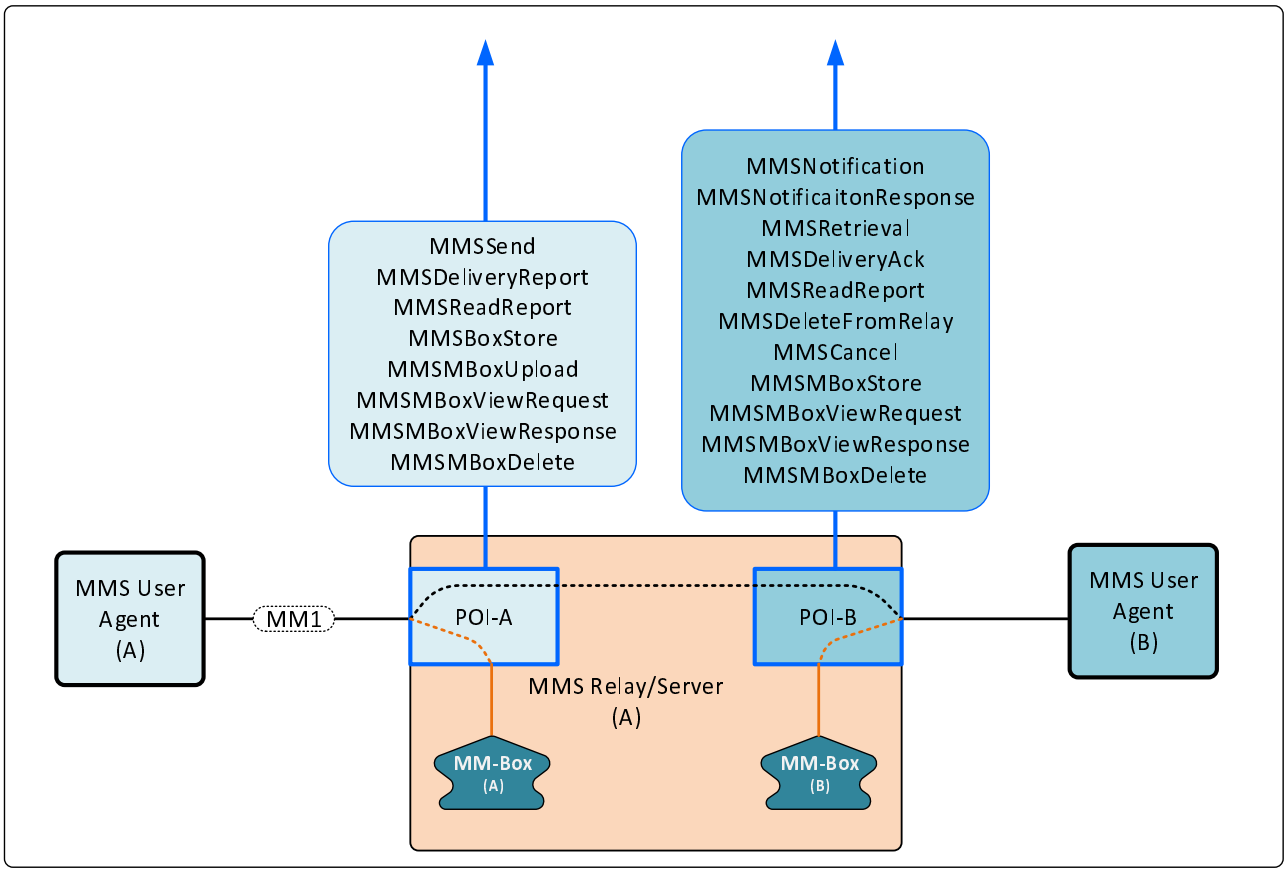


Figure 4.4.3.6.2-1: Single MMS Relay/Server serving originating and terminating targets

The MM1 messages that flow between the MMS Relay/Server and the target MMS User Agent are used to generate the corresponding xIRIs.

Use of the MMBox is optional on a per MMS User Agent level. In other words, either one or both, or neither of the two originating MMS User Agents may have the MMBox. When used, the MMBox specific operations (which either of the two MMS User Agents may perform in a mutually exclusive manner) along with the associated LI aspects are illustrated in 4.4.3.2.2.

4.4.3.6.3 Originating MMS User Agent is the target

4.4.3.6.3.1 General

In all the illustrations, when MM message is sent by the originating MMS User Agent, if requested so, a copy of the MM message is stored in the MMBox. Therefore, the storing of the MM message at the MMBox does not happen if the store indication is not sent in the m-send-req message.

Even though the trigger for the generation of xIRI MMSSend is the m-send-conf message that the MMS-Relay/Server

Since the sending of the m-delivery-ind and m-read-orig-ind are conditional, the generation of xIRIs MMSDeliveryReport and MMSReadReport are also conditional.

4.4.3.6.3.2 Case 1 - Immediate retrieval

Figure 4.4.3.6.3.2-1 gives an overview of the MMS LI messages applicable when single MMS Relay/Server is serving both originating and terminating MMS User Agent with originating MMS User Agent being the target.

The MMS User Agent (B) upon receiving the notification automatically retrieves the MM message (immediate retrieval).

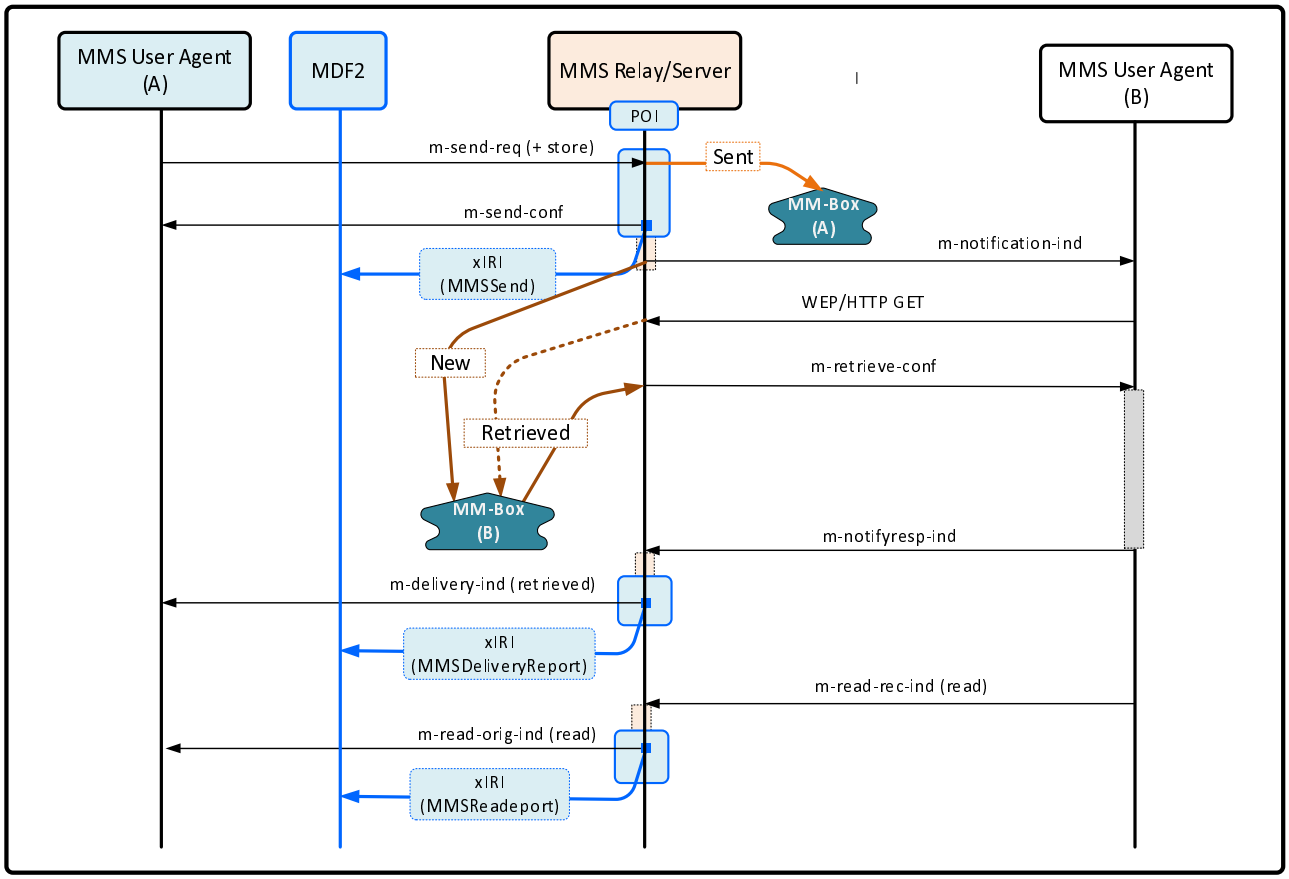


Figure 4.4.3.6.3.2-1: Single MMS Relay/Server – MMS User Agent (A) is the target (immediate retrieval)

In this illustration, the terminating MMS User Agent automatically retrieves the MM message.

4.4.3.6.3.3 Originating MMS User Agent is the target (deferred retrieval)

Figure 4.4.3.6.3.3-1 gives an overview of the MMS LI messages applicable when single MMS Relay/Server is serving both originating and terminating MMS User Agent with originating MMS User Agent being the target.

The MMS User Agent (B) receives the notification and subsequently retrieves the MM message (deferred retrieval).

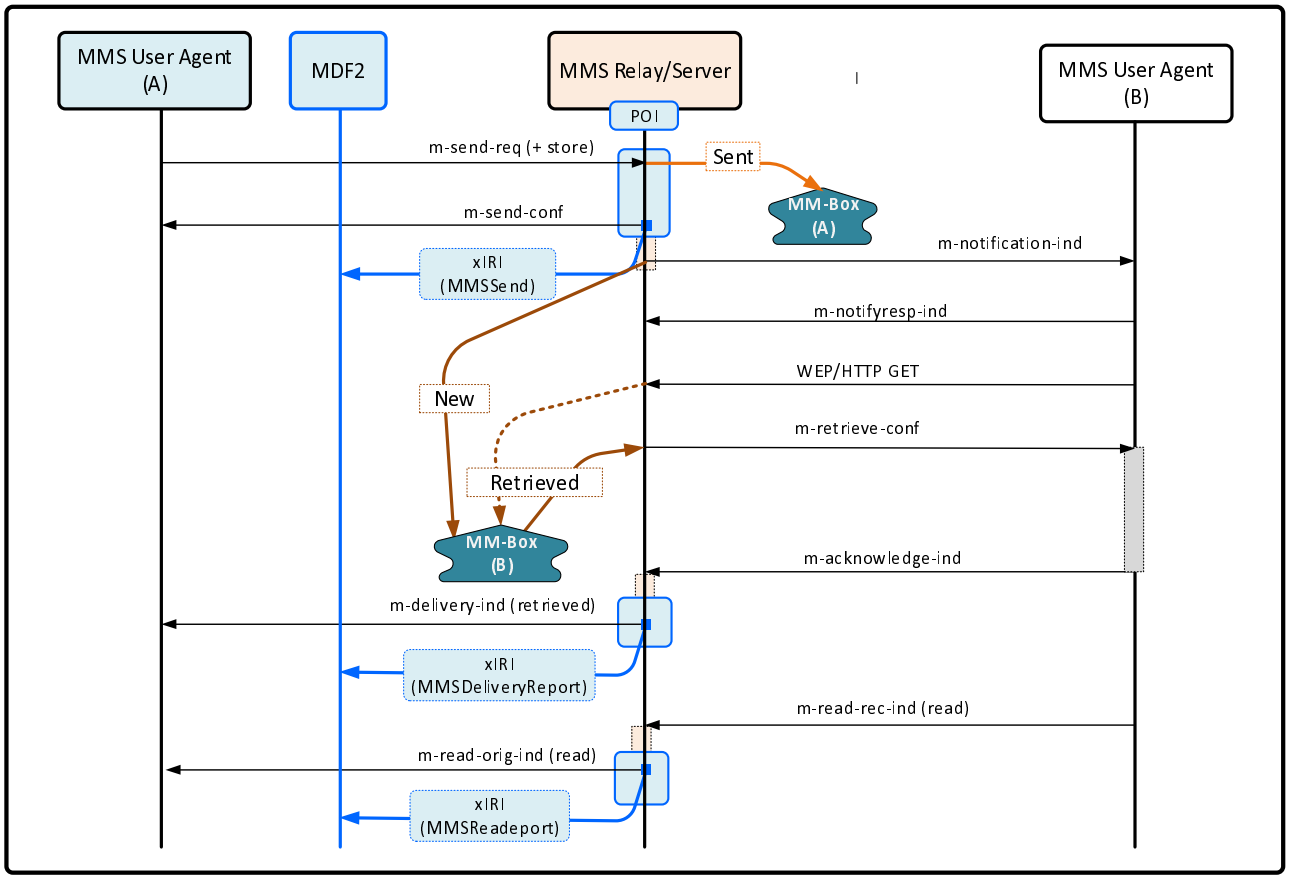


Figure 4.4.3.6.3.3-1: Single MMS Relay/Server – MMS User Agent (A) is the target (deferred retrieval)

In this illustration, the terminating MMS User Agent upon receiving a notification of the incoming MM message retrieves the MM message with a time-gap, referred to as deferred retrieval.

4.4.3.6.3.4 Originating MMS User Agent is the target (deletion)

Figure 4.4.3.6.3.4-1 gives an overview of the MMS LI messages applicable when single MMS Relay/Server is serving both originating and terminating MMS User Agent with originating MMS User Agent being the target.

The MMS User Agent (B) upon receiving the notification and deletes the MM message. This case is applicable only for a deferred retrieval case.

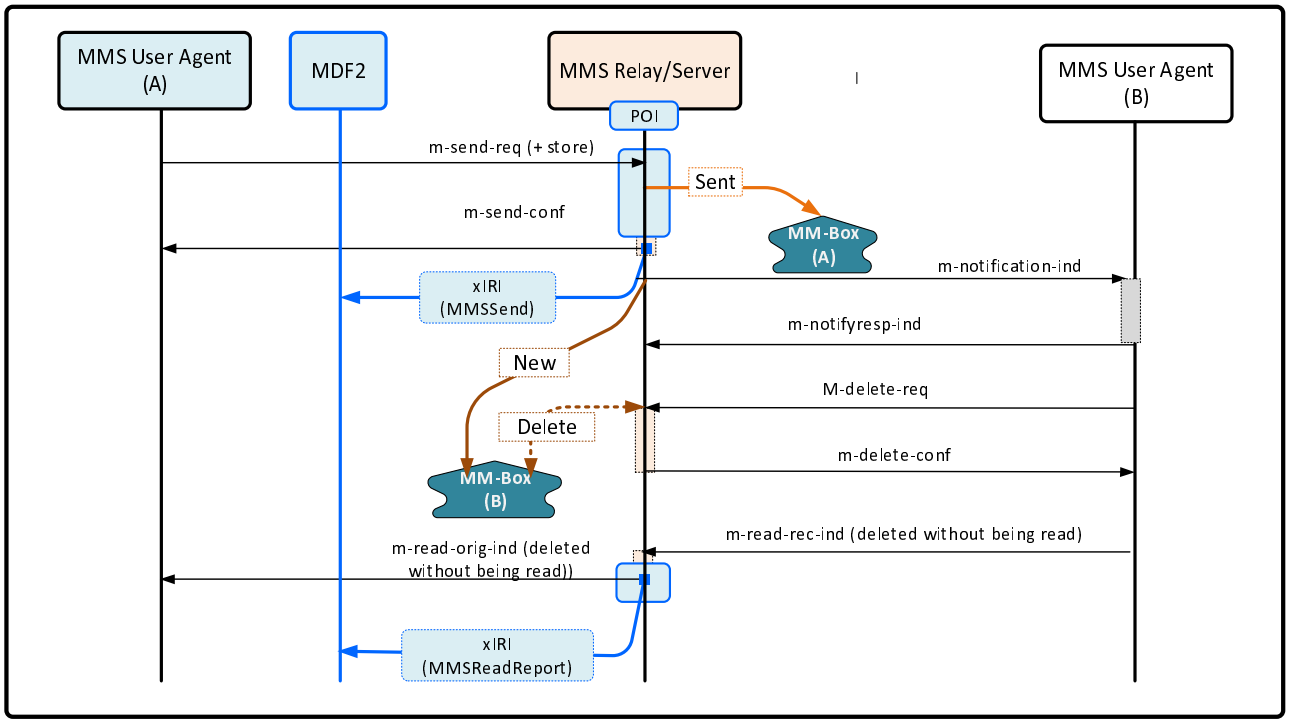


Figure 4.4.3.6.3.4-1: Single MMS Relay/Server – MMS User Agent (A) is the target (deletion)

In this illustration, the terminating MMS User Agent, upon receiving a notification of the incoming MM message deletes the MM message without a retrieval.

4.4.3.6.4 Terminating MMS User Agent is the target (immediate retrieval)

4.4.3.6.4.1 General

In all the illustrations, when MM message is sent by the originating MMS User Agent, if requested so, a copy of the MM message is stored in the MMBox. Therefore, the storing of the MM message at the MMBox does not happen if the store indication is not sent in the m-send-req message.

Since the sending m-read-rec-ind are conditional, the generation of xIRI MMSReadReport is also conditional.

4.4.3.6.4.2 Case 1 - Immediate retrieval

Figure 4.4.3.6.4.2-1 gives an overview of the MMS LI messages applicable when single MMS Relay/Server is serving both originating and terminating MMS User Agent with terminating MMS User Agent being the target.

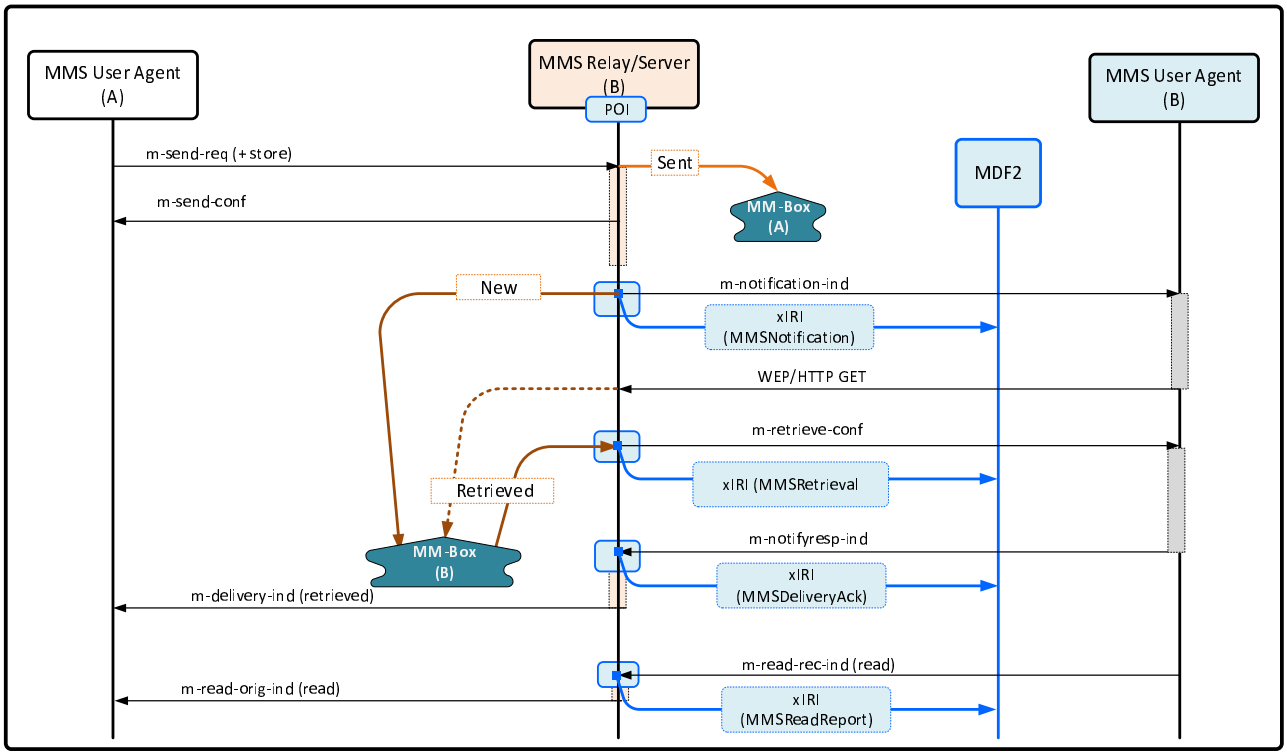


Figure 4.4.3.6.4.2-1: Single MMS Relay/Server – MMS User Agent (B) is the target (immediate retrieval)

In this illustration, the terminating MMS User Agent automatically retrieves the MM message.

4.4.3.6.4.3 Case 2 – deferred retrieval

Figure 4.4.3.6.4.3-1 gives an overview of the MMS LI messages applicable when single MMS Relay/Server is serving both originating and terminating MMS User Agent with terminating MMS User Agent being the target.

The MMS User Agent (B) receives the notification and subsequently retrieves the MM message (deferred retrieval).

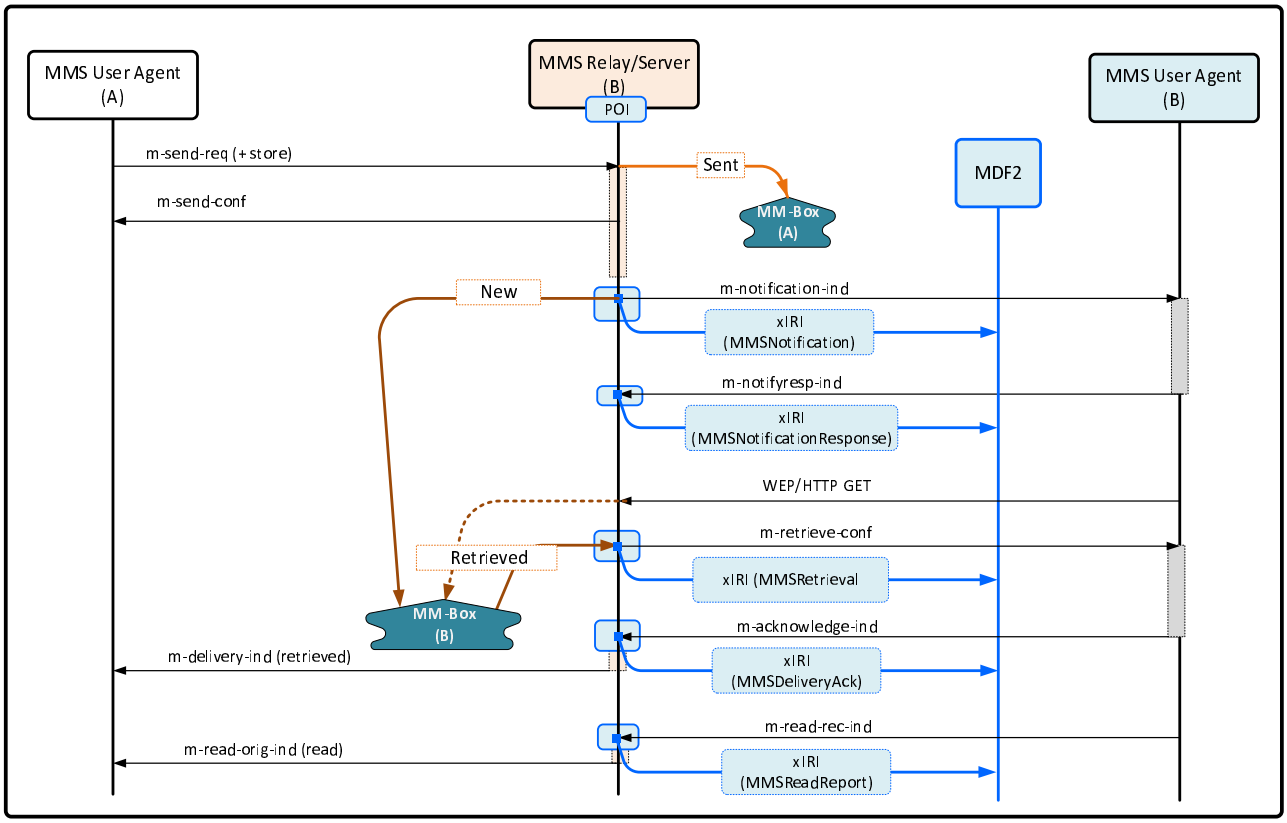


Figure 4.4.3.6.4.3-1: Single MMS Relay/Server – MMS User Agent (B) is the target (deferred retrieval)

In this illustration, the terminating MMS User Agent upon receiving a notification of the incoming MM message retrieves the MM message with a time-gap, referred to as deferred retrieval.

4.4.3.6.4.4 Case 3 - deletion

Figure 4.4.3.6.4.4-1 gives an overview of the MMS LI messages applicable when single MMS Relay/Server is serving both originating and terminating MMS User Agent with terminating MMS User Agent being the target.

The MMS User Agent (B) upon receiving the notification and deletes the MM message. This case is applicable only for a deferred retrieval case.

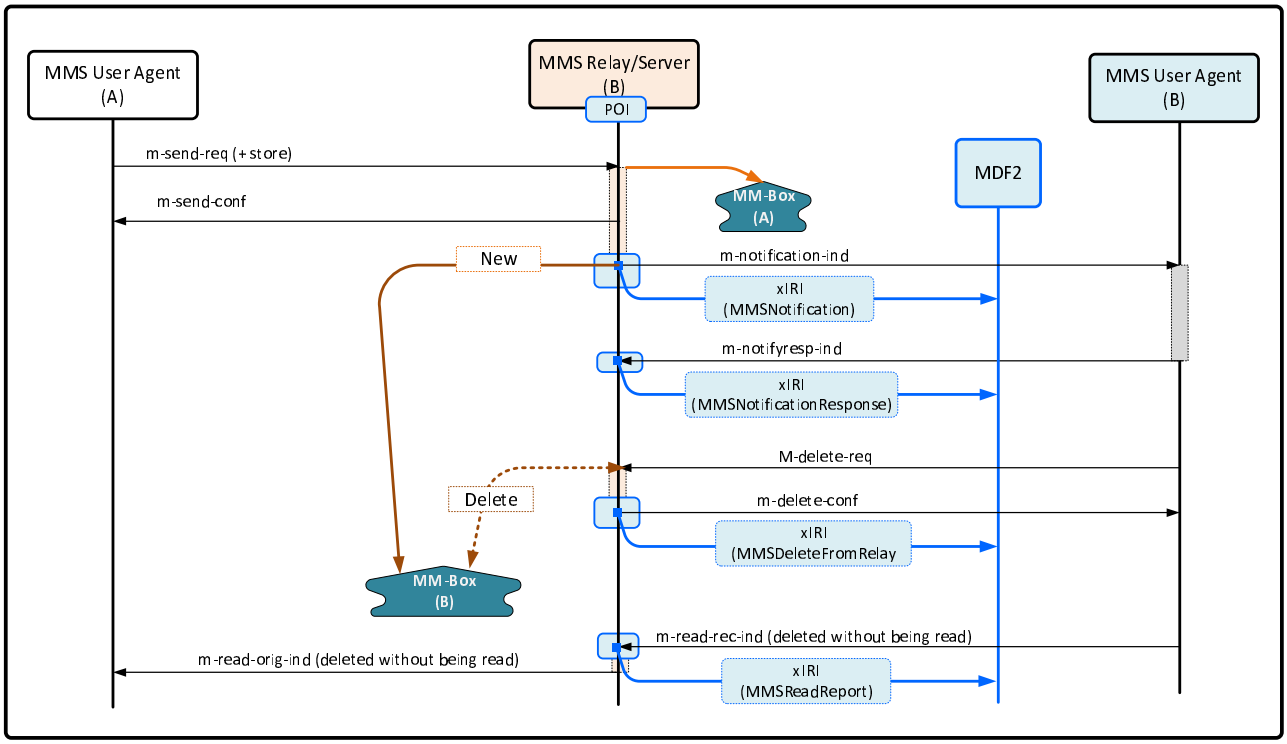


Figure 4.4.3.6.4.4-1: Single MMS Relay/Server – MMS User Agent (B) is the target (deletion)

In this illustration, the terminating MMS User Agent, upon receiving a notification of the incoming MM message deletes the MM message without a retrieval.

4.4.3.7 Single MMS Relay/Server: redirection case

4.4.3.7.1 General

4.4.3.7.1.1 Scope

In the illustrations, originating, redirecting and redirected-to MMS User Agents are served by the same MMS Relay/Server. Seven cases are illustrated in the subsequent clauses:

- Retrieval case, originating MMS User Agent is the target (immediate retrieval).
- Retrieval case, redirecting MMS User Agent is the target (immediate retrieval).
- Retrieval case, redirected-to MMS User Agent is the target (immediate retrieval).
- Retrieval case, redirected-to MMS User Agent is the target (deferred retrieval).
- Deletion case, originating MMS User Agent is the target.
- Deletion retrieval case, redirecting MMS User Agent is the target.
- Deletion retrieval case, redirected-to MMS User Agent is the target.

The retrieval, deletion refers to the actions taken at the final destination, i.e. the redirected-to MMS User Agent. There is no impact due to retrieval method for MMS LI with originating MMS User Agent as a target or with the redirecting MMS User Agent as a target.

The case of Email Client or MSISDN based Email Address as the target Ids or target non-local ID do not come into play in these illustrations, since all users are local to the MMS Relay Server.

4.4.3.7.1.2 MMBox specific steps

In the illustrations, MMBox interactions are shown as and when the MM message pass through. Each MMS User Agent has own inboxes in the MMBox. In the illustrations, separate MMBoxes are shown for each MMS User Agent to infer that individuality.

If the MMBox is not supported or not setup by any of the MMS User Agents, then MMBox related steps shown for that MMS User Agent in the subsequent clauses are not applicable. In that case, the MMS Relay/Server stores the MM message locally and a reference then would be to the internally stored MM message.

Specific descriptions that apply to a particular illustration noted in the respective clause.

When the MMBox is used, each MMS User Agent may also perform a few MMBox specific operations as illustrated in clause 4.4.3.2.2.

Originating MMS User Agent

When MM message is sent by the originating MMS User Agent, if requested so, a copy of the MM message is stored in the MMBox with the status showing "sent", if an MMBox is supported for the originating MMS User Agent. Therefore, the storing of the MM message at the MMBox does not happen if the store indication is not sent in the m-send-req message, even if the MMBox is supported for the originating MMS User Agent.

Redirecting MMS User Agent

When the MMBox is supported, the incoming MM message is automatically stored in the MMBox, if setup and configured so, by the redirecting MMS User Agent. The status of thus stored MM message would be "New". In that case, redirecting MMS User Agent would point to the MM message stored in the MMBox in the forwarding request. Once forwarded, the status of the MM message in the MMBox is changed to "forwarded".

If the MMBox is supported, but not configured to store the incoming MM message automatically, then the MMS User Agent may request the MMS Relay/Server to store the MM message in the MMBox, when a forwarding request is sent. The status of the thus stored MM message would be "forwarded".

Redirected-to MMS User Agent

When the MMBox is supported, the incoming MM message is automatically stored in the MMBox , if setup and configured so, by the redirected-to MMS User Agent. The status of thus stored MM message would be "New". In that case, the redirected-to MMS User Agent would point to the MM message stored in the MMBox for any subsequent actions (e.g. retrieval, deletion, forwarding). The status of the MM message changes according to the action taken (e.g. "Retrieved", "Forwarded").

If the MMBox is supported, but not configured to store the incoming MM message automatically, then the redirected-to MMS User Agent may request the MMS Relay/Server to store the MM message in the MMBox using the m-mbox-store-req message as illustrated in clause 4.4.3.2.2.

4.4.3.7.1.3 Delivery reports

Delivery report to the originating MMS User Agent

The sending of delivery report to the originating MMS User Agent is conditional. The delivery report is sent only if the originating MMS User Agent has requested such a report and the redirecting MMS User Agent permits the sending of such a report.

Delivery report from the redirected-to MMS domain

The sending of delivery report to the redirecting MMS User Agent is conditional. The delivery report is sent only if the redirecting MMS User Agent has requested such a report and the redirected-to MMS User Agent permits the sending of such a report.

4.4.3.7.1.4 Read reports

A read report is not sent to the originating MMS User Agent.

A read report may be sent by the redirected-to MMS User Agent and delivered to the redirecting MMS User Agent, if the redirecting MMS User Agent requests for such a report, and the redirected-to MMS User Agent sends such a report.

4.4.3.7.2 Overview

Figure 4.4.3.7.2-1 gives an overview where originating, redirecting and redirected-to MMS User Agents are served by the same MMS Relay/Server. In the illustration, all three MMS User Agents are the targets.

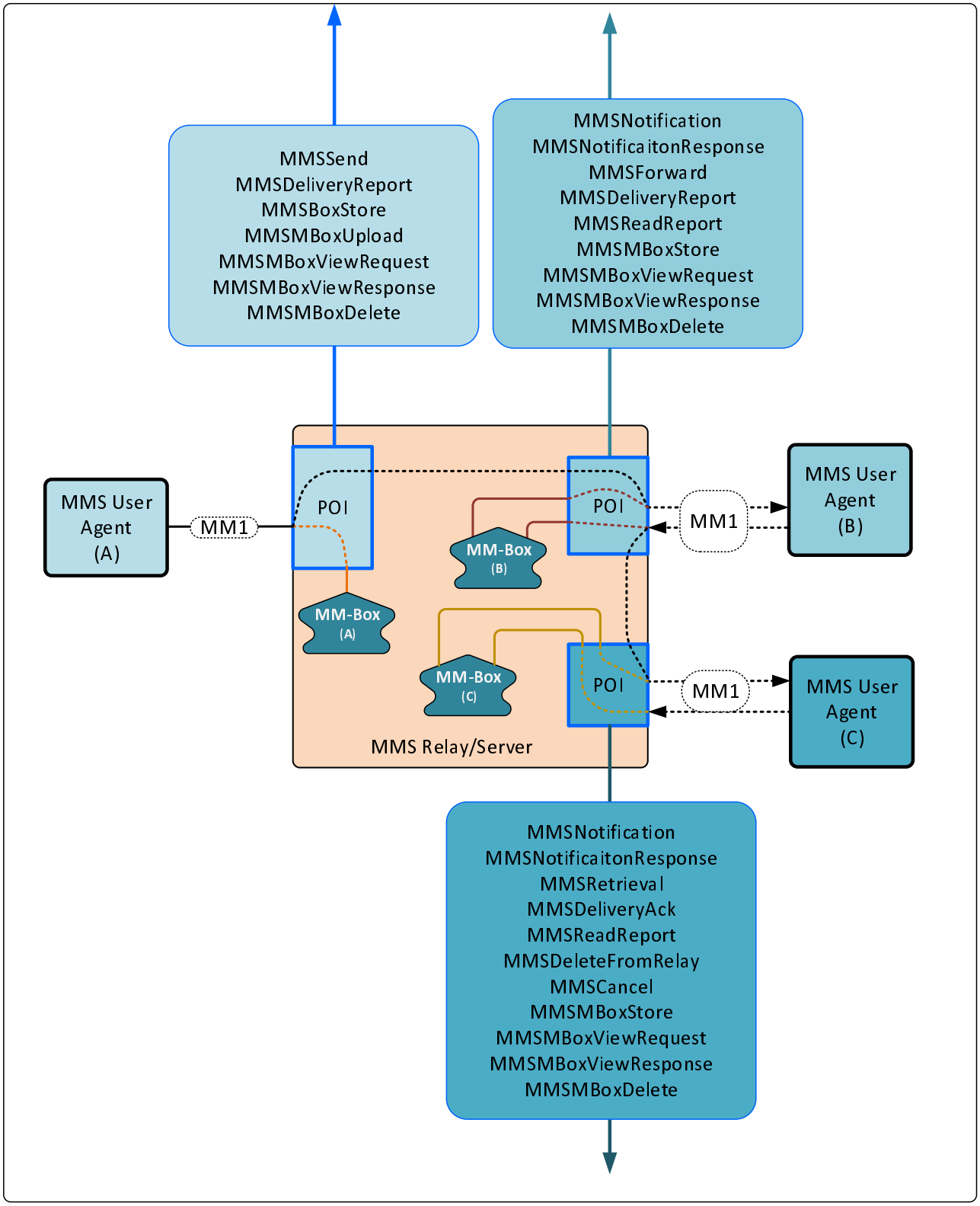


Figure 4.4.3.7.2-1: Single MMS Relay/Server serving originating and terminating targets

The MM1 messages that flow between the MMS Relay/Server and the target MMS User Agent are used to generate the corresponding xIRIs.

Use of the MMBox is optional on a per MMS User Agent level. In other words, either one, or two, or three, or none of the three MMS User Agents may have the MMBox. When used, the MMBox specific operations (which either of the three MMS User Agents may perform in a mutually exclusive manner) along with the associated LI aspects are illustrated in 4.4.3.2.2.

4.4.3.7.3 MM message retrieval by the redirected-to MMS User Agent

4.4.3.7.3.1 General

In the illustrations single MMS Relay/Server serves all the three MMS User Agents. Four scenarios are illustrated:

- MMS User Agent (A) is the target. MMS User Agent (C) retrieves the MM message automatically.
- MMS User Agent (B) is the target. MMS User Agent (C) retrieves the MM message automatically.
- MMS User Agent (C) is the target. MMS User Agent (C) retrieves the MM message automatically.
- MMS User Agent (C) is the target. MMS User Agent (C) retrieves the MM message after a time-gap (deferred retrieval).

The retrieval, automatic or deferred has no impact on the MMS LI when either originating MMS User Agent (A) or the redirecting MMS User Agent (B) is a target.

4.4.3.7.3.2 Originating MMS User Agent (A) is the target

In figure 4.4.3.7.3.2-1, the originating MMS User Agent (A) is the target and sends an MM message to MMS User Agent (B) who in turn forwards the same to the MM User Agent (C). The MMS User Agent (C) retrieves the MM message automatically.

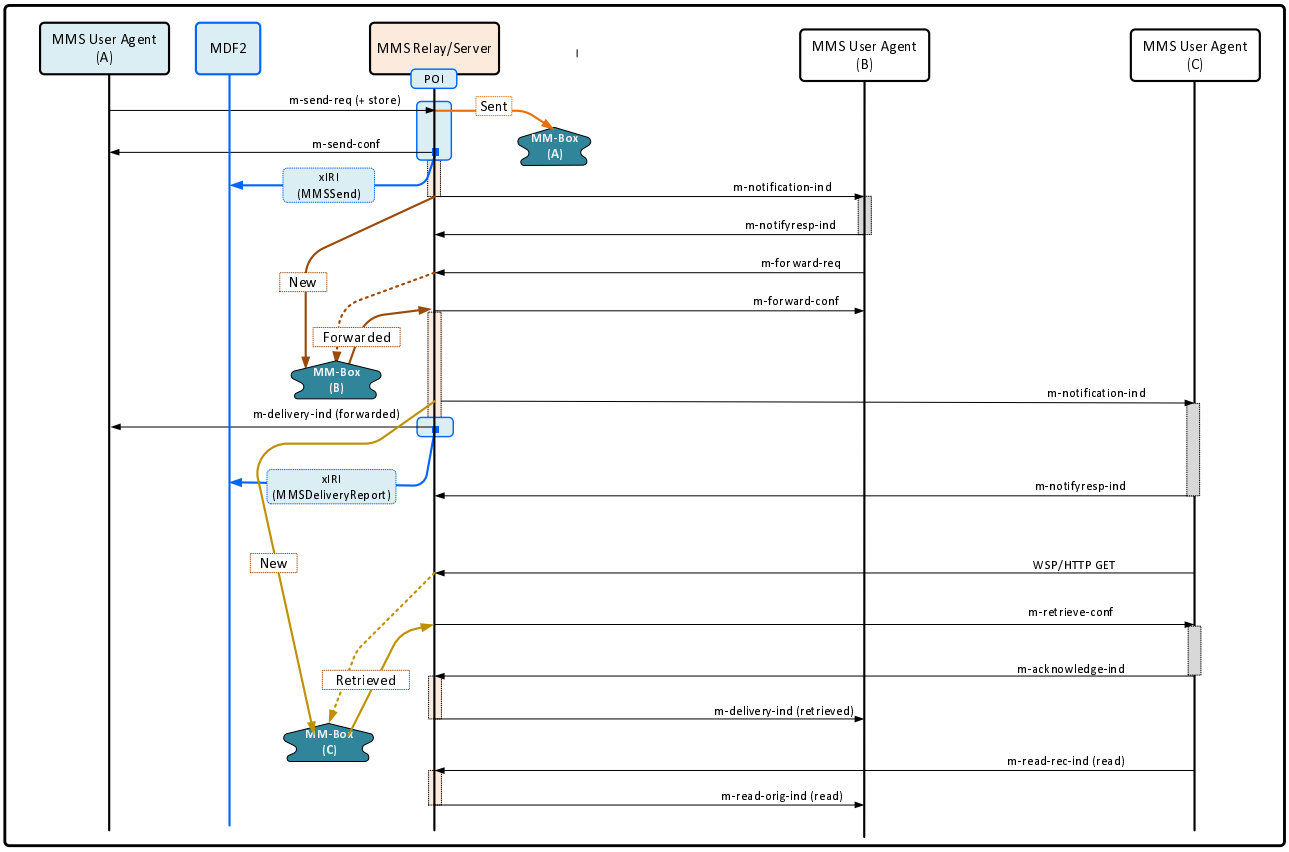


Figure 4.4.3.7.3.2-1: Single MMS Relay/Server – MMS User Agent (A) is the target

When MM message is sent by the originating MMS User Agent, if requested so, a copy of the MM message is stored in the MMBox. Therefore, the storing of the MM message at the MMBox does not happen if the store indication is not sent in the m-send-req message.

Even though the trigger for the generation of xIRI MMSSend is the m-send-conf message that the MMS-Relay/Server sends to the MMS User Agent (A), the xIRI includes information from the m-send-req as well.

Since the sending of the m-delivery-ind (forwarded) is conditional, the generation of xIRIs MMSDeliveryReport is also conditional.

4.4.3.7.3.3 Redirecting MMS User Agent (B) is the target

In figure 4.4.3.7.3.3-1, the originating MMS User Agent (A) sends an MM message to MMS User Agent (B) who in turn forwards the same to the MM User Agent (C). In this illustration, the MMS User Agent (B) is the target. The MMS User Agent (C) retrieves the MM message automatically.

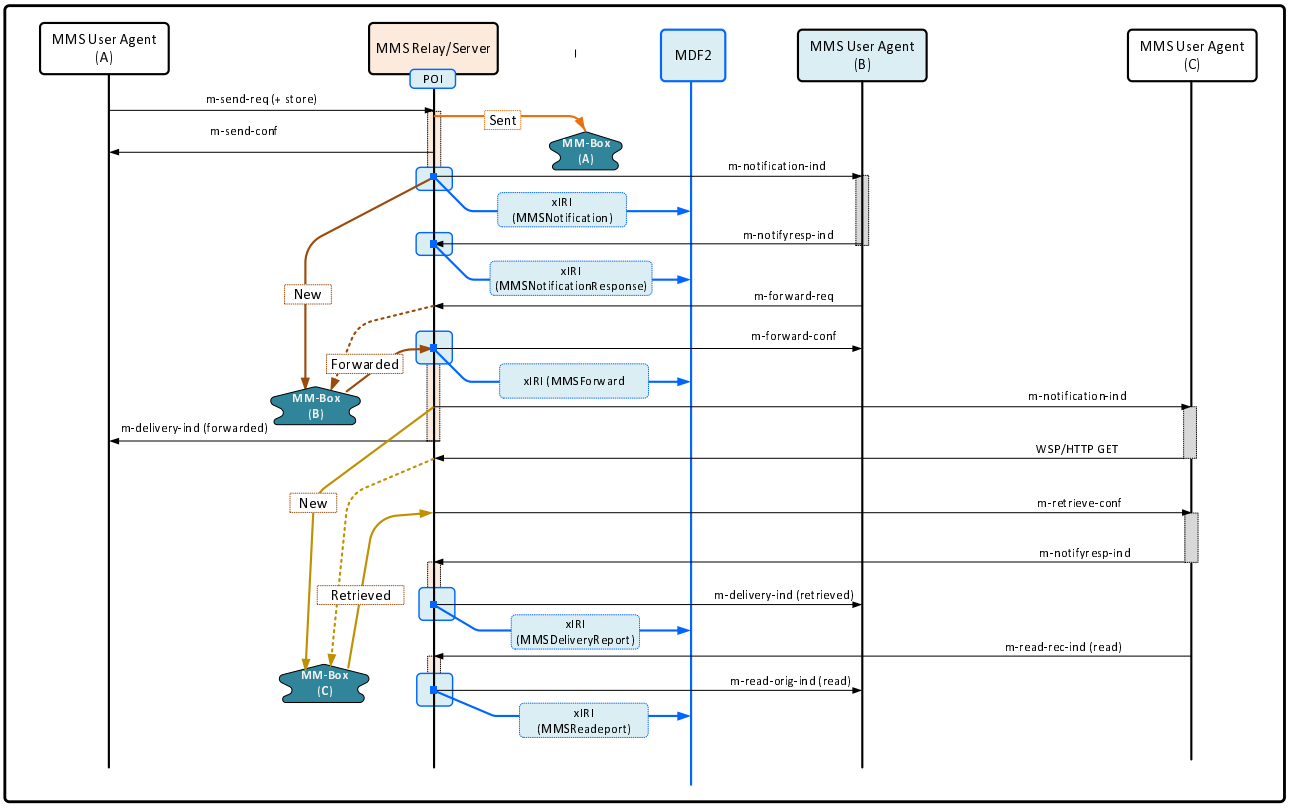


Figure 4.4.3.7.3.3-1: Single MMS Relay/Server – MMS User Agent (B is the target

When MM message is sent by the originating MMS User Agent, if requested so, a copy of the MM message is stored in the MMBox. Therefore, the storing of the MM message at the MMBox does not happen if the store indication is not sent in the m-send-req message.

Even though the trigger for the generation of xIRI MMSForward is the m-forward-conf message that the MMS-Relay/Server sends to the MMS User Agent (B), the xIRI includes information from retrieved MM message as well

Since the sending of the m-delivery-ind and m-read-orig-ind to the MMS User Agent (B) are conditional, the generation of xIRIs MMSDeliveryReport and MMSReadReport are also conditional.

4.4.3.7.3.4 Redirected-to MMS User Agent (C) is the target (immediate retrieval)

In figure 4.4.3.7.3.4-1 the originating MMS User Agent (A) sends an MM message to MMS User Agent (B) who in turn forwards the same to the MM User Agent (C). In this illustration, the MMS User Agent (C) is the target. The MMS User Agent (C) retrieves the MM message automatically.

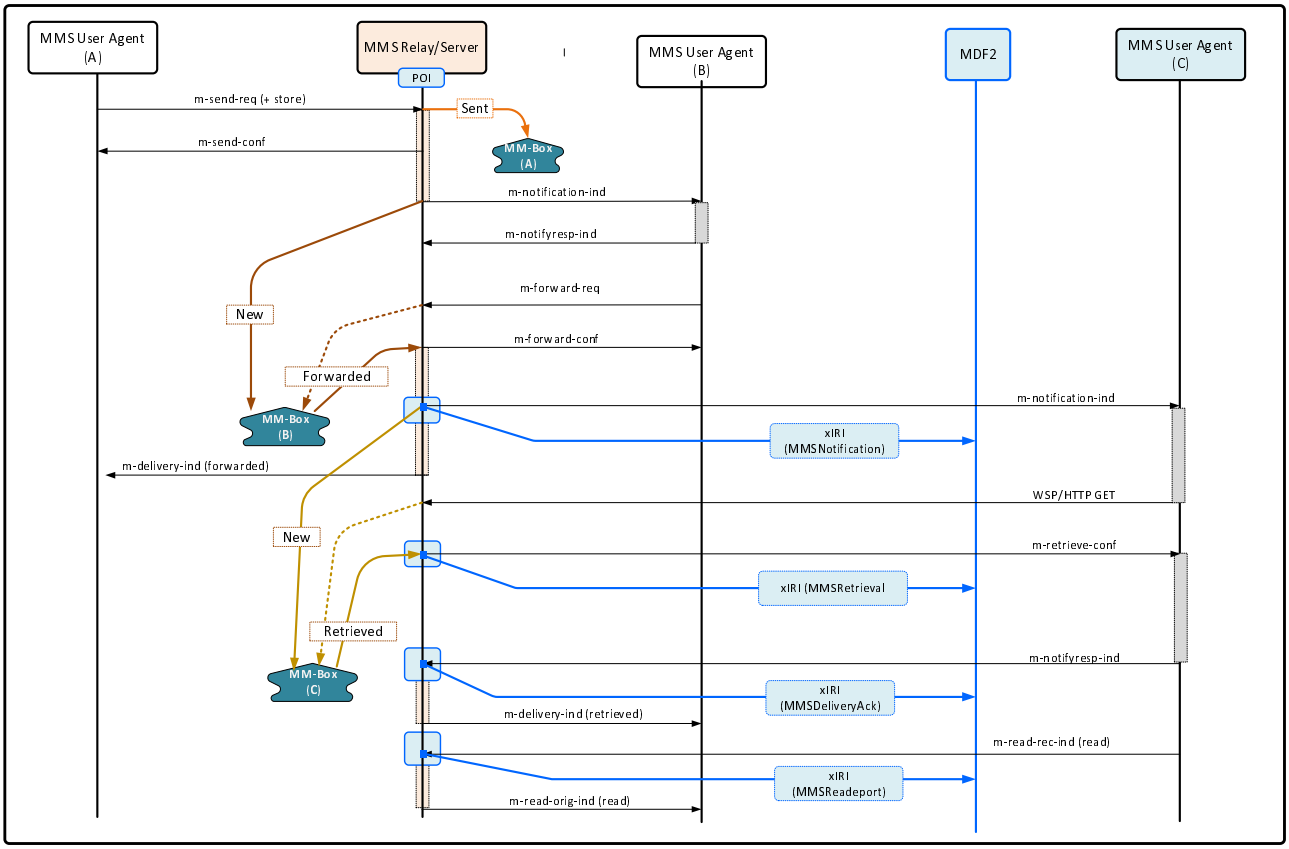


Figure 4.4.3.7.3.4-1: Single MMS Relay/Server – MMS User Agent (C) is the target (immediate)

When MM message is sent by the originating MMS User Agent, if requested so, a copy of the MM message is stored in the MMBox. Therefore, the storing of the MM message at the MMBox does not happen if the store indication is not sent in the m-send-req message.

The MMSDeliveryAck is sent when the whe MMS Relay/Server sends the notifyresp-ind to the target MMS User Agent (C).

The trigger for the xIRIMMSReadReport is MMS Relay Server receiving the m-read-rec-ind from the target MMS User Agent (C). However, MMS User Agent (C) sending of the m-read-rec-ind is conditional, and therefore, the generation of xIRI MMSReadReport is conditional.

4.4.3.7.3.5 Redirected-to MMS User Agent (C) is the target (deferred retrieval)

In figure 4.4.3.7.3.5-1 the originating MMS User Agent (A) sends an MM message to MMS User Agent (B) who in turn forwards the same to the MM User Agent (C). In this illustration, the MMS User Agent (C) is the target. The MMS User Agent (C) retrieves the MM message after a time-gap (deferred retrieval).

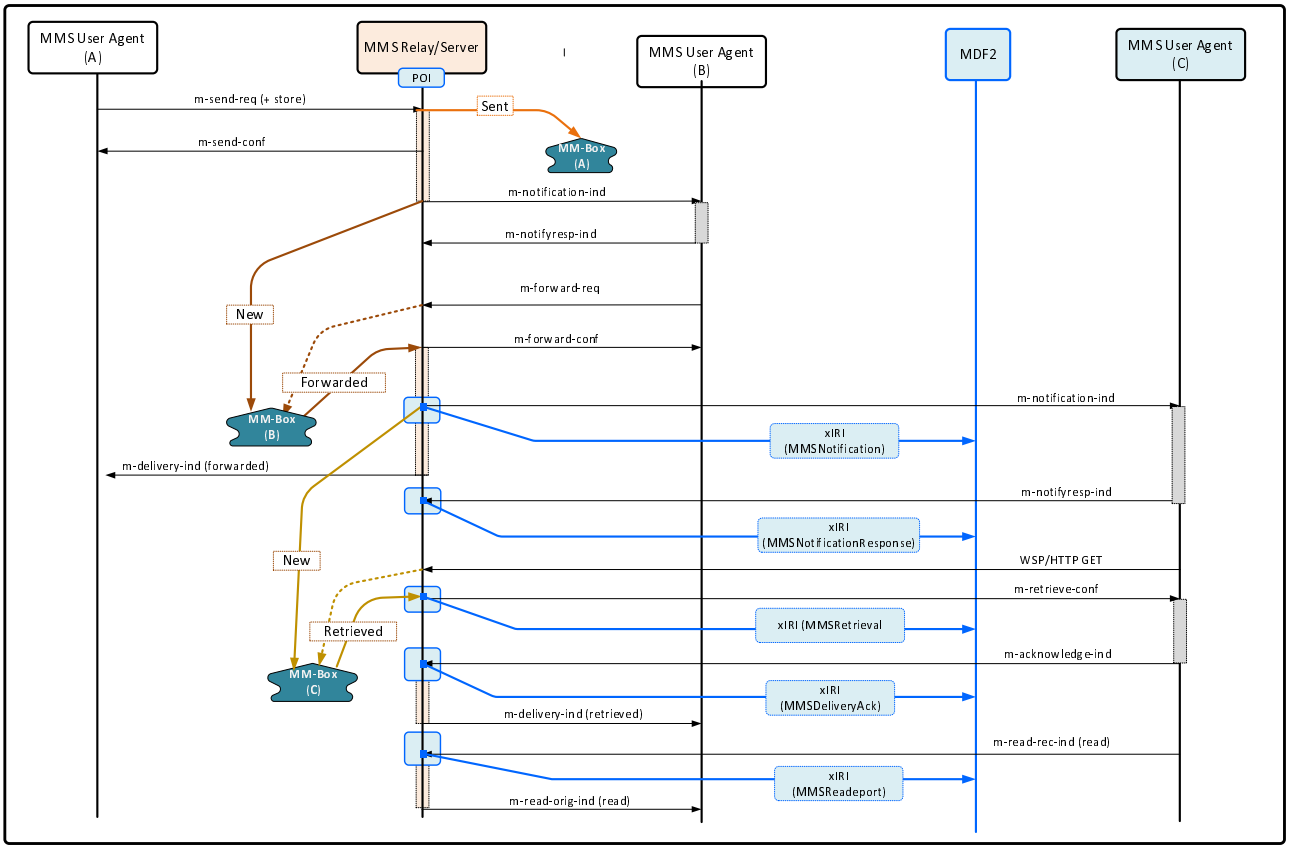


Figure 4.4.3.7.3.5-1: Single MMS Relay/Server – MMS User Agent (C) is the target (deferred)

When MM message is sent by the originating MMS User Agent, if requested so, a copy of the MM message is stored in the MMBox. Therefore, the storing of the MM message at the MMBox does not happen if the store indication is not sent in the m-send-req message.

The MMSDeliveryAck is sent when the whe MMS Relay/Server sends the m-acknowledge-ind to the target MMS User Agent (C).

The trigger for the xIRIMMSReadReport is MMS Relay Server receiving the m-read-rec-ind from the target MMS User Agent (C). However, MMS User Agent (C) sending of the m-read-rec-ind is conditional, and therefore, the generation of xIRI MMSReadReport is conditional.

4.4.3.7.4 Deletion of MM message by the redirected-to MMS User Agent

4.4.3.7.4.1 General

In the illustrations single MMS Relay/Server serves all the three MMS User Agents. In the illustration, the redirected-to MMS User Agent (C) upon receiving the notification deletes the MM message without a prior retrieval. Three scenarios are illustrated:

- MMS User Agent (A) is the target.
- MMS User Agent (B) is the target.
- MMS User Agent (C) is the target.

Other than the status of the MM message in the read-report sent to the redirecting MMS user Agent, the MMS LI has no impact due to the deletion of the MM message by MMS User Agent (C) when MMS User Agent (A) or MMS User Agent B) are the targets.

4.4.3.7.4.2 Originating MMS User Agent (A) is the target

In figure 4.4.3.7.4.2-1, the originating MMS User Agent (A) is the target and sends an MM message to MMS User Agent (B) who in turn forwards the same to the MM User Agent (C). MMS User Agent (C) deletes the MM message without a prior retrieval. In this illustration, MMS user Agent (A) is the target.

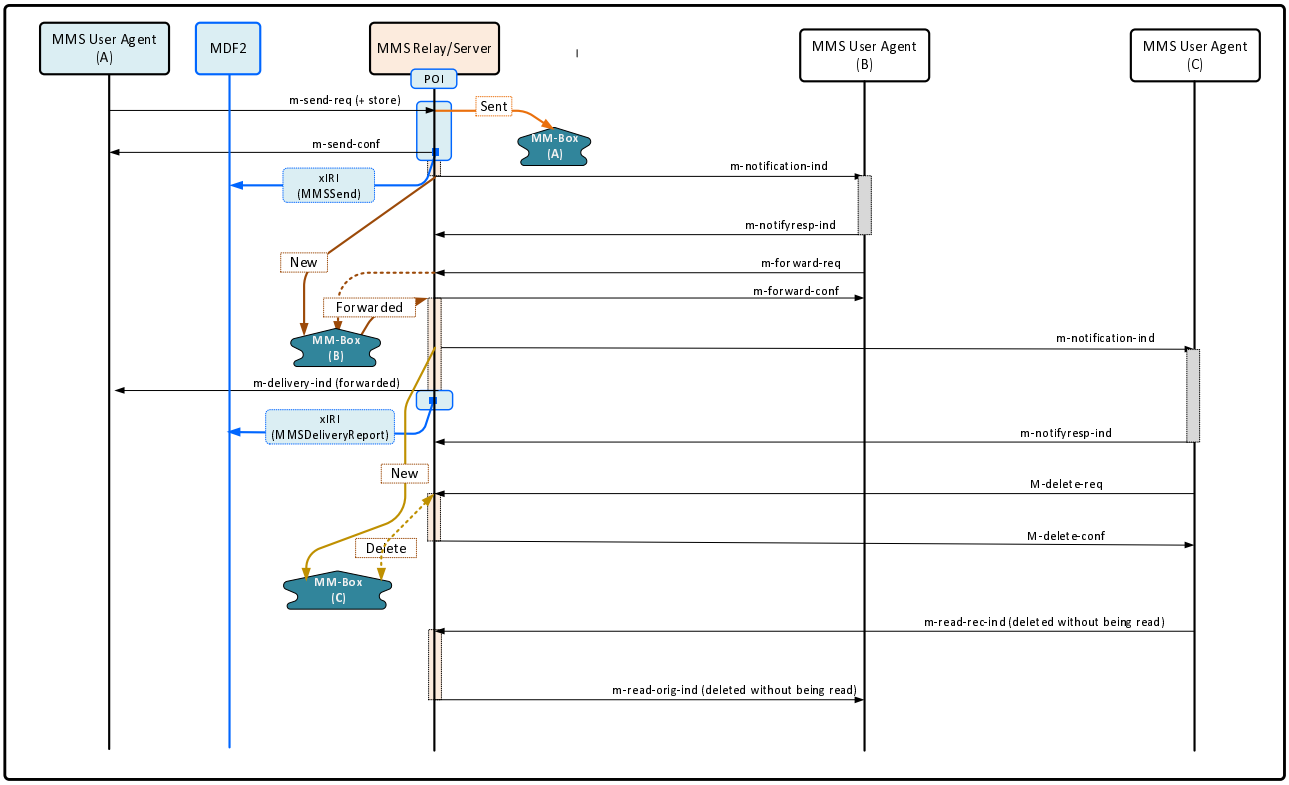


Figure 4.4.3.7.4.2-1: Single MMS Relay/Server – MMS User Agent (A) is the target (delete)

When MM message is sent by the originating MMS User Agent, if requested so, a copy of the MM message is stored in the MMBox. Therefore, the storing of the MM message at the MMBox does not happen if the store indication is not sent in the m-send-req message.

Even though the trigger for the generation of xIRI MMSSend is the m-send-conf message that the MMS-Relay/Server sends to the MMS User Agent (A), the xIRI includes information from the m-send-req as well.

Since the sending of the m-delivery-ind (forwarded) is conditional, the generation of xIRIs MMSDeliveryReport is also conditional.

The deletion of MM message before retrieval has no impact on the MMS LI when MMS User Agent (A) is the target.

4.4.3.7.4.3 Redirecting MMS User Agent (B) is the target

In figure 4.4.3.7.4.3-1, the originating MMS User Agent (A) is the target and sends an MM message to MMS User Agent (B) who in turn forwards the same to the MM User Agent (C). MMS User Agent (C) deletes the MM message without a prior retrieval. In this illustration, the MMS User Agent (B) is the target.

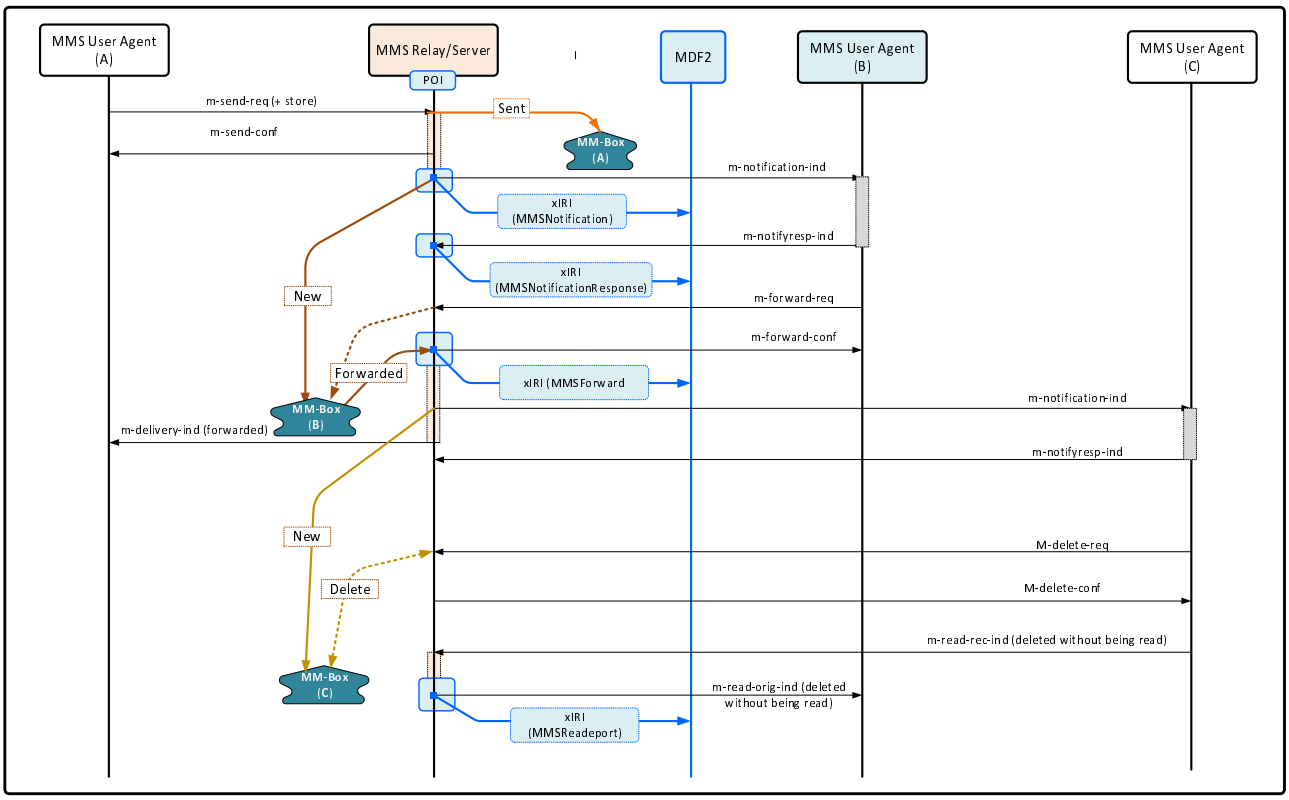


Figure 4.4.3.7.4.3-1: Single MMS Relay/Server – MMS User Agent (B) is the target (delete)

When MM message is sent by the originating MMS User Agent, if requested so, a copy of the MM message is stored in the MMBox. Therefore, the storing of the MM message at the MMBox does not happen if the store indication is not sent in the m-send-req message.

Even though the trigger for the generation of xIRI MMSForward is the m-forward-conf message that the MMS-Relay/Server sends to the MMS User Agent (B), the xIRI includes information from retrieved MM message as well.

If a read report is requested by the redirecting MMS User Agent (B), then the MMS User Agent (C) may send a m-read-rec-ind with "deleted without being read". The xIRI MMSReadReport is generated when the MMS Relay/Server sends the m-read-orig-ind message to the MMS User Agent (B), the target.

Since the reception of m-read-rec-ind from MMS User Agent (C) is conditional, the sending of m-read-orig-ind to MMS User Agent (B) is also conditional. Therefore, the generation of MMSReadReport is also conditional.

4.4.3.7.4.4 Redirected-to MMS User Agent (C) is the target

In figure 4.4.3.7.4.4-1, the originating MMS User Agent (A) is the target and sends an MM message to MMS User Agent (B) who in turn forwards the same to the MM User Agent (C). MMS User Agent (C) deletes the MM message without a prior retrieval. In this illustration, the MMS User Agent (C) is the target.

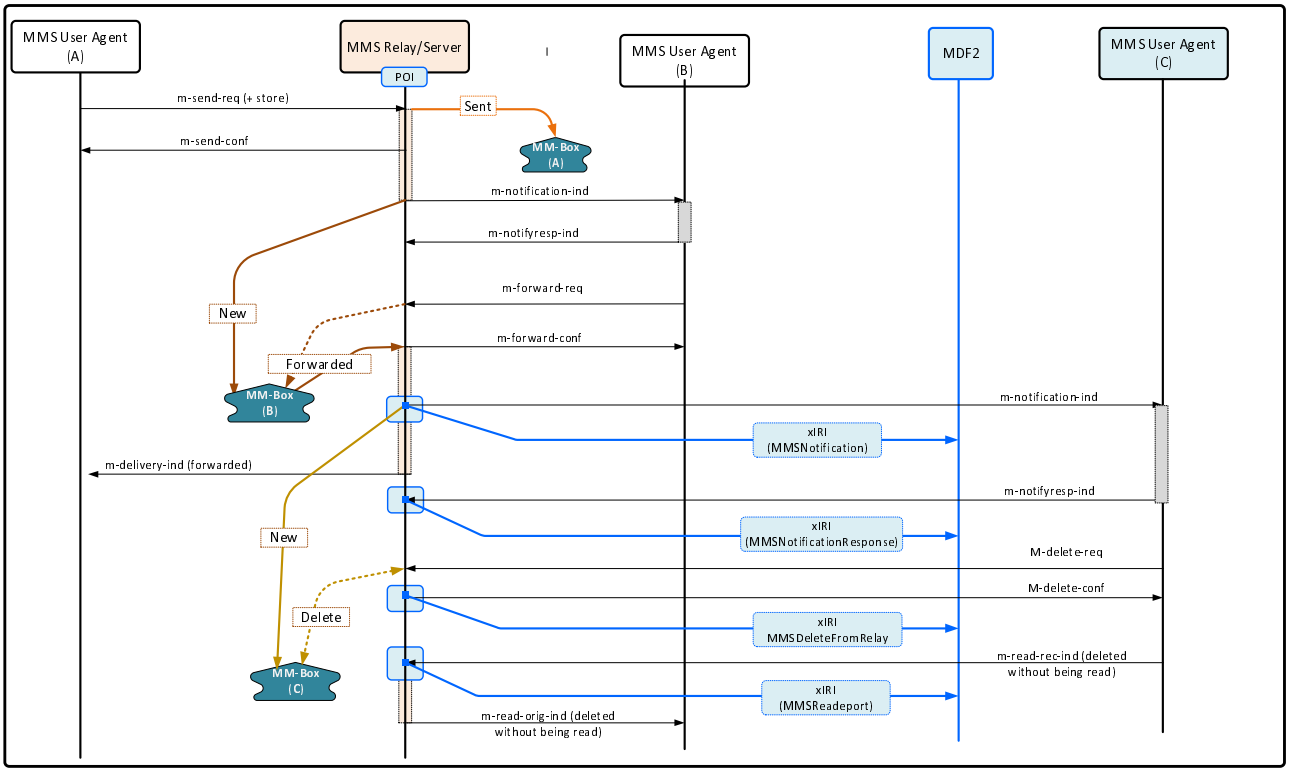


Figure 4.4.3.7.4.4-1: Single MMS Relay/Server – MMS User Agent (C) is the target (delete)

When MM message is sent by the originating MMS User Agent, if requested so, a copy of the MM message is stored in the MMBox. Therefore, the storing of the MM message at the MMBox does not happen if the store indication is not sent in the m-send-req message.

The MMSDeliveryAck is sent when the whe MMS Relay/Server sends the notifyresp-ind to the target MMS User Agent (C).

If a read report is requested by the redirecting MMS User Agent (B), then the MMS User Agent (C) may send a m-read-rec-ind with "deleted without being read". The xIRI MMSReadReport is generated when the MMS Relay/Server receives the m-read-rec-ind message.

Since the reception of m-read-rec-ind from MMS User Agent (C) is conditional, the generation of MMSReadReport is also conditional.

Annex A:
Change history

Change history							
Date	**Meeting**	**TDoc**	**CR**	**Rev**	**Cat**	**Subject/Comment**	**New version**
2024-07	SA3LI#94	S3i240511				Initial version	0.0.1
2024-11	SA3LI#95	S3i240753				Revised as agreed by SA3LI	0.0.2
2026-07	SA3LI#102	S3i260363				Agreed for Change Control	0.0.3
2026-08	SA#113	SP-260734				Submitted to SA#113 Plenary for Approval	1.0.0
2026-09	SA#113	-				Approved by SA#113 Plenary for Change Control	19.0.0

History

Version	Date	Status
V19.0.0	October 2026	Publication